

Supplementary Material for “Baseline Configuration and Random-Sampling Controls in Metaheuristic Comparisons for Wind Farm Layout Optimization”

Prince Solanki, Pulkit Dwivedi, Vanita Garg, and Vinay Shukla

Tables, figures, sections, algorithms and equations of this document are numbered S1, S2, $\dots$; references without the prefix S (e.g., Table II or (4)) point to the main text. The sections follow the order of the main text: details of the evaluation protocol and of the implementation (Section S-I), of the effect of the baseline setting and the main comparison (Section S-II), of the component analysis, the practical equivalence and the robustness of the statistical conclusions (Section S-III), of the tests beyond the benchmark (Section S-IV) and of the sensitivity to the modeling assumptions, including the direction resolution of the wind roses (Section S-V), followed by the theoretical details (Section S-VI), the algorithm diagnostics (Section S-VII), the model derivations (Section S-VIII), a note on published Kusiak–Song results (Section S-IX) and a map from each result to the script that produces it (Section S-X). Unless stated otherwise, all results are based on 30 seed-paired runs of 6,030 evaluations (objective calls, including the initial population and, for MS-SLSQP, the finite-difference gradient calls) with random initialization; the method names are those of the main text, and PSO denotes the constriction setting (7) unless the old setting is named. All Horns Rev 1 results use the corrected direction binning of Section S-I-G. The results hold for the discretized wind roses of the benchmark (24 bins of 15°) and of Horns Rev 1 (5° bins); Section S-V-C re-evaluates the final layouts with 1° bins. All tables of Sections S-II–S-V are generated from the published run data by `analysis/mpce_results.py`, except those of Section S-III-A (`analysis/mpce_inference_extra.py`), of Section S-II-B (`analysis/mpce_csweep.py`) and of Section S-V-C (`analysis/mpce_direction.py`); the diagnostics of Section S-VII come from instrumented reruns (`analysis/mpce_diagnostics.py`), and every number of Section S-VI is recomputed by `analysis/make_theory_figures.py`.

S-I Evaluation Protocol and Implementation Details

This section gives the data, the pseudocode and the settings needed to reproduce the protocol of Section V: the two wind data sets of the 68 benchmark cases, the salp-swarm methods compared as Phase-1 alternatives in the component analysis, the details of the ranking and the tests (Sections S-I-A–S-I-C), and the implementation details that the main text does not state: iteration counts, MS-SLSQP, the feasibility-preserving initialization, the Horns Rev 1 model and the feasibility tolerance (Sections S-I-D–S-I-H). All settings are those of the code in `analysis/` (Section S-X).

S-I-A Wind Data

Table S1 lists the two wind data sets of the Kusiak–Song benchmark [1, 4].

Table S1: Wind Data Sets I and II: probability p_l of direction bin l ($[15(l-1), 15l)^\circ$, direction the wind blows towards, counter-clockwise from east) and Weibull scale ψ ; $\zeta = 2$ in all bins. The most probable bin is $l = 7$ (90° – 105° towards, i.e., wind from 165° – 180° in the meteorological convention $\theta_{\text{met}} = 270^\circ - \theta$) for Data Set I and $l = 13$ (180° – 195° towards, i.e., wind from 75° – 90°) for Data Set II.

Bin l	1	2	3	4	5	6	7	8	9	10	11	12
DS I: p_l ($\psi = 13$)	0.000	0.010	0.010	0.010	0.010	0.200	0.600	0.010	0.010	0.010	0.010	0.010
DS II: ψ	7.0	5.0	5.0	5.0	5.0	4.0	5.0	6.0	7.0	7.0	8.0	9.5
DS II: p_l	0.0002	0.0080	0.0227	0.0242	0.0225	0.0339	0.0423	0.0290	0.0617	0.0813	0.0994	0.1394
Bin l	13	14	15	16	17	18	19	20	21	22	23	24
DS I: p_l ($\psi = 13$)	0.010	0.010	0.010	0.010	0.010	0.010	0.010	0.010	0.010	0.010	0.010	0.000
DS II: ψ	10.0	8.5	8.5	6.5	4.6	2.6	8.0	5.0	6.4	5.2	4.5	3.9
DS II: p_l	0.1839	0.1115	0.0765	0.0080	0.0051	0.0019	0.0012	0.0010	0.0017	0.0031	0.0097	0.0317

S-I-B Salp Swarm Algorithm and Its Laplacian Variant

Algorithm S1 summarizes SSA [5] and LX-SSA [6] as used in Phase 1 of SSA-VNS and LX-SSA-VNS and as stand-alone methods (Section III-B). A follower moves to the midpoint of its current position and the *new* position of the preceding salp (already updated in the same iteration); in LX-SSA it also forms the Laplace candidate (10), evaluates both candidates and keeps the better one. The new population is then evaluated as a whole, so a kept LX-SSA follower is evaluated twice; with the N_p candidate evaluations of the $N_p/2$ followers, one LX-SSA iteration therefore costs $2N_p$ evaluations and one SSA iteration N_p . At 6,030 evaluations, stand-alone SSA runs for $T = 200$ and LX-SSA for $T = 100$ iterations; in the hybrids with $\omega = 0.5$, Phase 1 runs for $T = 100$ (SSA) and $T = 50$ (LX-SSA) iterations, i.e., 3,030 evaluations each, and ξ_1 decreases over these T iterations. Both methods use the code of the previous study, as stand-alone methods and inside the hybrids (Section V).

Algorithm S1 SSA and LX-SSA (lines marked LX: LX-SSA only); evaluations per line in brackets

```

1: Draw and evaluate  $N_p$  salps; sort them by  $F_p$  and set the best as food source  $\mathbf{H}$ .  $\triangleright [N_p]$ 
2: for  $t = 1$  to  $T$  do
3:   Update  $\xi_1 = 2 \exp[-(4t/T)^2]$ .
4:   for salp  $i = 1, \dots, N_p$  do
5:     if  $i \leq N_p/2$  then
6:       Update leader  $i$  using (9) (new position  $\tilde{\mathbf{X}}^i$ ).
7:     else
8:       Form the follower candidate  $\mathbf{Z}^i = \frac{1}{2}(\mathbf{X}^i + \tilde{\mathbf{X}}^{i-1})$  and set  $\tilde{\mathbf{X}}^i \leftarrow \mathbf{Z}^i$ .
9:       LX: form  $\mathbf{Y}^i$  using (10), box-repair  $\mathbf{Y}^i$  and  $\mathbf{Z}^i$ , evaluate both and set  $\tilde{\mathbf{X}}^i$  to the better one.  $\triangleright [2$ 
        per follower]
10:    end if
11:  end for
12:  Box-repair  $\tilde{\mathbf{X}}$ , set  $\mathbf{X} \leftarrow \tilde{\mathbf{X}}$ , evaluate  $F_p$  for all salps, sort them, and update  $\mathbf{H}$ .  $\triangleright [N_p]$ 
13: end for
14: return  $\mathbf{H}$ .
```

S-I-C Ranking and Test Details

For the paired run-level tests, a run is scored by a prioritized composite endpoint, feasibility first and then the benchmark objective: a feasible run is scored by its objective, an infeasible run below every feasible run, and infeasible runs are ordered among themselves by their minimum spacing (larger is better; boundary violations are not used). Paired differences with $|d| \leq 10^{-9}$ are treated as zero and dropped, both in the Wilcoxon signed-rank test and in the rank-biserial correlation r_{rb} . Case means are rounded to 10^{-6} objective units before ranking, so that exact ties (e.g., all methods at the wake-free optimum for $N = 2$) are not broken by floating-point noise. Holm’s correction for the run-level tests is applied within each case over the comparisons of one table; the main-comparison table (Table III) and the ablation table (Table IV) are different families, so the same pair of methods (e.g., PSO-VNS vs. PSO) can have slightly different W/T/L tallies in the two tables. The W/T/L counts are descriptive tallies. In the case-mean Wilcoxon test, a case in which only one of the two methods has at least 15 feasible runs enters as a difference in its favor that is larger than every observed difference, and a case in which neither has is dropped (Section V). The 68 cases are not independent (nested N within six farm and wind combinations), so all p values describe this benchmark, not a population of farms.

S-I-D Iteration Counts and Phase-1 Budgets

With $N_p = 30$ and $B = 6,030$, stand-alone PSO, SSA and DE run for $(B - N_p)/N_p = 200$ iterations and LX-SSA for $(B - N_p)/(2N_p) = 100$. A hybrid with split ω runs $T_1 = \text{round}((\omega B - N_p)/c)$ Phase-1 iterations, where c is the number of evaluations per iteration (N_p for PSO and SSA, $2N_p$ for LX-SSA) and Python’s round-half-to-even rule applies; for $\omega = 0.5$ the argument is 99.5 for PSO and SSA, which is rounded to $T_1 = 100$ (3,030 evaluations), and LX-SSA gets $T_1 = 50$ (3,030 evaluations). For the split study, $\omega = 0.25$ gives 49 PSO iterations (1,500 evaluations), $\omega = 0.75$ gives 150 (4,530 evaluations) and $\omega = 0.9$ gives 180 (5,430 evaluations). RS-VNS evaluates $\text{round}(\omega B) = 3,015$ random layouts (including the seeded initial population of 30), about the same effort as the 3,030 Phase-1 evaluations of the swarm hybrids; its random layouts, like all random initial layouts, are uniform in the bounding square $[-r, r]^{2N}$, so all N turbines lie inside the circle only with probability $(\pi/4)^N$, and a sample that also satisfies the spacing constraint is far rarer (Table S51). Both random-sampling controls switch to VNS after these 3,015 evaluations (the swarm hybrids after 3,030). The disc-sampling control RSD-VNS evaluates the same 3,015 layouts; only its 2,985 layouts after the common initial population are

drawn uniformly in the farm disc instead (Section S-III). DE/rand/1/bin (scale factor 0.5, crossover rate 0.9) clips the mutant to the box before crossover and replaces a member immediately when its trial vector is better (asynchronous one-to-one selection).

S-I-E MS-SLSQP

MS-SLSQP does not minimize F_p : it minimizes the wake loss (without penalty) under explicit inequality constraints, using SciPy’s SLSQP [8] with the bounds $[-r, r]$ on every coordinate, at most 200 iterations per start, $\text{ftol} = 10^{-9}$ and forward-difference gradients with a step of 1 m. The constraints are scaled, $(\ell_{ij}^2 - \ell_{\min}^2)/\ell_{\min}^2 \geq 0$ for every pair and $(r^2 - x_i^2 - y_i^2)/r^2 \geq 0$ for every turbine (for Horns Rev 1, the signed distance of every turbine to every edge of the parallelogram divided by the half-width of the search box, 943 m). The 30 initial layouts are evaluated with F_p and used as starts in increasing order of F_p ; after the 30 initial layouts, every further start is a new random layout drawn from the same generator (uniform in the box, or feasible under the feasibility-preserving initialization), until the budget is used up. Every objective call, including the finite-difference calls, counts as an evaluation. MS-SLSQP returns the best evaluated layout (by wake loss) whose scaled constraints are all $\geq -10^{-9}$; the reported feasibility of this layout is then judged with the common tolerance of 10^{-6} m (Section V). These are standard, untuned settings; since the top-hat Jensen wake makes the wake loss discontinuous in the coordinates, finite-difference gradients are unreliable, which handicaps MS-SLSQP on this benchmark.

S-I-F Feasibility-Preserving Initialization

Algorithm S2 generates each of the N_p initial layouts of the feasible-start study (Section VIII-A); it uses no objective evaluations, so it does not count toward the budget. With spacing $s = \ell_{\min}$, it minimizes the smooth packing penalty

$$P(\mathbf{z}) = \frac{1}{\tilde{s}^4} \left[\frac{1}{2} \sum_{i < j} \max(0, \tilde{s}^2 - \ell_{ij}^2)^2 + \sum_i b_i \right], \quad \tilde{s} = 1.0005 s, \quad (\text{S1})$$

with $b_i = \max(0, x_i^2 + y_i^2 - \tilde{r}^2)^2$, $\tilde{r} = (1 - 10^{-7}) r$, for the circular farms and $b_i = \tilde{s}^2 \sum_e \max(0, \delta_{ie})^2$, with δ_{ie} the signed distance of turbine i outside edge e , for the Horns Rev 1 parallelogram. The slightly inflated spacing and shrunken radius leave a margin, so that the result passes the feasibility test. All random draws come from the seeded global generator, so the runs of all methods with the same seed start from the same feasible initial population and stay paired; under this initialization the later random restarts of MS-SLSQP also use Algorithm S2. RS-VNS and RSD-VNS were not run with this initialization, since they would need one packing solve for each of their 3,015 random layouts.

Algorithm S2 Feasibility-preserving initialization of one layout (N turbines, spacing s , circle of radius r or convex polygon)

- 1: **for** try = 1 to 200 **do**
 - 2: Draw N positions uniformly in the site (circle: uniform in the disc; polygon: rejection sampling from its bounding box).
 - 3: Minimize P (S1) from these positions with L-BFGS-B (analytic gradient; at most 3,000 iterations, $\text{ftol} = 10^{-15}$, $\text{gtol} = 10^{-12}$).
 - 4: Circle: move every turbine that lies outside the circle radially back onto it.
 - 5: **if** every pairwise distance is $\geq s - 10^{-6}$ m and every turbine lies within 10^{-6} m of the site **then**
 - 6: **return** the layout.
 - 7: **end if**
 - 8: **end for**
 - 9: Stop with an error.
-

S-I-G Horns Rev 1 Model

The Horns Rev 1 block (Section VIII-B) uses the V80 power and thrust curves and the 12-sector Weibull wind climate of PyWake’s Horns Rev 1 example [9], with the sector frequencies normalized to one. The boundary is the parallelogram through the four corner turbines of the installed 16-turbine block, enlarged by 0.1% about the centre of the block so that the installed turbines (integer coordinates) lie inside it; in F_p (4), the boundary violation of a turbine is its distance (m) outside the parallelogram, and the spacing violation is $\ell_{\min} - \ell_{ij}$ (m) with $\ell_{\min} = 4D = 320$ m. The search box is the square of half-width 943 m (the largest absolute coordinate of the parallelogram) centred on the block; the same half-width is the r of the VNS radii. The wake model is the Jensen top-hat wake with $K = 0.04$, a hub-centre in-wake test (turbine i is waked by j if it lies downstream of j

and within $R + Kx$ of its axis, where x is the downstream distance; there is no rotor-overlap fraction), the thrust coefficient at the free-stream speed (which gives a somewhat lower wake loss than PyWake; Section S-IV-B) and root-sum-square superposition. The wind speed is discretized in 1-m/s bins centred at 3, 4, ..., 25 m/s, each with the Weibull probability of its ± 0.5 m/s interval. The wind direction is discretized in 5° bins centred at $2.5^\circ, 7.5^\circ, \dots, 357.5^\circ$ (meteorological convention); every 30° sector (centred at $0^\circ, 30^\circ, \dots, 330^\circ$) contains exactly six bins, each carrying the Weibull parameters of its sector and one sixth of its frequency.

Corrected direction binning. An earlier version of the model centred the bins at $0^\circ, 5^\circ, \dots, 355^\circ$ and assigned them to sectors by rounding half to even, so that the sectors received seven and five bins alternately and the frequencies summed to 1.003. This was corrected as described above, and all Horns Rev 1 runs of this paper (all methods, budgets and initializations) were repeated with the corrected model; the installed and wake-free AEP values and all Horns Rev 1 results in the main text and in this document come from the corrected model. The check of the model against PyWake is described in Section S-IV-B.

S-I-H Feasibility Tolerance

The penalty F_p treats any positive violation as infeasible, whereas every reported feasibility (final layouts, convergence curves, ranking rule) uses the tolerance of 10^{-6} m on both the spacing and the boundary, in all studies. Because the penalty term is $(1 + \mu g)^2$ with $\mu = 10^{10}$, the comparison of F_p values is a feasibility-first rule only for violations above $(\sqrt{L} - 1)/\mu$, where L is the wake loss of the feasible layout compared with; for the benchmark cases this is at most $4.6 \cdot 10^{-8}$ (m for the spacing, m^2 for the boundary; Lemma S6). A smaller violation adds little to F_p and can beat a feasible layout. Conversely, violations between about $5 \cdot 10^{-8}$ and 10^{-6} m are heavily penalized in F_p but counted as feasible in the reporting; the penalty's effective tolerance is thus much stricter than the reporting tolerance.

S-II Baseline Setting and Main Comparison: Details

This section documents the runs behind the effect of the baseline setting (Section VI-A) and gives the per-case results, the full case-level statistics, the coordinates of the best layouts and further figures of the main comparison of the eight methods (Section VI-B).

S-II-A Old and Corrected PSO Setting

Table I ranks the method pool of the previous version of this study (LX-SSA-VNS, SSA-VNS, LX-SSA, SSA, PSO, DE, VNS and MS-SLSQP) twice: once with PSO in the *old setting* $w = 0.7$, $c_1 = c_2 = 2$, and once with the *corrected* constriction setting (7) used for PSO everywhere else in this paper. The old setting satisfies the order-1 (mean) stability condition but violates the order-2 (variance) condition (8) of [7] ($c_1 + c_2 = 4$ against the bound 3.50 for $w = 0.7$), so that, under stagnation and without bounds, the second moments of the particle positions are unbounded (Proposition S2). Both conditions are derived under stagnation and for an unbounded search space, and in our code the velocities are kept when a position is clipped to the box (Lemma S3); the old setting is run without velocity clamping. The instrumented runs of Section S-VII-A show what happens in practice: the swarm of the old setting contracts much less and moves faster, clips more coordinates and has far fewer feasible particle evaluations late in the run (2.6% against 48.4% in iterations 101–200), and it is feasible in only 86.0% of its runs (Section VI-A); its infeasible final layouts mostly violate only the spacing constraint (Table S52). Section S-II-B varies the coefficients across the order-2 boundary and the bound handling. The old-setting runs come from the PSO code of the previous study (`analysis/authors_optimizers.py`, run by `analysis/full_grid_experiments.py`); the corrected runs use the same code, in which only the three coefficients are changed. Both settings thus share the population of 30, the 200 iterations, the zero initial velocities, the box repair and the immediate update of the personal and global bests, and both are run on the same 68 cases with the same penalized objective (4), the same budget of 6,030 evaluations and the same seeds 1–30. Because every optimizer draws its initial population first after seeding, a run with the old setting starts from the same initial population as the run with the corrected setting and as the runs of all other methods with the same seed, so the comparison of the two settings is seed-paired. The two columns of Table I share the runs of all methods except PSO and use the same feasibility-aware ranking rule (Table II); every change in the ranking therefore comes from the PSO setting alone. The old setting was run only on the 68 benchmark cases at 6,030 evaluations (and in the coefficient sweep of Section S-II-B), so the change in the ranking is shown at this budget only. At larger budgets the ranking changes: at 120,030 evaluations SSA-VNS has a better average rank on the six largest cases than PSO (4.17 vs. 5.83; Section VIII-A). All other results of the paper use the corrected setting. Table S2 gives the per-case comparison of the two settings.

Table S2: Per-case ranks (1 = best of 8; feasibility-aware rule) of PSO, SSA-VNS and LX-SSA-VNS in the method pool of Table I with the old PSO setting ($w = 0.7$, $c_1 = c_2 = 2$) and with the constriction setting (68 cases, 6,030 evaluations).

DS	r	N	Old setting			Constriction		
			PSO	SSA-VNS	LX-SSA-VNS	PSO	SSA-VNS	LX-SSA-VNS
I	500	2	4	4	4	4	4	4
I	500	3	4	3	5	3	4	5
I	500	4	4	2	3	1	3	4
I	500	5	1	5	2	1	5	2
I	500	6	2	1	4	1	2	4
I	500	7	3	6	5	1	6	5
I	500	8	6	3	5	1	4	6
I	500	9	7	3	4	1	4	5
I	500	10	8	2	3	2	3	4
I	750	2	4	4	4	4	4	4
I	750	3	4	3	2	3	4	2
I	750	4	4	3	2	1	4	3
I	750	5	3	1	2	1	2	3
I	750	6	3	1	2	1	2	3
I	750	7	4	1	2	1	2	3
I	750	8	4	1	3	1	2	4
I	750	9	7	1	2	1	2	3
I	750	10	7	1	4	1	2	5
I	750	11	7	2	1	1	3	2
I	750	12	7	3	2	1	4	3
I	1000	2	4	4	4	4	4	4
I	1000	3	3	3	3	3	3	3
I	1000	4	4	1	2	1	2	3
I	1000	5	5	1	2	1	2	3
I	1000	6	4	1	2	1	2	3
I	1000	7	4	1	2	1	2	3
I	1000	8	5	1	2	1	2	3
I	1000	9	6	1	2	1	2	3
I	1000	10	6	1	2	1	2	3
I	1000	11	7	1	3	1	2	4
I	1000	12	7	1	3	1	2	4
I	1000	13	7	1	2	1	2	3
I	1000	14	7	2	1	1	3	2
I	1000	15	7	1	2	1	2	3
II	500	2	4	4	4	4	4	4
II	500	3	4	3	2	4	3	2
II	500	4	1	3	4	1	3	4
II	500	5	2	3	1	1	3	2
II	500	6	4	1	2	1	2	3
II	500	7	6	2	3	1	3	5
II	500	8	7	2	3	1	3	4
II	500	9	7	2	3	1	3	4
II	500	10	8	2	4	1	3	5
II	750	2	4	4	4	4	4	4
II	750	3	4	1.5	3	4	1.5	3
II	750	4	3	4	2	3	4	2
II	750	5	3	1	2	1	2	3
II	750	6	3	1	2	1	2	3
II	750	7	4	1	2	1	2	3
II	750	8	6	1	2	1	2	3
II	750	9	7	1	2	1	2	3
II	750	10	7	2	4	1	3	5
II	750	11	6	2	4	1	3	5
II	750	12	7	3	2	1	4	3
II	1000	2	4	4	4	4	4	4
II	1000	3	5	2.5	2.5	3	3	3
II	1000	4	4	2	3	2	3	4
II	1000	5	4	2	1	1	3	2
II	1000	6	4	1	2	1	2	3
II	1000	7	3	2	1	1	3	2
II	1000	8	5	1	2	1	2	3
II	1000	9	7	1	2	1	2	3
II	1000	10	7	1	2	1	2	3
II	1000	11	7	1	3	1	2	4
II	1000	12	7	1	3	1	2	4
II	1000	13	7	1	2	1	2	3
II	1000	14	6	1	2	1	2	3
II	1000	15	7	1	2	1	2	3

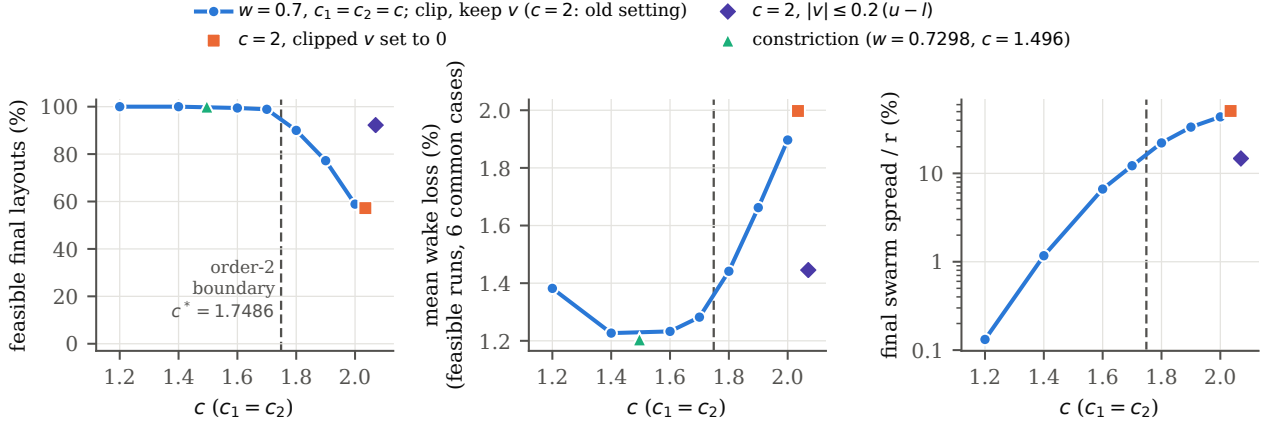


Figure S1: Stand-alone PSO with $w = 0.7$ and $c_1 = c_2 = c$ on the 12 split cases $\times$ 30 seeds (6,030 evaluations, random starts; line with circles), the old setting ($c = 2$) with two other bound-handling rules (clipped velocity components set to zero; velocity clamping $|v| \leq v_{\max} = 0.2(u-l)$; drawn slightly to the right of $c = 2$), and the constriction setting ($w = 0.7298, c = 1.49618$; stored runs of the main comparison, spread not logged). The dashed vertical line is the order-2 stability boundary $c^* = 12(1-w^2)/(7-5w) = 1.7486$ of Proposition S1 (stagnation, no bounds). Left: feasible final layouts (% of 360 runs). Middle: mean wake loss of the feasible runs, averaged over the 6 cases in which every setting has at least 15 feasible runs. Right: median final swarm spread (mean over particles of the RMS turbine distance to the global best), relative to the farm radius (log scale). All settings start from the same seeded initial swarm. Values: Table S3.

S-II-B Coefficient Sweep and Bound Handling

The comparison of two settings cannot show whether the order-2 boundary marks where this PSO fails, nor separate the step size from the bound handling. Stand-alone PSO was therefore run with $w = 0.7$ and $c_1 = c_2 = c$ from 1.2 to 2.0 on the 12 cases of the split study $\times$ 30 seeds (360 runs per setting, 6,030 evaluations, random starts; Fig. S1 and Table S3); the point $c = 2.0$ reproduces the stored runs of the old setting bit for bit. On the order-2 stable side, $c \leq 1.7$, at least 98.9% of the final layouts are feasible and all 12 cases are qualified; beyond the boundary $c^* = 1.7486$, feasibility falls to 90.0% at $c = 1.8$, 77.2% at 1.9 and 58.9% at 2.0. The degradation begins in the step from 1.7 to 1.8, which contains c^* , and accelerates beyond it; it is an onset, not a jump at c^* . The final swarm spread grows smoothly with c , and the mean wake loss already rises from $c = 1.6$ to 1.7, on the stable side. The old setting thus lies beyond the order-2 boundary, consistent with its loss of feasibility, and the boundary marks approximately where this PSO starts to fail; the sweep does not show that crossing it causes the failure.

The old setting is run without velocity clamping, and a clipped coordinate keeps its velocity (Lemma S3). Setting the velocity components of clipped coordinates to zero does not rescue it (57.2% feasible). Velocity clamping, $|v| \leq 0.2(u-l)$, raises feasibility to 92.2%, about the level of $c = 1.8$ without clamping, but the clamped old setting remains worse than the constriction setting (lower feasibility, and a higher mean wake loss in all 12 cases). The deficit of the old setting therefore comes mainly from its large steps, not from the bound handling.

Table S3: PSO coefficient sweep and bound handling (12 split cases $\times$ 30 seeds, 6,030 evaluations, random starts). Feasible: final layouts feasible (% of 360 runs); $N \geq 10$: the same for the 180 runs with $N = 10, 12, 15$; qualified: cases with at least 15 of 30 feasible runs; L : mean wake loss (%) of the feasible runs, averaged over the 6 cases in which every setting is qualified; L_{12} : the same over all 12 cases (only for settings qualified in all 12); rank: average rank over the 12 cases among the ten settings (feasibility-aware rule of the main comparison: settings with fewer than 15 feasible runs rank below all qualified ones); spread: median final swarm spread relative to the radius (%); clipped: coordinates leaving the box per iteration (%; iterations 1–200); feasible evals: particle evaluations with a feasible layout in iterations 101–200 (%). The row $c = 2.0$ reproduces the stored runs of the old setting bit for bit; the constriction row is the stored PSO data of the main comparison (dynamics not logged). Order-2: whether (w, c) satisfies the order-2 condition of Proposition S1 ($c < 1.7486$ at $w = 0.7$), which assumes no velocity clamping. For every setting and seed, the final layout is feasible in Data Set I if and only if it is feasible in Data Set II (same geometry; the penalty dominates while a layout is infeasible), so the feasibility columns rest on 6 geometries $\times$ 30 seeds.

w	c	Bound handling	Order-2	Feasible (%)	$N \geq 10$ (%)	Qualified	L (%)	L_{12} (%)	Rank	Spread (%)	Clipped (%)	Feas. evals (%)
0.7	1.2	clip, keep v	yes	100.0	100.0	12	1.38	3.05	5.42	0.1	0.5	80.4
0.7	1.4	clip, keep v	yes	100.0	100.0	12	1.23	2.77	2.50	1.2	0.9	69.9
0.7	1.6	clip, keep v	yes	99.4	99.2	12	1.23	2.77	2.33	6.6	1.9	41.5
0.7	1.7	clip, keep v	yes	98.9	98.3	12	1.28	2.93	3.58	12.2	2.9	25.4
0.7	1.8	clip, keep v	no	90.0	85.0	12	1.44	3.37	6.17	22.1	4.3	13.1
0.7	1.9	clip, keep v	no	77.2	65.8	10	1.66	–	7.83	33.4	6.3	5.6
0.7	2.0	clip, keep v	no	58.9	38.3	6	1.90	–	9.17	43.6	8.7	2.6
0.7	2.0	clip, $v \leftarrow 0$	no	57.2	35.8	6	2.00	–	9.83	50.8	8.8	2.5
0.7	2.0	$ v \leq v_{\max}$, clip	no	92.2	88.3	12	1.45	3.42	6.42	14.7	1.6	11.8
0.7298	1.496	clip, keep v	yes	100.0	100.0	12	1.20	2.69	1.75	–	–	–

S-II-C Per-Case Results

Tables S4–S9 give the mean and standard deviation of the benchmark objective of every method in every case of the main comparison.

Table S4: Data Set I, 500-m farm: benchmark objective, mean (SD) over the feasible runs of 30 seed-paired runs at 6,030 evaluations. Bold: highest mean among the methods with at least 15 feasible runs. A superscript gives the number of feasible runs when fewer than 30; “–” means no feasible run.

N	Wake-free	PSO-VNS	PSO	SSA-VNS	SSA	LX-SSA	DE	VNS	MS-SLSQP
2	28091.5	28091.5 (0.0)	28091.5 (0.0)	28091.5 (0.0)	28091.5 (0.0)	28091.5 (0.0)	28091.5 (0.0)	28091.5 (0.0)	28086.7 (4.5)
3	42137.2	42131.3 (8.5)	42128.7 (8.8)	42128.5 (10.7)	42114.9 (10.3)	42114.8 (14.2)	42130.7 (8.8)	42131.6 (9.7)	42119.9 (11.0)
4	56182.9	56114.4 (20.7)	56110.0 (25.6)	56101.8 (25.3)	56086.8 (19.8)	56078.8 (22.6)	56092.8 (29.8)	56106.1 (21.7)	56057.0 (55.8)
5	70228.7	70051.2 (28.5)	70053.1 (29.3)	69967.2 (159.2)	69994.0 (34.0)	69972.6 (72.0)	69960.5 (78.6)	69876.9 (199.8)	69876.4 (139.3)
6	84274.4	83918.5 (66.9)	83925.6 (55.8)	83780.8 (213.2)	83535.6 (372.5)	83544.9 (465.3)	83326.7 (761.5)	83299.3 (524.5)	83636.4 (227.9)
7	98320.2	97588.7 (292.3)	97569.6 (325.2)	96529.9 (794.9)	96757.0 (692.4)	96380.6 (1013.8)	95903.3 (1316.3) ¹⁵	96967.1 (590.6)	97049.0 (477.4)
8	112365.9	110777.2 (718.5)	110957.4 (615.4)	109316.4 (1117.4)	109172.6 (1362.4)	108258.3 (1704.3) ²⁸	107632.7 (1592.8) ²	109493.5 (2035.2)	110233.3 (571.9)
9	126411.6	123153.2 (1132.3)	123499.5 (1006.4)	121985.3 (1201.8)	120043.1 (2755.8) ²⁵	119134.2 (2666.4) ²¹	– ⁰	122259.2 (1232.0)	123307.1 (814.5)
10	140457.4	135160.5 (1708.4)	135396.0 (1790.4)	134321.3 (1482.4) ²⁶	132616.3 (2328.0) ¹²	130967.4 (3191.0) ¹³	130994.8 (0.0) ¹	133567.5 (1820.9)	135642.4 (920.1)

Table S5: Data Set I, 750-m farm: benchmark objective, mean (SD) over the feasible runs of 30 seed-paired runs at 6,030 evaluations. Bold: highest mean among the methods with at least 15 feasible runs. A superscript gives the number of feasible runs when fewer than 30; “–” means no feasible run.

N	Wake-free	PSO-VNS	PSO	SSA-VNS	SSA	LX-SSA	DE	VNS	MS-SLSQP
2	28091.5	28091.5 (0.0)	28091.5 (0.0)	28091.5 (0.0)	28091.5 (0.0)	28091.5 (0.0)	28091.5 (0.0)	28091.5 (0.0)	28091.0 (1.6)
3	42137.2	42137.2 (0.0)	42137.0 (1.0)	42137.0 (1.0)	42134.4 (4.1)	42134.2 (4.8)	42136.8 (1.5)	42137.2 (0.0)	42133.3 (5.0)
4	56182.9	56170.9 (10.1)	56170.9 (11.3)	56162.3 (13.6)	56153.8 (17.9)	56149.9 (14.0)	56146.9 (17.5)	56167.0 (13.8)	56139.3 (11.7)
5	70228.7	70169.6 (23.7)	70164.8 (25.1)	70150.9 (25.4)	70122.2 (26.3)	70118.1 (28.6)	70083.9 (31.7)	70125.8 (47.0)	70091.5 (35.7)
6	84274.4	84139.4 (24.3)	84130.1 (30.7)	84095.6 (36.9)	84035.2 (58.4)	84033.5 (58.3)	83948.2 (74.1)	84029.2 (141.3)	83965.7 (117.2)
7	98320.2	98064.7 (54.7)	98059.9 (46.4)	97910.8 (164.8)	97875.2 (169.3)	97840.0 (167.7)	97479.1 (285.1)	97774.4 (232.6)	97663.0 (196.3)
8	112365.9	111921.0 (119.4)	111924.1 (87.8)	111683.9 (282.2)	111506.5 (334.5)	111374.0 (460.8)	110595.8 (445.6)	111411.4 (493.5)	111287.8 (290.8)
9	126411.6	125552.7 (330.8)	125639.1 (260.7)	125207.2 (539.4)	124756.2 (610.5)	124698.0 (960.4)	121674.9 (2276.5) ²³	124780.0 (716.8)	124787.6 (424.2)
10	140457.4	139122.6 (345.7)	139212.9 (276.9)	138304.3 (685.9)	137817.6 (1160.1)	137748.6 (1116.3)	134937.7 (998.5) ⁴	138170.4 (764.7)	138182.1 (569.0)
11	154503.1	152343.5 (565.1)	152626.6 (443.6)	151286.3 (1117.4)	150404.1 (1042.2) ²⁹	149863.8 (1886.6) ²⁹	148555.7 (1134.8) ³	151004.2 (1339.0)	151173.5 (598.0)
12	168548.8	165526.4 (774.4)	165826.3 (766.5)	164279.2 (1284.1)	162903.5 (1685.8)	161630.5 (1988.2) ²⁸	– ⁰	163712.4 (1575.6)	164478.5 (634.3)

Table S6: Data Set I, 1000-m farm: benchmark objective, mean (SD) over the feasible runs of 30 seed-paired runs at 6,030 evaluations. Bold: highest mean among the methods with at least 15 feasible runs. A superscript gives the number of feasible runs when fewer than 30; “–” means no feasible run.

N	Wake-free	PSO-VNS	PSO	SSA-VNS	SSA	LX-SSA	DE	VNS	MS-SLSQP
2	28091.5	28091.5 (0.0)	28091.5 (0.0)	28091.5 (0.0)	28091.5 (0.0)	28091.5 (0.0)	28091.5 (0.0)	28091.5 (0.0)	28091.3 (0.7)
3	42137.2	42137.2 (0.0)	42137.2 (0.0)	42137.2 (0.0)	42136.7 (1.6)	42136.7 (1.4)	42137.2 (0.0)	42137.2 (0.0)	42136.6 (1.8)
4	56182.9	56181.4 (3.1)	56181.8 (3.1)	56180.0 (3.8)	56169.4 (9.2)	56172.9 (7.8)	56166.9 (8.4)	56177.3 (8.6)	56165.3 (10.2)
5	70228.7	70209.8 (10.8)	70208.0 (13.1)	70202.1 (15.9)	70174.0 (19.7)	70183.3 (21.0)	70148.7 (22.4)	70193.3 (24.5)	70159.6 (39.9)
6	84274.4	84214.5 (19.7)	84213.1 (24.2)	84175.9 (36.0)	84149.9 (29.7)	84142.7 (23.5)	84096.1 (24.0)	84170.0 (49.5)	84064.6 (113.6)
7	98320.2	98192.1 (32.2)	98178.1 (38.0)	98128.4 (59.9)	98084.7 (47.0)	98078.4 (82.1)	97942.3 (67.9)	98064.6 (118.7)	97930.4 (157.1)
8	112365.9	112142.9 (48.9)	112125.1 (48.4)	112008.3 (131.2)	111947.1 (125.7)	111924.6 (164.7)	111608.3 (168.9)	111862.5 (246.1)	111736.9 (163.4)
9	126411.6	126003.5 (103.1)	125993.3 (89.8)	125842.0 (277.4)	125761.1 (168.4)	125701.5 (265.4)	124996.3 (282.9)	125589.7 (312.6)	125450.0 (305.1)
10	140457.4	139874.2 (102.1)	139834.3 (108.2)	139561.7 (359.4)	139348.0 (433.0)	139354.2 (388.5)	137969.0 (623.0)	139290.3 (447.7)	138991.7 (381.3)
11	154503.1	153621.0 (192.7)	153621.2 (168.9)	153184.6 (292.3)	152825.1 (385.8)	152736.8 (558.8)	150189.7 (1068.9) ²⁹	153010.5 (393.8)	152538.4 (405.6)
12	168548.8	167262.1 (383.9)	167223.9 (330.8)	166530.1 (760.0)	166091.9 (816.6)	165814.4 (827.9)	162390.4 (1765.9) ¹⁷	166510.6 (669.8)	166171.0 (462.2)
13	182594.6	180623.7 (496.4)	180638.4 (495.0)	179973.5 (642.1)	179101.5 (1028.9)	178913.6 (967.4)	174763.4 (1389.6) ⁴	179762.6 (1091.2)	179275.1 (644.9)
14	196640.3	194092.5 (813.0)	194082.6 (688.3)	193260.8 (939.8)	192235.7 (1504.8)	191046.6 (1754.6)	– ⁰	192874.6 (1745.6)	192657.7 (605.3)
15	210686.1	207308.1 (894.1)	207263.9 (720.0)	207119.3 (639.3)	204777.6 (1420.5)	203681.8 (1899.5) ²⁹	– ⁰	206486.7 (1093.6)	205876.8 (798.9)

Table S7: Data Set II, 500-m farm: benchmark objective, mean (SD) over the feasible runs of 30 seed-paired runs at 6,030 evaluations. Bold: highest mean among the methods with at least 15 feasible runs. A superscript gives the number of feasible runs when fewer than 30; “–” means no feasible run.

N	Wake-free	PSO-VNS	PSO	SSA-VNS	SSA	LX-SSA	DE	VNS	MS-SLSQP
2	14630.8	14630.8 (0.0)	14630.8 (0.0)	14630.8 (0.0)	14630.8 (0.0)	14630.8 (0.0)	14630.8 (0.0)	14630.8 (0.0)	14626.0 (5.6)
3	21946.1	21909.3 (15.9)	21904.5 (13.8)	21907.1 (18.9)	21900.9 (11.1)	21899.7 (9.5)	21901.2 (5.5)	21932.3 (17.2)	21892.0 (45.9)
4	29261.5	29124.8 (63.8)	29142.9 (54.2)	29067.5 (60.7)	29028.7 (93.8)	29004.2 (87.4)	29044.2 (61.5)	29070.8 (94.1)	28993.5 (96.1)
5	36576.9	36100.0 (100.0)	36118.0 (117.9)	35958.0 (130.6)	35918.9 (74.0)	35880.1 (168.6)	35789.6 (193.8)	35878.2 (160.9)	35914.9 (144.5)
6	43892.3	42851.5 (166.9)	42857.9 (163.5)	42638.7 (198.9)	42444.3 (224.5)	42512.1 (227.9)	42257.5 (374.9)	42560.3 (202.5)	42601.8 (178.6)
7	51207.6	49332.5 (195.5)	49337.3 (189.8)	49008.6 (239.7)	48887.9 (304.1)	48626.5 (340.5)	48203.9 (767.8) ¹⁵	49032.1 (314.4)	48959.7 (202.6)
8	58523.0	55727.9 (300.0)	55668.1 (312.6)	55274.9 (503.7)	54900.5 (471.3)	54699.5 (620.7) ²⁸	54451.3 (618.7) ²	55182.8 (738.6)	55427.1 (303.9)
9	65838.4	61754.3 (344.3)	61643.9 (481.7)	61380.0 (541.4)	60581.0 (844.8) ²⁵	59963.3 (652.2) ²¹	– ⁰	61166.8 (574.8)	61516.6 (326.2)
10	73153.8	67773.5 (477.6)	67472.2 (637.3)	67106.6 (779.5) ²⁵	65880.7 (761.6) ¹²	65868.7 (586.0) ¹³	64240.2 (0.0) ¹	67056.0 (749.1)	67297.9 (402.5) ²⁹

Table S8: Data Set II, 750-m farm: benchmark objective, mean (SD) over the feasible runs of 30 seed-paired runs at 6,030 evaluations. Bold: highest mean among the methods with at least 15 feasible runs. A superscript gives the number of feasible runs when fewer than 30; “–” means no feasible run.

N	Wake-free	PSO-VNS	PSO	SSA-VNS	SSA	LX-SSA	DE	VNS	MS-SLSQP
2	14630.8	14630.8 (0.0)	14630.8 (0.0)	14630.8 (0.0)	14630.8 (0.0)	14630.8 (0.0)	14630.8 (0.0)	14630.8 (0.0)	14630.2 (1.8)
3	21946.1	21945.5 (2.5)	21945.0 (3.9)	21946.1 (0.0)	21940.0 (8.2)	21937.3 (8.1)	21942.5 (5.8)	21946.1 (0.0)	21938.3 (11.3)
4	29261.5	29225.9 (22.5)	29221.2 (22.7)	29210.5 (34.5)	29182.5 (36.2)	29201.4 (25.7)	29195.3 (26.2)	29227.3 (29.9)	29175.2 (48.2)
5	36576.9	36473.4 (36.9)	36480.6 (37.9)	36417.1 (67.3)	36332.0 (91.4)	36366.8 (74.3)	36277.7 (105.9)	36389.7 (97.1)	36267.4 (108.2)
6	43892.3	43596.3 (87.7)	43611.5 (96.3)	43500.1 (119.6)	43419.5 (153.1)	43377.4 (175.7)	43142.2 (144.8)	43367.4 (201.2)	43264.0 (140.5)
7	51207.6	50584.2 (207.2)	50592.0 (149.5)	50370.1 (169.6)	50264.8 (202.5)	50095.4 (193.6)	49696.6 (156.3)	50180.2 (303.8)	50174.3 (220.0)
8	58523.0	57411.1 (173.3)	57404.8 (185.9)	57033.9 (298.6)	56806.2 (303.6)	56659.2 (332.0)	56033.1 (331.4)	56855.9 (391.5)	56837.8 (340.9)
9	65838.4	64016.2 (259.2)	63991.1 (213.9)	63632.1 (366.8)	63291.6 (402.5)	63252.7 (342.8)	61970.8 (763.0) ²³	63217.0 (515.4)	63472.4 (272.8)
10	73153.8	70558.3 (205.2)	70484.8 (208.8)	70000.5 (393.6)	69520.3 (512.6)	69480.7 (385.7)	68103.2 (423.7) ⁴	69930.2 (355.0)	70128.5 (305.6)
11	80469.2	76912.3 (421.8)	76804.4 (395.8)	76265.0 (512.1)	75742.5 (677.1) ²⁹	75181.6 (681.1) ²⁹	74596.6 (476.3) ³	76099.7 (566.3)	76420.9 (338.9)
12	87784.5	82968.9 (469.9)	82874.2 (531.0)	82500.8 (492.0)	81822.1 (625.7)	81341.8 (792.1) ²⁸	– ⁰	82368.6 (603.1)	82792.2 (462.3)

Table S9: Data Set II, 1000-m farm: benchmark objective, mean (SD) over the feasible runs of 30 seed-paired runs at 6,030 evaluations. Bold: highest mean among the methods with at least 15 feasible runs. A superscript gives the number of feasible runs when fewer than 30; “–” means no feasible run.

N	Wake-free	PSO-VNS	PSO	SSA-VNS	SSA	LX-SSA	DE	VNS	MS-SLSQP
2	14630.8	14630.8 (0.0)	14630.8 (0.0)	14630.8 (0.0)	14630.8 (0.0)	14630.8 (0.0)	14630.8 (0.0)	14630.8 (0.0)	14630.4 (1.2)
3	21946.1	21946.1 (0.0)	21946.1 (0.0)	21946.1 (0.0)	21945.6 (1.8)	21945.3 (2.8)	21946.1 (0.0)	21946.1 (0.0)	21943.9 (3.7)
4	29261.5	29254.5 (7.9)	29251.3 (10.2)	29247.6 (12.2)	29232.5 (18.5)	29228.0 (14.9)	29225.9 (12.1)	29253.1 (11.7)	29220.8 (35.1)
5	36576.9	36514.3 (28.7)	36519.7 (32.4)	36510.6 (35.3)	36485.9 (37.5)	36468.2 (46.1)	36402.8 (39.4)	36498.4 (49.1)	36426.0 (58.3)
6	43892.3	43776.8 (48.3)	43789.7 (42.7)	43736.8 (52.7)	43626.4 (74.5)	43656.6 (73.3)	43488.5 (69.7)	43686.1 (101.2)	43554.1 (85.6)
7	51207.6	50952.8 (83.2)	50969.1 (79.2)	50800.0 (120.6)	50731.8 (118.8)	50727.3 (119.3)	50442.7 (136.0)	50696.0 (203.8)	50596.6 (137.4)
8	58523.0	58050.4 (138.4)	58051.3 (154.0)	57883.4 (180.0)	57739.3 (142.0)	57645.1 (186.6)	57210.0 (191.8)	57588.6 (311.3)	57528.0 (226.3)
9	65838.4	65052.9 (170.2)	65065.3 (177.0)	64730.4 (233.0)	64476.4 (203.0)	64428.7 (273.1)	63790.1 (227.2)	64430.8 (311.5)	64357.7 (268.5)
10	73153.8	71906.7 (236.9)	71902.4 (238.0)	71541.9 (328.8)	71139.4 (336.8)	71099.4 (266.9)	70190.1 (286.9)	71342.3 (339.4)	71142.5 (257.2)
11	80469.2	78633.2 (289.7)	78478.9 (290.3)	78133.1 (465.6)	77697.5 (425.8)	77561.6 (498.0)	75960.9 (577.6) ²⁹	77948.1 (483.3)	77762.2 (436.7)
12	87784.5	85355.3 (330.7)	85172.5 (311.4)	84647.1 (650.8)	84049.6 (660.3)	83981.2 (515.0)	82211.6 (905.2) ¹⁷	84485.7 (606.8)	84385.5 (437.1)
13	95099.9	91904.0 (428.6)	91685.0 (453.5)	91179.4 (556.4)	90267.0 (482.6)	90303.5 (674.2)	88665.1 (496.0) ⁴	90787.8 (806.9)	90948.3 (429.2)
14	102415.3	98223.0 (410.2)	97889.6 (465.1)	97590.4 (615.0)	96936.5 (669.3)	96229.6 (1019.3)	– ⁰	97079.1 (980.3)	97371.5 (411.6)
15	109730.7	104794.8 (531.9)	104375.6 (501.6)	104035.3 (475.4)	102746.3 (827.8)	102327.4 (838.7) ²⁹	– ⁰	103777.6 (794.4)	103901.2 (406.2)

S-II-D Case-Level Statistics and Best Layouts

Table S10 extends Table II by the full case-level statistics, and Table S11 lists the coordinates of the best layouts for the largest N of each farm (shown in Fig. S9); when PSO-VNS is itself the best method of a farm, Table S11 has a single entry for it.

Table S10: Case-level analysis over the 68 benchmark cases. Ranks of the mean feasible benchmark objective within each case (1 = best); a method with fewer than 15 feasible runs out of 30 in a case is ranked below all other methods, by its number of feasible runs (“<15”: number of such cases). Friedman $\chi_F^2 = 319.3$ (7 d.f.), $p = 4.56 \times 10^{-65}$; Iman–Davenport $F_F = 136.5$, $p = 6.71 \times 10^{-109}$. p_z : Holm-adjusted p of the average-rank z test against PSO-VNS (best-ranked method: PSO-VNS). Because mean-rank post hoc tests depend on the pool of compared methods [11], pairwise two-sided Wilcoxon signed-rank tests on the per-case mean wake losses (%) are also given: p_W (Holm-adjusted over the 7 comparisons), matched-pairs rank-biserial correlation r_{rb} (positive = PSO-VNS better) and mean difference $\overline{\Delta L}$ of the wake loss (percentage points, PSO-VNS minus method, over the cases where both methods have at least 15 feasible runs). “Best”: cases in which the method alone ranks first; “Feas.”: percentage of feasible runs.

Method	Avg. rank	Best	Feas. (%)	<15	p_z	p_W	r_{rb}	$\overline{\Delta L}$ (pp)
PSO-VNS	1.82	29	100.0	0	–	–	–	–
PSO	1.97	25	100.0	0	0.713	0.296	+0.16	–0.018
SSA-VNS	3.45	0	99.6	0	2.04×10^{-4}	5.69×10^{-11}	+1.00	–0.330
VNS	4.40	3	100.0	0	2.42×10^{-9}	1.26×10^{-10}	+0.98	–0.465
SSA	5.38	0	97.6	2	8.32×10^{-17}	4.55×10^{-11}	+1.00	–0.611
MS-SLSQP	5.88	1	100.0	0	1.84×10^{-21}	4.18×10^{-11}	+0.96	–0.435
LX-SSA	6.06	0	96.9	2	3.33×10^{-23}	4.55×10^{-11}	+1.00	–0.765
DE	7.04	0	71.4	18	1.04×10^{-34}	5.59×10^{-11}	+1.00	–0.887

Table S11: Coordinates (m, farm center at the origin) of the best layout of all methods and of the best PSO-VNS layout for the largest N of each farm (6,030 evaluations); objective in benchmark units.

DS	r (m)	N	Method	Objective	Coordinates (x_i, y_i)
I	500	10	MS-SLSQP	138291.2	(469.2, 168.9); (127.1, -90.8); (-183.2, 21.4); (-365.2, -324.6); (-493.9, 32.0); (-320.1, 380.7); (273.4, 411.2); (-38.6, -356.1); (270.8, -412.5); (-6.8, 274.6)
I	500	10	PSO-VNS	137994.8	(485.3, -32.1); (-346.0, 302.7); (322.3, -382.2); (-376.7, -328.3); (317.9, 385.7); (142.8, 131.7); (-59.3, -101.1); (-46.5, 497.1); (-497.9, -0.5); (-88.7, -476.6)
I	750	12	PSO-VNS	167040.0	(-71.5, 110.7); (192.8, -653.4); (741.7, 104.8); (395.5, 126.3); (65.5, -302.0); (-728.9, 172.1); (185.8, 655.0); (-226.9, 453.5); (-531.3, -385.6); (534.1, 524.5); (542.8, -343.4); (-223.2, -715.4)
I	1000	15	PSO	208908.8	(-466.8, -48.4); (923.5, 23.2); (684.6, 693.4); (-230.6, 303.0); (524.3, -694.5); (137.7, 67.9); (517.2, 119.0); (137.7, 964.0); (153.5, -960.7); (-232.5, -607.3); (675.4, -366.5); (-684.6, -361.8); (-980.6, 75.4); (-48.5, -210.1); (-687.0, 528.7)
I	1000	15	PSO-VNS	208720.0	(594.1, 135.8); (-268.3, -387.2); (786.8, -517.5); (-79.4, 263.7); (960.7, 115.7); (-852.8, -274.3); (-998.2, 26.9); (104.5, -420.5); (-552.9, -78.4); (342.1, -881.7); (97.7, 888.8); (-84.5, -912.6); (-515.9, 610.4); (-707.3, 234.6); (355.1, 465.8)
II	500	10	PSO-VNS	68621.3	(499.3, -17.9); (-357.0, 305.5); (302.3, -322.3); (-362.6, -295.2); (336.9, 352.5); (46.0, 150.6); (-45.5, -150.2); (-65.8, 495.6); (-499.5, -2.6); (-102.9, -475.6)
II	750	12	PSO-VNS	83996.8	(-159.3, -214.1); (492.9, 564.5); (-391.5, 639.1); (121.1, -739.2); (-44.2, 619.1); (-662.7, 13.0); (-641.7, -309.1); (-465.3, 325.3); (-348.1, -575.4); (665.0, -320.3); (98.2, 250.2); (724.3, 15.3)
II	1000	15	PSO-VNS	105675.2	(-19.0, 118.7); (-84.9, -328.5); (-602.6, 187.3); (639.7, -523.0); (-942.8, -326.2); (216.5, -974.4); (928.9, -325.3); (-386.7, 906.8); (349.5, 906.0); (-310.8, 528.7); (993.7, 2.3); (322.4, -676.0); (469.3, 84.1); (701.3, 536.4); (-435.2, -798.8)

S-II-E Additional Figures

Fig. S2 shows the average ranks of Table II and Fig. S3 the mean wake loss versus the number of turbines. Figs. S4–S9 show the convergence, the distribution of the final wake loss, the feasibility and the best layouts of the eight methods of the main comparison.

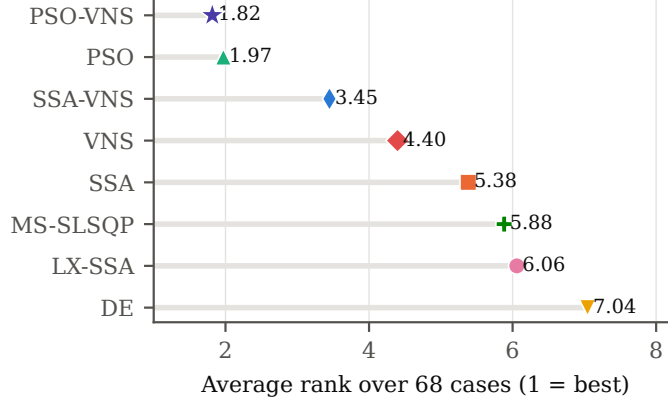


Figure S2: Average rank of the eight methods of the main comparison over the 68 benchmark cases (feasibility-aware ranking of Table II; 1 = best).

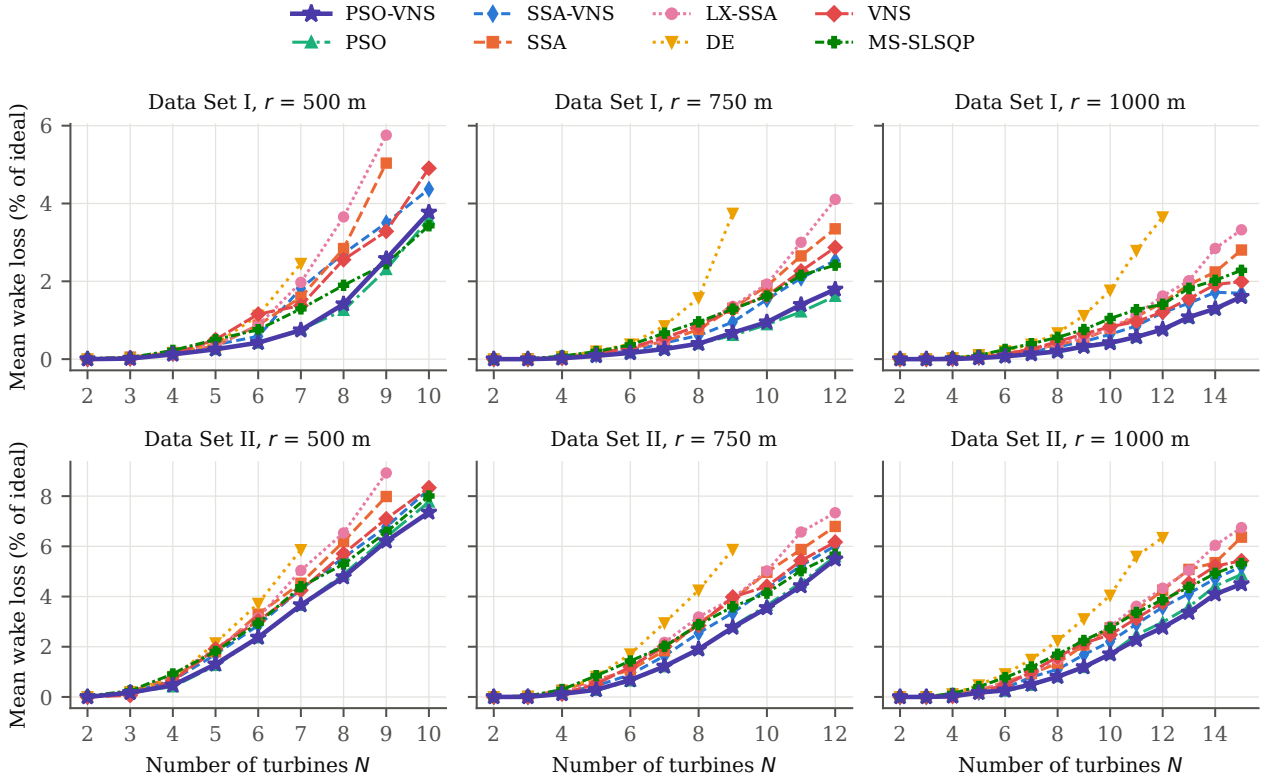


Figure S3: Mean wake loss (% of the wake-free benchmark objective) of the feasible runs versus the prescribed number of turbines N , for each farm and wind data set (30 runs of 6,030 evaluations, random initialization). A point is drawn only when the method has at least 15 feasible runs in that case, so a curve (mostly DE) stops where the method has too few feasible runs.

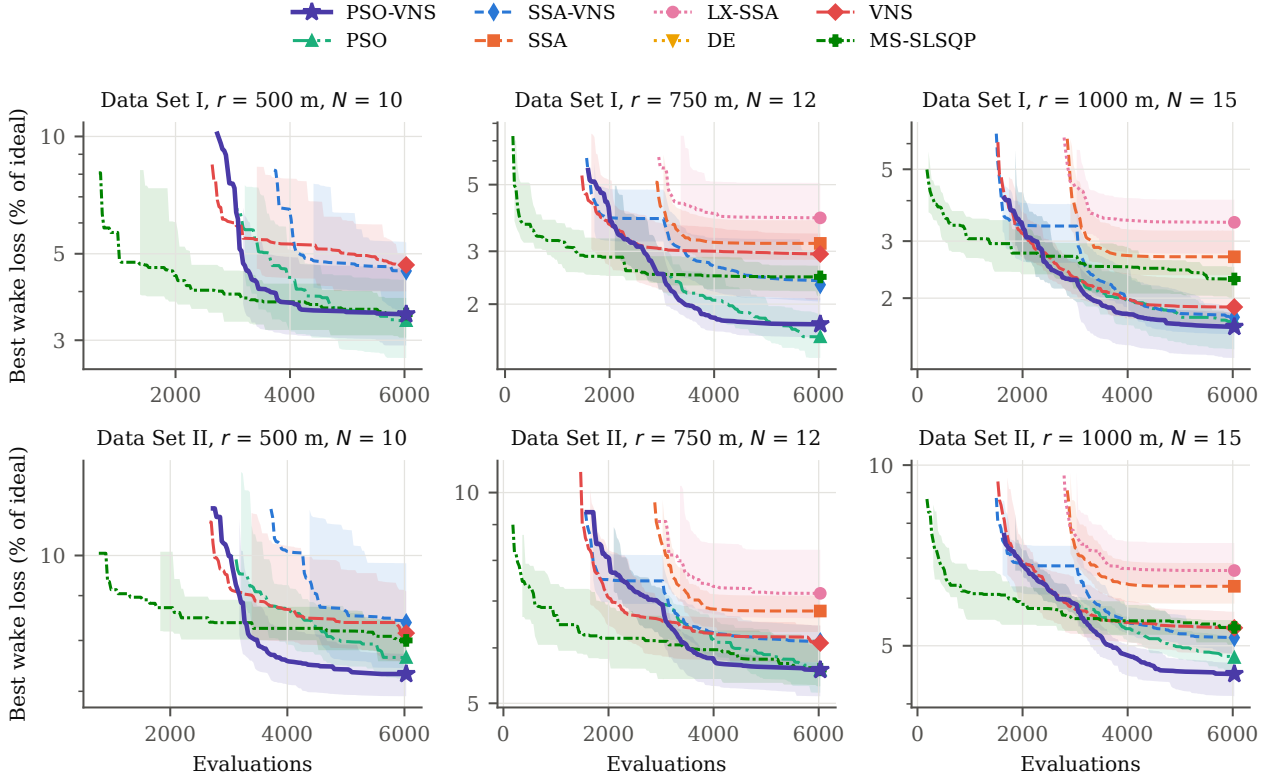


Figure S4: Convergence of the eight methods of the main comparison for the largest N of each farm: median (line) and interquartile range (band) of the best feasible wake loss over 30 runs versus the number of evaluations (objective-function calls), logarithmic vertical axis. The median is drawn once at least 15 of the 30 runs have found a feasible layout and the upper quartile once at least 23 have, so curves start late where early layouts are infeasible; a method that never reaches this share (DE in the largest cases) has no curve in the panel.

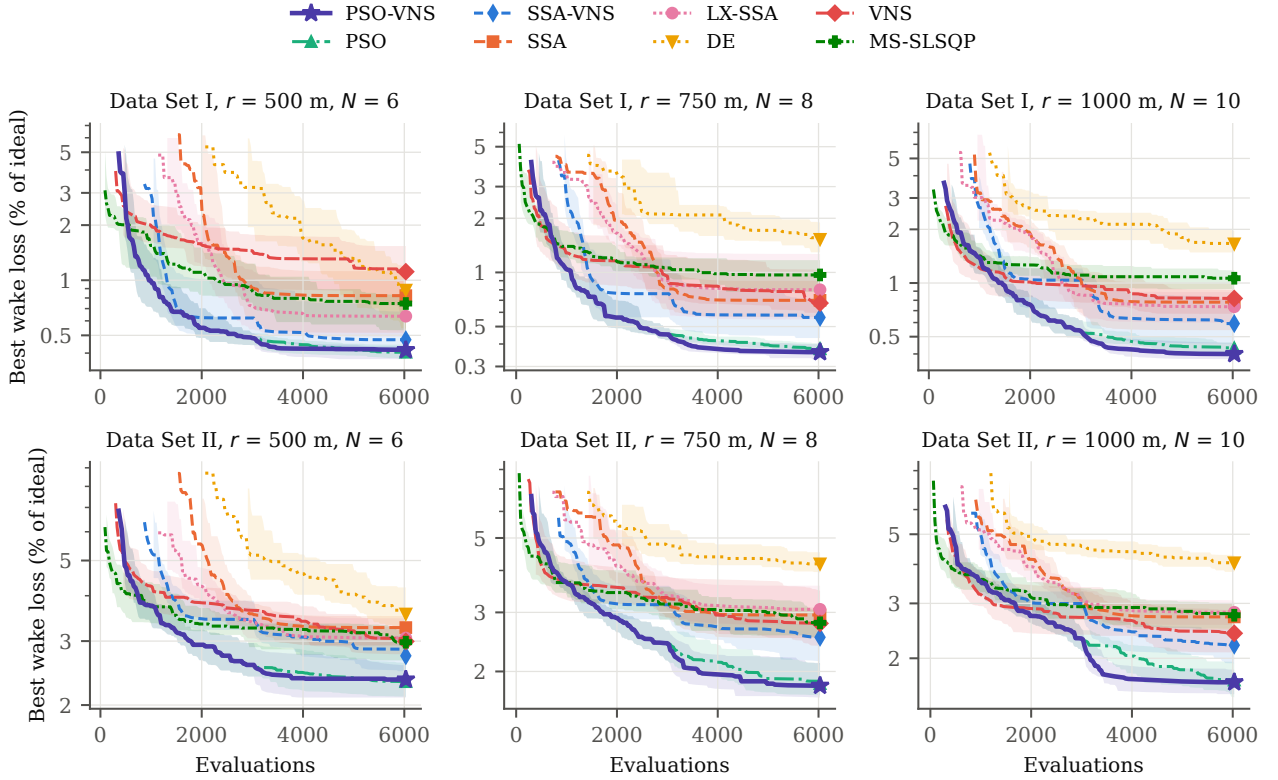


Figure S5: As Fig. S4, for the middle N of each farm ($N = 6, 8$ and 10 for $r = 500, 750$ and 1000 m).

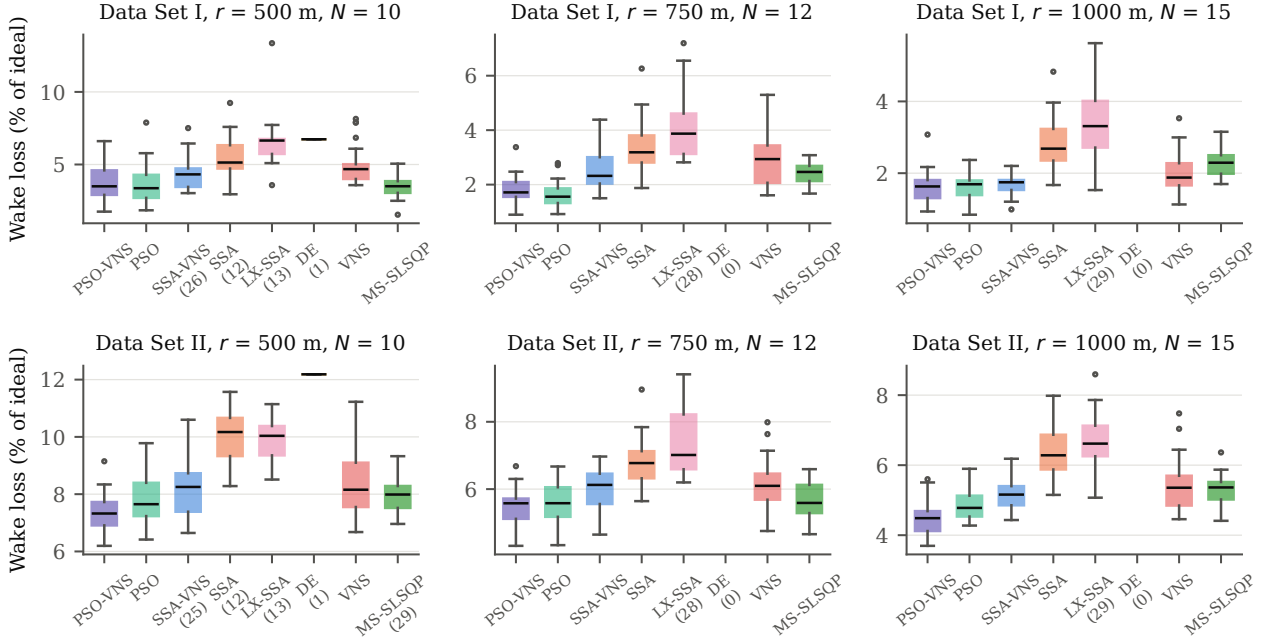


Figure S6: Distribution of the wake loss of the feasible runs (30 runs of 6,030 evaluations, random initialization) for the largest N of each farm (number of feasible runs in parentheses when fewer than 30).

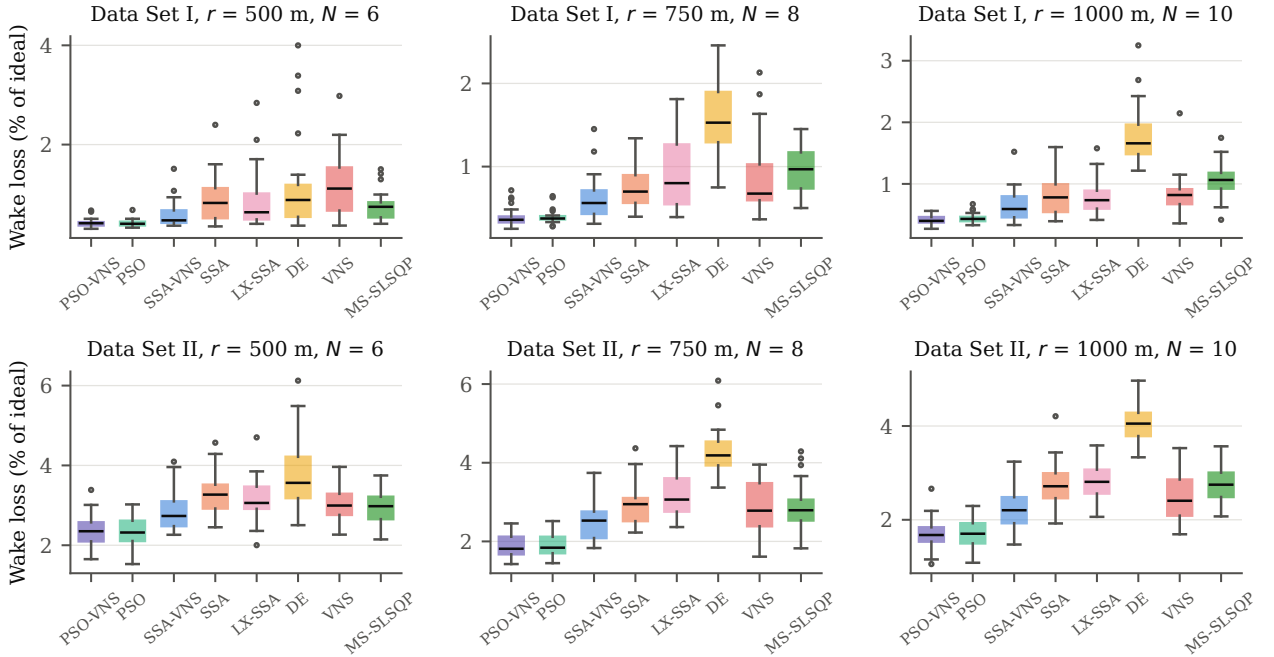


Figure S7: As Fig. S6, for the middle N of each farm.

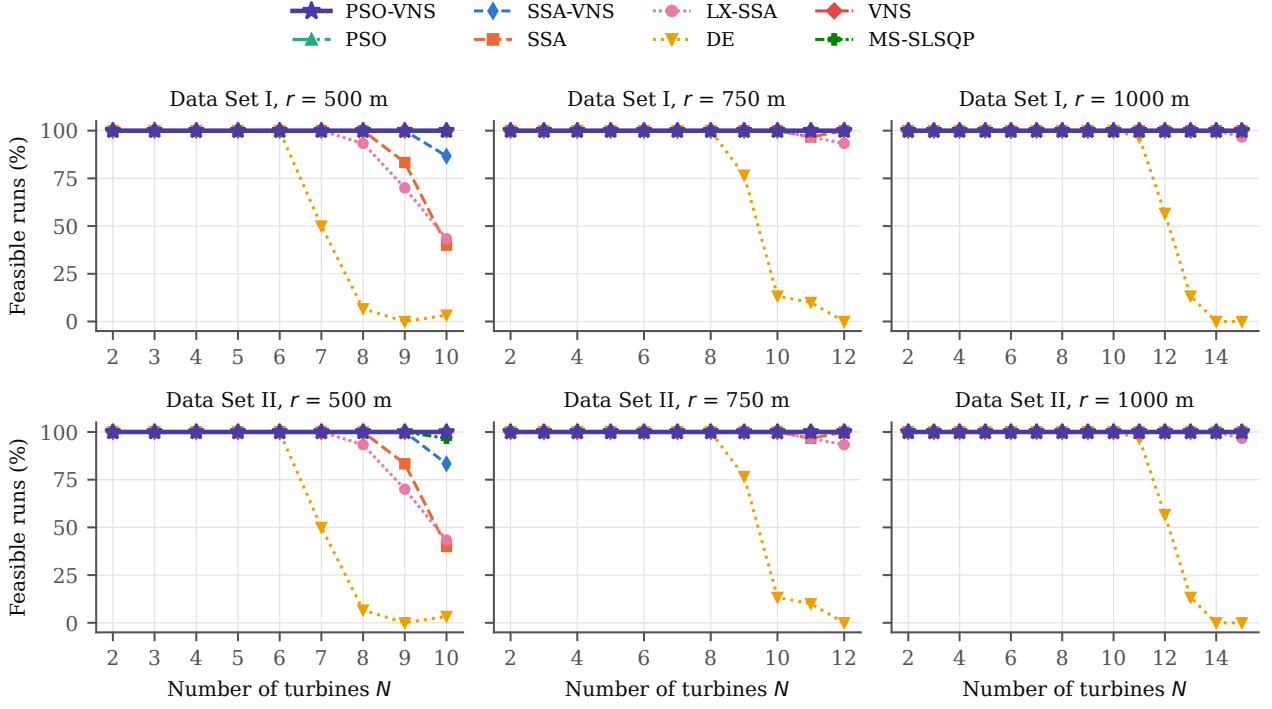


Figure S8: Percentage of feasible final layouts of the eight methods of the main comparison (30 seed-paired runs of 6,030 evaluations with random initialization per case) versus the number of turbines N , for each farm and wind data set.

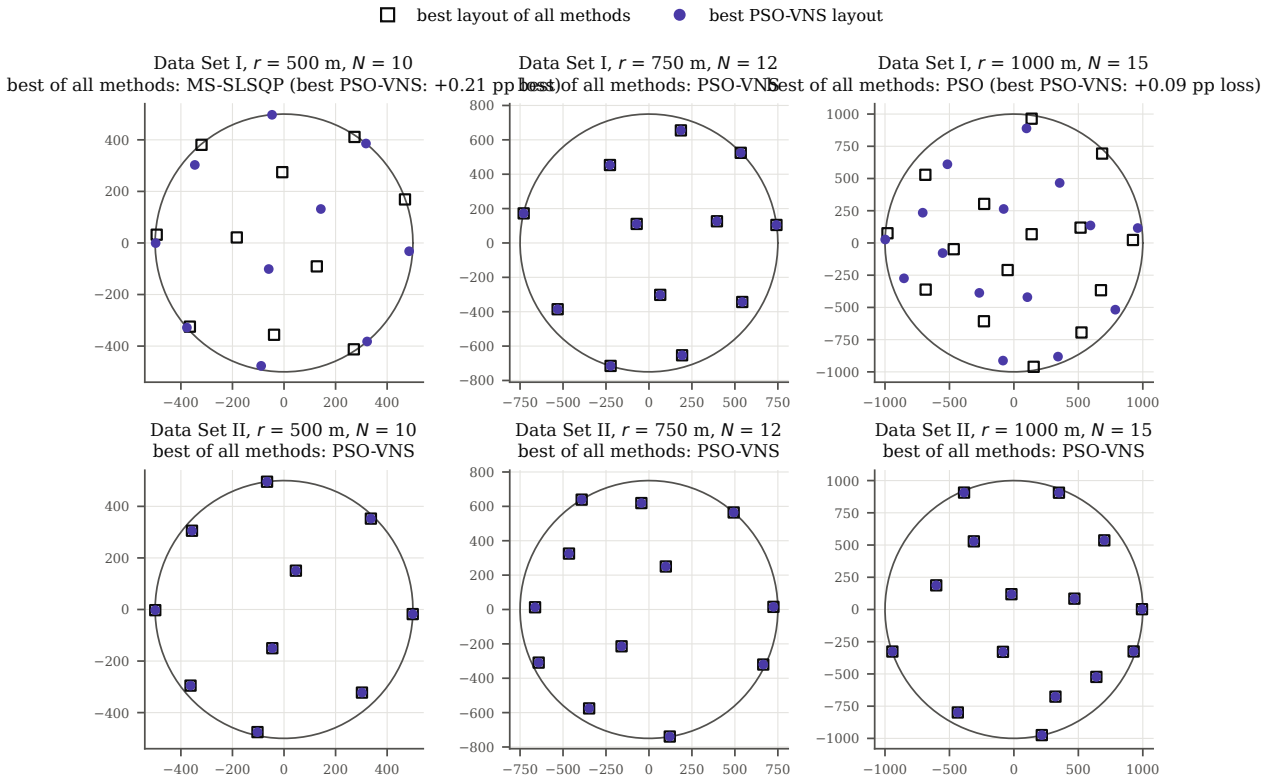


Figure S9: Best layout of all methods (squares) and best PSO-VNS layout (dots) for the largest N of each farm (one marker set when PSO-VNS is itself the best method); the coordinates are listed in Table S11.

S-III Component Analysis: Details

The component analysis of Section VII (Table IV) uses the same 68 cases and runs as the main comparison, plus LX-SSA-VNS and two random-sampling controls, RS-VNS and RSD-VNS. It ranks nine variants (Friedman test with 8 degrees of freedom) and tests fifteen planned contrasts in every case, Holm-adjusted over the contrasts of the case (Section S-I-C).

Random-sampling controls. Both controls replace Phase 1 by 3,015 random layouts of equal cost, the seeded initial population of 30 shared with all methods (so their runs stay seed-paired) followed by 2,985 further random layouts, and then run the same VNS phase from the best of them (lowest F_p); they search nothing in Phase 1. RS-VNS draws these layouts uniformly in the bounding square (Section S-I-D). It is deliberately a weak control, so a Phase-1 method that is better than RS-VNS passes only a minimum bar. A square sample is feasible with probability at most $(\pi/4)^N$ (Proposition S5), and far less because of the spacing constraint (Table S51). In 730 RS-VNS runs, Phase 1 draws no feasible layout at all (Table S55), although all of them draw layouts with every turbine inside the farm (730 of 730), so the spacing constraint is binding in all of them; paired by seed with the disc-sampling control, 242 of these runs obtain a feasible sample in the disc, whereas 488 have none under either sampling. The weakness of RS-VNS thus lies mainly in the spacing constraint and only in part in the square. RSD-VNS is the stronger control: it draws every turbine uniformly in the farm disc (angle uniform in $[0, 2\pi)$, distance $r\sqrt{U}$ from the centre with U uniform in $[0, 1]$), so that a sample can violate only the spacing constraint; budget, seeds and VNS phase are those of RS-VNS. A geometry-only replay of the random streams of all runs, verified against the stored convergence curves, gives 19.3% feasible Phase-1 samples for RSD-VNS against 9.1% for RS-VNS, and 494 against 730 runs without any feasible Phase-1 sample; at the switch, the best layout is feasible in 75.7% against 64.2% of the runs (Table S14). Both controls switch to VNS after 3,015 evaluations; their feasibility at the switch is read at the last stored checkpoint before it (call 3,000), which for RS-VNS gives the same runs as at 3,015, whereas for RSD-VNS the last 15 samples can make a few more runs feasible.

What the controls show. Against the square-sampling control, an SSA phase gives only a small gain and an LX-SSA phase none (Section VII). Against the disc control, the SSA phase gives no gain: SSA-VNS is never significantly better than RSD-VNS at run level (W/T/L 0/66/2), and the case means even favor the disc control ($\Delta\bar{L} = +0.024$ pp; higher mean loss in 41 and lower in 18 cases, $p_{\text{Holm}} = 0.014$), but not at the level of farm clusters (RSD-VNS lower in 4 of 6, cluster-robust $p = 0.55$; Section S-III-A). LX-SSA-VNS is worse than RSD-VNS (W/T/L 1/62/5, $\Delta\bar{L} = +0.087$ pp, $p_{\text{Holm}} = 8.2 \times 10^{-6}$; RSD-VNS lower in 5 of 6 clusters, cluster-robust $p = 0.087$), whereas a PSO phase clearly beats both controls (PSO-VNS against RSD-VNS: W/T/L 45/23/0, $\Delta\bar{L} = -0.306$ pp, $p_{\text{Holm}} = 4.0 \times 10^{-10}$). RSD-VNS itself is better than RS-VNS (W/T/L 6/62/0, $\Delta\bar{L} = -0.093$ pp, $p_{\text{Holm}} = 4.2 \times 10^{-7}$) and has the third-best average rank of the nine variants (3.77), after PSO-VNS and PSO and ahead of SSA-VNS (4.46). At 6,030 evaluations from random starts, salp-swarm phases therefore add nothing of practical size beyond random sampling in the farm: a gain of the SSA phase over disc sampling larger than the margin is ruled out at the seed and case levels (*Practical equivalence* below), and the LX-SSA phase is worse; the small gain of SSA-VNS over RS-VNS reflects only the weakness of the square-sampling control. LX-SSA as published (Laplace step and twice the evaluations per iteration, hence half the iterations) is worse than SSA (Section VII); these contrasts do not separate the Laplace step from its cost.

VNS phase. At 6,030 evaluations, Phase 2 of all hybrids is essentially one compass-search descent from the switch point (Section S-VII-B): in PSO-VNS, the shaking-plus-local-search cycles take 21.5% of the Phase-2 evaluations (pooled over the runs; median run 4.7%) and give 0.5% of the Phase-2 gain (2.5% including the cycle cut off by the budget). The component analysis therefore measures the value of a local-search (compass-search) phase after Phase 1, and the neighbourhood structure of VNS can matter only for small N .

Table S13 gives the per-case values of LX-SSA-VNS, RS-VNS and RSD-VNS and the per-case outcomes of the fifteen contrasts. Table S14 gives the feasibility and the wake loss of the Phase-1 result at the switch to VNS.

Budget split. Table S12 (moved here from the main text) summarizes the budget-split study of Section VII, in which the split $\omega = 0.5$ of PSO-VNS, fixed before the experiments, is compared with $\omega = 0.25$, 0.75 and 0.9 and with the end point $\omega = 1$, plain PSO at the same budget (five settings): $\omega = 0.75$ has the lowest mean wake loss and the best average rank, and plain PSO has a higher mean wake loss than $\omega = 0.75$, so the best tested split is not the PSO end point. Table S16 gives the per-case values. The split $\omega = 0.9$ differs significantly from $\omega = 0.75$ in no case (W/T/L of $\omega = 0.75$: 0/12/0; case means $p = 0.15$) and has a higher mean loss (2.553% vs. 2.543%), so $\omega = 0.75$ remains the best of the five settings.

Practical equivalence. A non-significant test does not show that two methods are equally good (Section S-VI-D). Table S15 therefore tests practical equivalence of the case means with two one-sided tests (TOST [16]) on the 90% bootstrap interval and adds the Bayesian signed-rank test with a region of practical equivalence [15]. One margin of ± 0.05 pp of wake loss is applied to every pair. It is a convention, set after the primary analysis and not derived from the data; in benchmark energy it corresponds to 4.1 MWh/yr per turbine in Data Set I and 2.1 MWh/yr in Data Set II. The table therefore reports, for every pair, the minimal margin $m_{\min}$ at which

Table S12: Budget Split of PSO-VNS (12 Cases, 30 Seeds)

ω	Mean loss (%)	Avg. rank	0.5 vs. ω (W/T/L)
0.25	2.796	4.75	5/7/0
0.5	2.643	3.25	—
0.75	2.543	1.42	0/8/4
0.9	2.553	1.83	0/8/4
1 (PSO alone)	2.693	3.75	3/9/0

$\omega = 0.75$ vs. $\omega = 1$ (PSO alone): 9/3/0 (W/T/L from the $\omega = 0.75$ side)
 $\omega = 0.9$ vs. $\omega = 0.75$: 0/12/0 (W/T/L from the $\omega = 0.9$ side)

ω : share of the 6,030 evaluations given to PSO ($\omega = 0.25, 0.5, 0.75, 0.9$; $\omega = 1$: PSO alone, same seeds); cases: middle and largest N of each farm and data set. Avg. rank among the five settings. W/T/L: cases in which the default $\omega = 0.5$ is significantly better / not different / worse (run-level Wilcoxon, Holm-adjusted over the six comparisons of each case).

Table S13: Component analysis per case (68 cases, 6,030 evaluations, 30 seed-paired runs): mean wake loss (%) of the feasible runs of LX-SSA-VNS, RS-VNS and RSD-VNS (“-”: fewer than 15 feasible runs), and run-level outcome of each planned contrast (+: first variant significantly better, -: significantly worse, .: no significant difference; Wilcoxon signed-rank, Holm-adjusted over the 15 contrasts of the case). C1: PSO-VNS vs. PSO; C2: SSA-VNS vs. SSA; C3: LX-SSA-VNS vs. LX-SSA; C4: PSO-VNS vs. VNS; C5: PSO-VNS vs. RS-VNS; C6: SSA-VNS vs. RS-VNS; C7: LX-SSA-VNS vs. RS-VNS; C8: PSO-VNS vs. RSD-VNS; C9: SSA-VNS vs. RSD-VNS; C10: LX-SSA-VNS vs. RSD-VNS; C11: RSD-VNS vs. RS-VNS; C12: PSO-VNS vs. SSA-VNS; C13: PSO-VNS vs. LX-SSA-VNS; C14: SSA-VNS vs. LX-SSA-VNS; C15: LX-SSA vs. SSA.

DS	r	N	LX-SSA-VNS	RS-VNS	RSD-VNS	C1	C2	C3	C4	C5	C6	C7	C8	C9	C10	C11	C12	C13	C14	C15
I	500	2	0.000	0.000	0.000	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
I	500	3	0.033	0.027	0.027	.	+	.	.	.	.	.	.	.	.	.	.	.	.	.
I	500	4	0.151	0.143	0.126	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
I	500	5	0.320	0.362	0.310	.	.	.	+	+	.	.	.	.	.	.	+	+	.	.
I	500	6	0.770	1.168	0.758	.	+	.	+	+	+	+	.	.	.	.	+	+	.	.
I	500	7	1.692	1.955	1.647	.	.	.	+	+	.	.	.	.	.	.	+	+	.	.
I	500	8	2.867	2.707	2.664	.	.	.	+	+	.	.	.	.	.	.	+	+	.	.
I	500	9	3.586	4.198	3.679	.	+	+	+	+	.	.	.	.	.	.	+	+	.	.
I	500	10	4.888	4.879	5.696	.	+	+	+	+	.	.	.	.	+	.	+	+	.	.
I	750	2	0.000	0.000	0.000	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
I	750	3	0.000	-0.000	-0.000	.	+	+	.	.	.	.	.	.	.	.	.	.	.	.
I	750	4	0.031	0.032	0.026	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
I	750	5	0.114	0.103	0.093	.	.	.	+	+	.	.	.	.	.	.	.	+	.	.
I	750	6	0.241	0.218	0.207	.	+	+	+	+	.	.	.	.	.	.	+	+	.	.
I	750	7	0.446	0.462	0.382	.	.	.	+	+	.	.	.	.	.	.	.	.	.	.
I	750	8	0.767	0.924	0.629	.	.	.	+	+	.	.	.	.	.	.	+	+	.	.
I	750	9	0.989	1.215	1.079	.	+	+	+	+	.	.	.	.	.	.	+	+	.	.
I	750	10	1.706	1.661	1.615	.	.	.	+	+	.	.	.	.	.	.	+	+	.	.
I	750	11	2.032	2.067	1.952	.	+	+	+	+	.	.	.	.	.	.	+	+	.	.
I	750	12	2.431	2.404	2.327	.	+	+	+	+	.	.	.	.	.	.	+	+	.	.
I	1000	2	0.000	0.000	0.000	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
I	1000	3	-0.000	-0.000	-0.000	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
I	1000	4	0.006	0.008	0.007	.	+	+	.	.	.	.	.	.	.	.	.	.	.	.
I	1000	5	0.044	0.047	0.042	.	+	.	+	+	.	.	.	.	.	.	.	+	.	.
I	1000	6	0.118	0.104	0.092	.	.	+	+	+	.	.	.	.	.	.	+	+	.	.
I	1000	7	0.212	0.202	0.175	.	+	.	+	+	.	.	.	.	.	.	+	+	.	.
I	1000	8	0.334	0.344	0.241	.	.	.	+	+	.	.	.	.	.	.	+	+	.	.
I	1000	9	0.498	0.502	0.447	.	.	.	+	+	.	.	.	.	.	.	+	+	.	.
I	1000	10	0.742	0.641	0.613	+	.	.	+	+	.	.	.	.	.	.	+	+	.	.
I	1000	11	1.004	0.983	0.818	.	+	.	+	+	.	.	.	.	.	.	+	+	.	.
I	1000	12	1.220	1.011	1.039	.	.	+	+	+	.	.	.	.	.	.	+	+	.	.
I	1000	13	1.546	1.448	1.431	.	+	+	+	+	.	.	.	.	.	.	+	+	.	.
I	1000	14	1.640	1.671	1.555	.	+	+	+	+	.	.	.	.	.	.	+	+	.	.
I	1000	15	1.777	1.946	1.854	.	+	+	+	.	.	.	.	.	.	.	+	+	.	.
II	500	2	0.000	0.000	0.000	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
II	500	3	0.169	0.163	0.158	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
II	500	4	0.705	0.762	0.648	.	.	.	.	+	.	.	.	.	.	.	.	.	.	.
II	500	5	1.660	1.742	1.587	.	.	.	+	+	.	.	.	.	.	.	+	+	.	.
II	500	6	2.861	3.024	2.785	.	+	.	+	+	.	.	.	.	.	.	+	+	.	.
II	500	7	4.399	4.335	4.229	.	.	+	+	+	.	.	.	.	.	.	+	+	.	.
II	500	8	5.619	5.572	5.607	+	+	+	+	+	.	.	.	.	.	.	+	+	.	.
II	500	9	6.909	7.092	7.089	.	+	+	+	+	.	.	.	.	.	.	+	+	.	.
II	500	10	8.548	8.652	8.288	+	+	+	+	+	.	.	.	.	.	.	+	+	.	.
II	750	2	0.000	0.000	0.000	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
II	750	3	0.005	0.005	-0.000	.	+	+	.	.	.	.	.	.	.	.	.	.	.	.
II	750	4	0.133	0.178	0.146	.	+	.	.	.	.	.	.	.	.	.	.	.	.	.
II	750	5	0.479	0.448	0.486	.	+	.	+	+	.	.	.	.	.	.	+	+	.	.
II	750	6	0.909	0.960	0.959	.	.	.	+	+	.	.	.	.	.	.	+	+	.	.
II	750	7	1.735	1.816	1.605	.	.	.	+	+	.	.	.	.	.	.	+	+	.	.
II	750	8	2.561	2.667	2.220	.	+	+	+	+	.	.	.	.	.	.	+	+	.	.
II	750	9	3.587	3.676	3.274	.	+	.	+	+	.	.	.	.	.	.	+	+	.	.
II	750	10	4.603	4.380	4.298	.	+	.	+	+	.	.	.	.	.	.	+	+	.	.
II	750	11	5.599	5.281	4.956	.	+	+	+	+	.	.	.	.	.	.	+	+	.	.
II	750	12	5.946	5.888	5.942	.	+	+	+	.	.	.	.	.	.	.	+	+	.	.
II	1000	2	0.000	0.000	0.000	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
II	1000	3	-0.000	-0.000	-0.000	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
II	1000	4	0.054	0.037	0.030	.	+	+	.	.	.	.	.	.	.	.	.	.	.	.
II	1000	5	0.180	0.182	0.198	.	.	+	.	.	.	.	.	.	.	.	.	.	.	.
II	1000	6	0.446	0.409	0.383	.	+	.	+	+	.	.	.	.	.	.	+	+	.	.
II	1000	7	0.775	0.683	0.658	.	.	+	+	+	.	.	.	.	.	.	+	+	.	.
II	1000	8	1.197	1.161	1.105	.	+	+	+	+	.	.	.	.	.	.	+	+	.	.
II	1000	9	1.836	1.641	1.534	.	+	.	+	+	.	.	.	.	.	.	+	+	.	.
II	1000	10	2.295	2.289	1.999	.	+	+	+	+	.	.	.	.	.	.	+	+	.	.
II	1000	11	3.202	2.963	2.527	+	+	+	+	+	.	.	.	.	.	.	+	+	.	.
II	1000	12	3.764	3.469	3.129	+	+	+	+	+	.	.	.	.	.	.	+	+	.	.
II	1000	13	4.276	4.281	3.954	+	+	+	+	+	.	.	.	.	.	.	+	+	.	.
II	1000	14	4.803	4.647	4.479	+	+	+	+	+	.	.	.	.	.	.	+	+	.	.
II	1000	15	5.220	5.210	4.950	+	+	+	+	+	.	.	.	.	.	.	+	+	.	.

Table S14: Switch point of the two-phase variants (68 cases, 30 runs, 6,030 evaluations): number of Phase-1 evaluations (the VNS phase starts with the next evaluation), runs whose best layout is feasible at the switch (%; read at the last convergence checkpoint at or before the switch, i.e. at 3,000 evaluations for RS-VNS and RSD-VNS, whose last 15 Phase-1 samples are not observable in the stored curves), mean wake loss at the switch (%; feasible runs), mean and median share (%) of the wake loss left at the switch that the second phase removes, and runs infeasible at the switch that end feasible. “PSO continued”: PSO alone over the same evaluations.

Variant	Switch	Feas. (%)	Loss (%)	Mean share	Median share	Made feasible
PSO-VNS	3,030	98.7	1.63	30.3	22.6	26
SSA-VNS	3,030	95.7	2.01	36.7	29.2	79
LX-SSA-VNS	3,030	94.1	2.11	39.0	31.9	118
RS-VNS	3,015	64.2	1.32	61.9	59.5	718
RSD-VNS	3,015	75.7	1.53	56.1	51.6	492
PSO continued (no VNS)	3,030	98.7	1.63	29.2	20.6	26

equivalence holds, and Fig. S10 shows the TOST p value of PSO-VNS vs. PSO as a function of the margin, so that readers can apply their own threshold. The verdict depends on the level of inference (Table S26). On the fixed benchmark (seed level: the 68 cases fixed, the run-to-run variability random), the 90% CI of PSO-VNS – PSO is $[-0.026, -0.010]$ pp, a small but significant average advantage of PSO-VNS that lies inside the margin, so the two are equivalent at this level; with the cases treated as exchangeable (case level), the 90% CI is $[-0.041, 0.004]$ pp ($p_{\text{TOST}} = 0.011$, $m_{\min} = 0.041$ pp), so the two are practically equivalent at this level too; generalized over farm clusters (cluster-robust CR2 interval), the 90% CI $[-0.067, 0.031]$ pp extends beyond -0.05 pp, and equivalence cannot be shown ($m_{\min} = 0.067$ pp). The average hides case-specific differences in both directions (Tables S28 and S29): 19 of the 68 cases differ by more than the margin, 11 in favor of PSO-VNS (by up to 0.41 pp) and 8 in favor of PSO (by up to 0.27 pp). For the 48 cases with $N < 10$ the two methods are equivalent (90% CI $[-0.005, 0.021]$ pp), for the 20 cases with $N \geq 10$ they are not (mean -0.079 pp, 90% CI $[-0.142, -0.017]$ pp), and on the 44 cases with a PSO-VNS wake loss of at least 0.2% equivalence is not shown either (90% CI $[-0.060, 0.008]$ pp). The Bayesian signed-rank test gives a posterior probability > 0.99 that practical equivalence is the most probable of the three outcomes (a rounded Monte Carlo share); the posterior mean shares of the pairs of cases are 0.25 (PSO-VNS better by more than the margin), 0.62 (inside the margin) and 0.12 (PSO better by more than the margin; Table S27), so about one pair of cases in four favors PSO-VNS by more than the margin. LX-SSA-VNS and RS-VNS are equivalent at the seed and case levels ($m_{\min} = 0.035$ pp, $P_{\text{rope}} > 0.99$) but not at the cluster level ($m_{\min} = 0.062$ pp); no pair of Table S15 is equivalent at the cluster level. The small gain of SSA-VNS over the weak control RS-VNS could not be shown to be equivalent to zero at ± 0.05 pp (90% CI $[-0.101, -0.039]$ pp, $m_{\min} = 0.102$ pp); in the Bayesian analysis, “practically negligible” is the more probable outcome ($P_{\text{rope}} = 0.70$ vs. $P_A = 0.30$). Against the disc control, the question is one-sided: a gain of the SSA phase larger than the margin is ruled out at the seed and case levels (90% CIs of SSA-VNS – RSD-VNS $[0.002, 0.046]$ and $[-0.022, 0.062]$ pp, both with a lower limit above -0.05 pp) but not at the cluster level (CR2 $[-0.051, 0.099]$ pp); equivalence itself is not shown at the case level ($m_{\min} = 0.062$ pp), and the difference points against SSA-VNS: the posterior probability that “SSA-VNS better” is the most probable outcome is $P_A < 0.01$, against $P_{\text{rope}} = 0.77$ and $P_B = 0.23$ for “RSD-VNS better”. LX-SSA-VNS is not equivalent to RSD-VNS and worse than it (90% CI $[0.047, 0.126]$ pp, $P_B = 0.90$), whereas the gain of PSO-VNS over RSD-VNS exceeds the margin: its whole 90% CI $[-0.380, -0.239]$ pp lies below -0.05 pp ($P_A > 0.99$). The disc control is better than the square control (RSD-VNS vs. RS-VNS: 90% CI $[-0.127, -0.055]$ pp, $P_A = 0.95$). PSO-VNS does not differ from PSO in the spread of its runs either: the per-case standard deviation and the worst feasible run of the 30 runs do not differ significantly ($p = 0.55$ and $p = 0.80$). Its robustness advantage lies in reaching feasibility, and it rests on one site with random starts: 30/30 vs. 19/30 feasible runs on the Horns Rev 1 block (exact McNemar $p = 9.8 \times 10^{-4}$, Table S31; Section VIII-B), whereas on the 68 benchmark cases both methods are always feasible.

Table S15: Practical equivalence of the per-case mean wake losses at 6,030 evaluations. $\overline{\Delta L}$: mean difference of the case-mean wake losses, first minus second method (percentage points, pp; negative = first method better), over the n cases in which both methods have at least 15 feasible runs out of 30 (cases excluded: LX-SSA vs. SSA 2; SSA-VNS vs. SSA 2; LX-SSA-VNS vs. LX-SSA 2; PSO-VNS vs. SSA 2; PSO-VNS vs. LX-SSA 2; PSO-VNS vs. DE 18); 95% and 90% percentile bootstrap CIs (10,000 resamples of the cases, fixed seed). One equivalence margin $m = 0.05$ pp is applied to every comparison; it was set after the primary analysis. Two one-sided tests (TOST): the pair is equivalent at m (“Eq.”) if the 90% CI lies inside $(-m, m)$; p_{TOST} : bootstrap TOST p (share of bootstrap means beyond $\mp m$, add-one corrected; $\leq$: no bootstrap mean beyond the margin); $m_{\min}$: smallest margin at which equivalence holds, $\max(|\log_{90}|, |\log_{10}|)$, rounded up. Bayesian signed-rank test [15] on the case-mean differences with region of practical equivalence $(-m, m)$ (Dirichlet-process prior strength $s = 0.5$ at $z_0 = 0$, 50,000 Monte Carlo samples, fixed seed): posterior probability that the first method is better (P_A), that the two are practically equivalent (P_{rope}) and that the second is better (P_B).

Pair (first vs. second)	n	$\overline{\Delta L}$ (pp)	95% CI	90% CI	p_{TOST}	$m_{\min}$	Eq.	P_A	P_{rope}	P_B
<i>Component-analysis contrasts</i>										
SSA-VNS vs. RS-VNS	68	-0.068	[-0.108, -0.034]	[-0.101, -0.039]	0.839	0.102	no	0.30	0.70	<0.01
LX-SSA-VNS vs. RS-VNS	68	-0.005	[-0.041, 0.026]	[-0.034, 0.021]	0.008	0.035	yes	<0.01	>0.99	<0.01
PSO-VNS vs. RS-VNS	68	-0.399	[-0.495, -0.308]	[-0.479, -0.324]	1.000	0.479	no	>0.99	<0.01	<0.01
SSA-VNS vs. LX-SSA-VNS	68	-0.063	[-0.091, -0.038]	[-0.086, -0.042]	0.841	0.087	no	0.53	0.47	<0.01
LX-SSA vs. SSA	66	+0.153	[0.096, 0.217]	[0.104, 0.206]	1.000	0.207	no	<0.01	0.12	0.88
SSA-VNS vs. SSA	66	-0.294	[-0.383, -0.214]	[-0.368, -0.226]	1.000	0.368	no	>0.99	<0.01	<0.01
LX-SSA-VNS vs. LX-SSA	66	-0.394	[-0.528, -0.276]	[-0.505, -0.293]	1.000	0.506	no	>0.99	<0.01	<0.01
PSO-VNS vs. PSO	68	-0.018	[-0.045, 0.008]	[-0.041, 0.004]	0.011	0.041	yes	<0.01	>0.99	<0.01
PSO-VNS vs. VNS	68	-0.465	[-0.559, -0.371]	[-0.545, -0.386]	1.000	0.546	no	>0.99	<0.01	<0.01
SSA-VNS vs. RSD-VNS	68	+0.024	[-0.033, 0.068]	[-0.022, 0.062]	0.141	0.062	no	<0.01	0.77	0.23
LX-SSA-VNS vs. RSD-VNS	68	+0.087	[0.039, 0.133]	[0.047, 0.126]	0.936	0.126	no	<0.01	0.10	0.90
PSO-VNS vs. RSD-VNS	68	-0.306	[-0.396, -0.227]	[-0.380, -0.239]	1.000	0.381	no	>0.99	<0.01	<0.01
RSD-VNS vs. RS-VNS	68	-0.093	[-0.134, -0.049]	[-0.127, -0.055]	0.971	0.128	no	0.95	0.05	<0.01
<i>Main comparison, PSO-VNS vs. each method (PSO and VNS: see above)</i>										
PSO-VNS vs. SSA-VNS	68	-0.330	[-0.409, -0.255]	[-0.397, -0.266]	1.000	0.397	no	>0.99	<0.01	<0.01
PSO-VNS vs. SSA	66	-0.611	[-0.762, -0.471]	[-0.735, -0.493]	1.000	0.735	no	>0.99	<0.01	<0.01
PSO-VNS vs. LX-SSA	66	-0.765	[-0.964, -0.580]	[-0.930, -0.606]	1.000	0.930	no	>0.99	<0.01	<0.01
PSO-VNS vs. DE	50	-0.887	[-1.191, -0.604]	[-1.137, -0.648]	1.000	1.137	no	>0.99	<0.01	<0.01
PSO-VNS vs. MS-SLSQP	68	-0.435	[-0.519, -0.353]	[-0.505, -0.366]	1.000	0.505	no	>0.99	<0.01	<0.01

Table S16: Sensitivity of PSO-VNS to the budget split between the PSO and VNS phases (25%, 50%, 75%, 90% of the 6,030 evaluations for PSO; 100%: PSO alone, same seeds): mean benchmark objective of the feasible runs (superscript: feasible runs when fewer than 30) and Wilcoxon signed-rank p (30 seed-paired runs), Holm-adjusted over the 6 comparisons of each case.

DS	r	N	$\omega = 0.25$	0.5	0.75	0.9	1	p (0.5/0.25)	p (0.5/0.75)	p (0.5/0.9)	p (0.5/1)	p (0.75/1)	p (0.9/0.75)
I	500	6	83851.3	83918.5	83929.5	83932.0	83925.6	0.068	1.000	0.980	1.000	0.155	1.000
I	500	10	135131.6	135160.5	135421.8	135697.8	135396.0	0.868	0.173	0.010	0.561	0.868	0.868
I	750	8	111835.7	111921.0	111944.0	111948.6	111924.1	0.036	0.180	0.288	0.954	0.012	0.954
I	750	12	164894.0	165526.4	165991.2	165975.5	165826.3	0.010	0.016	0.003	0.063	0.063	0.984
I	1000	10	139700.6	139874.2	139883.7	139875.2	139834.3	0.003	1.000	1.000	0.002	2.24×10^{-8}	0.104
I	1000	15	207367.9	207308.1	207724.0	207620.2	207263.9	0.741	9.59×10^{-5}	3.96×10^{-4}	0.741	1.04×10^{-4}	0.252
II	500	6	42809.4	42851.5	42883.5	42879.3	42857.9	0.275	0.321	0.321	1.000	3.36×10^{-4}	1.000
II	500	10	67624.1	67773.5	67799.8	67755.1	67472.2	0.209	1.000	1.000	2.48×10^{-4}	2.65×10^{-5}	0.209
II	750	8	57358.0	57411.1	57448.4	57444.6	57404.8	0.303	0.457	0.616	0.764	0.004	0.764
II	750	12	82784.6	82968.9	83182.0	83086.9	82874.2	0.280	0.016	0.280	0.280	8.38×10^{-5}	0.280
II	1000	10	71769.3	71906.7	71976.0	71981.0	71902.4	0.046	0.027	0.007	1.000	0.007	1.000
II	1000	15	104349.6	104794.8	104784.9	104689.9	104375.6	9.27×10^{-4}	0.968	0.168	1.62×10^{-5}	3.69×10^{-7}	0.063

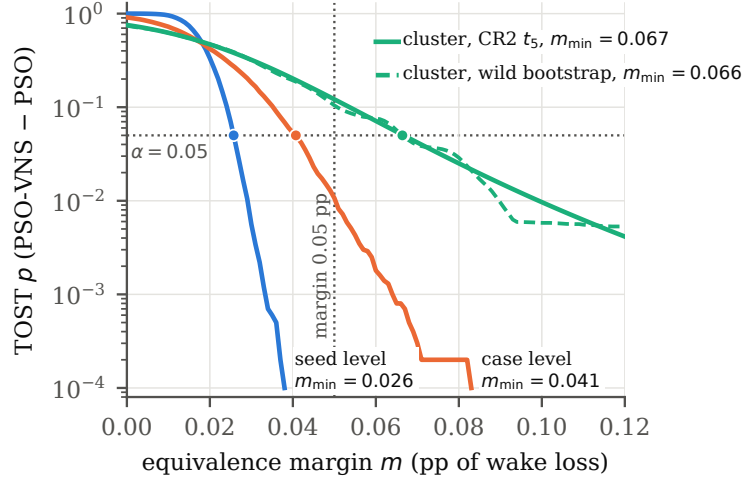


Figure S10: Equivalence curve of PSO-VNS vs. PSO: TOST p value versus the equivalence margin m (pp of wake loss) at three levels of inference (Table S26): seed level (fixed benchmark; bootstrap of the seed-paired runs within every case), case level (bootstrap over the 68 cases, as Table S15) and cluster level (six farm clusters; CR2 cluster-robust t with 5 d.f., and the restricted wild-cluster bootstrap). Dotted lines: $\alpha = 0.05$ and the margin $m = 0.05$ pp. The dots mark the minimal margins $m_{\min}$, at which a curve crosses α ; equivalence holds at every margin to the right of the dot. The bootstrap curves stop at their resolution limit $1/10,001$.

S-III-A Robustness of the Statistical Conclusions

Tables S17–S24 check whether the statistical conclusions of the main text depend on choices of the analysis. *Qualification threshold* (Tables S17 and S18): with each of the six thresholds 1, 10, 15, 20, 25 and 30 feasible runs of 30 (the paper uses 15), PSO-VNS is best-ranked, the complete rank order of the eight methods is the same, and all seventeen conclusions listed in the tables are unchanged.

Dependence between cases (Tables S19 and S20): the 68 cases form six (data set, radius) clusters with nested N . Every comparison that is significant on the case means favors the same method in all six clusters except SSA-VNS vs. RSD-VNS and LX-SSA-VNS vs. RSD-VNS, and leaving out any one cluster changes, of the nineteen verdicts, only that of SSA-VNS vs. RSD-VNS; PSO-VNS is best-ranked in every leave-one-cluster-out run. These cluster analyses concern significance verdicts; equivalence at the cluster level is not shown for any pair (Table S26 and Section S-III). The case-level result that SSA-VNS is worse than RSD-VNS does not hold at the cluster level: RSD-VNS has the lower mean in 4 of 6 clusters, the cluster-robust t test gives $p = 0.55$, the cluster-bootstrap interval $[-0.053, 0.084]$ pp contains zero, and one leave-one-cluster-out run loses significance (p up to 0.080); we therefore state only that the SSA phase gives no gain over disc sampling. LX-SSA-VNS is worse than RSD-VNS in 5 of 6 clusters with a cluster-bootstrap interval above zero ($[0.014, 0.158]$ pp) and no leave-one-cluster-out change, but the cluster-robust test is not significant ($p = 0.087$). PSO-VNS beats RSD-VNS, and RSD-VNS beats RS-VNS, in all six clusters (cluster-robust $p = 0.0061$ and 0.0043). The small gain of SSA-VNS over the weak control RS-VNS is found in all clusters, but with cluster-robust $p = 0.061$.

Multiplicity (Table S22): with one Holm correction over all 28 case-mean Wilcoxon tests of the main text (not over the other tests of the paper), 21 of the 23 unadjusted significant tests remain significant; the only ones lost are PSO-VNS vs. PSO, $N \geq 10$, both data sets and PSO-VNS ($\omega = 0.5$) vs. PSO-VNS ($\omega = 0.9$) (under Benjamini–Hochberg only PSO-VNS vs. PSO, $N \geq 10$, both data sets), whereas the Data Set II subgroup with $N \geq 10$ ($p_{\text{Holm}} \leq 0.018$) and all four contrasts with RSD-VNS ($p_{\text{Holm}} \leq 0.038$) survive. Thus $\omega = 0.9$ is better than $\omega = 0.5$ only unadjusted ($p_{\text{Holm}} = 0.19$), and it beats $\omega = 0.75$ at no level; the other split conclusions hold in all clusters (Table S21).

Subgroup and heterogeneity (Tables S23, S28 and S29; exploratory, stated after the primary analysis): the advantage of PSO-VNS over PSO grows with N only in Data Set II (mean within-farm Spearman $\rho = -0.66$, $p = 3.0 \times 10^{-4}$), not in Data Set I, where the trend has the opposite sign ($\rho = 0.34$, $p = 0.065$), and not over all cases once N is compared only within a farm ($p = 0.23$). With the density $\phi = N(\ell_{\min}/2r)^2$, the two data sets form a crossover: within Data Set II PSO-VNS gains with density (slope -0.52 pp per unit ϕ , $p = 1.0 \times 10^{-4}$), within Data Set I PSO gains (slope $+0.29$ pp, $p = 0.0012$; interaction $p = 0.01$), and 3 of the 6 cluster means of PSO-VNS – PSO exceed the margin, in opposite directions.

Energy (Table S24): over the 68 cases, the benchmark-energy gain of PSO-VNS over PSO is 0.020% of AEP (95% CI $[-0.008, 0.047]\%$), or 0.2 MWh/yr per turbine, against a margin of 4.1 (Data Set I) and 2.1 MWh/yr per turbine (Data Set II); in Data Set II with $N \geq 10$ it is 0.221%, and the gain of SSA-VNS over the weak

control RS-VNS (none against RSD-VNS) is 0.071%, against 0.413% for PSO-VNS over RS-VNS. These are differences in benchmark (model) energy, whose capacity factor is far above real sites.

Wake model (Table S25): the Jensen and Gaussian wake losses of the same final layouts differ by 0.38 pp on average, but this level shift largely cancels in paired comparisons: the per-case PSO-VNS – PSO difference changes by 0.031 pp on average between the two models, and its sign changes in 18 of the 68 cases. The model shift is therefore not used to justify the margin.

Estimates that match the tests (Table S30): the Hodges–Lehmann estimate, which belongs to the Wilcoxon signed-rank test, of SSA-VNS – RSD-VNS is +0.025 pp (95% CI [0.007, 0.058] pp), consistent with the significant Wilcoxon test, whereas the 95% CI of the mean contains zero; that of PSO-VNS – PSO is –0.003 pp (95% CI [–0.011, 0.003] pp), smaller in magnitude than the mean because a few large cases dominate the mean.

Table S17: Sensitivity of the case-level conclusions to the qualification threshold (a method is ranked in a case by the mean objective of its feasible runs only if at least this many of its 30 runs are feasible; the paper uses 15). Unq.: case-method cells of the eight-method comparison that do not qualify (PSO-VNS always qualifies, so each of them enters a case-mean test of PSO-VNS with an imputed maximal difference). Best-ranked method and average ranks of the Friedman ranking of Table II; p_z ns: methods not significantly different from PSO-VNS in the Holm-adjusted average-rank tests. Unadjusted two-sided Wilcoxon p on the per-case mean wake losses (with the same imputation rule at each threshold) for PSO-VNS vs. PSO and the component-analysis contrasts SSA-VNS vs. RS-VNS, LX-SSA-VNS vs. RS-VNS, SSA-VNS vs. LX-SSA-VNS and PSO-VNS vs. RS-VNS. Same: every conclusion of the paper holds at this threshold (best rank of PSO-VNS; only PSO not significantly different in rank; PSO-VNS vs. PSO not significant; SSA-VNS better than RS-VNS; LX-SSA-VNS not different from RS-VNS; SSA-VNS better than LX-SSA-VNS; PSO-VNS better than RS-VNS; each both unadjusted and Holm-adjusted within its table family), and so do those of the controls in Table S18.

Thresh.	Unq.	Best	Rank PSO-VNS	Rank PSO	p_z ns	PSO-VNS/ PSO	SSA-VNS/ RS-VNS	LX-SSA-VNS/ RS-VNS	SSA-VNS/ LX-SSA-VNS	PSO-VNS/ RS-VNS	Same
1	8	PSO-VNS	1.82	1.97	PSO	0.296	4.5×10^{-4}	0.718	5.4×10^{-6}	2.6×10^{-11}	yes
10	18	PSO-VNS	1.82	1.97	PSO	0.296	4.5×10^{-4}	0.718	5.4×10^{-6}	2.6×10^{-11}	yes
15 (paper)	22	PSO-VNS	1.82	1.97	PSO	0.296	4.5×10^{-4}	0.718	5.4×10^{-6}	2.6×10^{-11}	yes
20	26	PSO-VNS	1.82	1.97	PSO	0.296	4.5×10^{-4}	0.718	5.4×10^{-6}	2.6×10^{-11}	yes
25	30	PSO-VNS	1.82	1.97	PSO	0.296	4.5×10^{-4}	0.988	5.4×10^{-6}	2.6×10^{-11}	yes
30	47	PSO-VNS	1.82	1.97	PSO	0.296	0.001	0.634	1.5×10^{-5}	2.6×10^{-11}	yes

Table S18: Sensitivity of the Phase-6 control comparisons to the qualification threshold (as Table S17). Left: case-mean Wilcoxon p (68 cases, unadjusted) of the four component-analysis contrasts with the disc-sampling control RSD-VNS; mark: verdict at $\alpha = 0.05$ both unadjusted and Holm-adjusted over the 15 component-analysis contrasts (+: first method better, –: second better, ∙: not significant). Right: case-mean Wilcoxon p (12 split cases, unadjusted as in Table S12) of the budget-split tests of PSO-VNS (share ω of the evaluations for PSO; $\omega = 1$: PSO alone); mark: unadjusted verdict. Unq.: split case-setting cells that do not qualify (imputed maximal differences).

Thresh.	Unq.	SSA-VNS/ RSD-VNS	LX-SSA-VNS/ RSD-VNS	PSO-VNS/ RSD-VNS	RSD-VNS/ RS-VNS	$\omega = 0.5/\omega = 0.25$	$\omega = 0.5/\omega = 0.75$	$\omega = 0.5/\omega = 1$ (PSO)	$\omega = 0.75/\omega = 1$ (PSO)	$\omega = 0.5/\omega = 0.9$	$\omega = 0.9/\omega = 0.75$
1	0	0.005 –	1.4×10^{-6} –	3.6×10^{-11} +	5.3×10^{-8} +	0.001 +	0.001 –	0.424 ∙	4.9×10^{-4} +	0.027 –	0.151 ∙
10	0	0.005 –	1.4×10^{-6} –	3.6×10^{-11} +	5.3×10^{-8} +	0.001 +	0.001 –	0.424 ∙	4.9×10^{-4} +	0.027 –	0.151 ∙
15 (paper)	0	0.005 –	1.4×10^{-6} –	3.6×10^{-11} +	5.3×10^{-8} +	0.001 +	0.001 –	0.424 ∙	4.9×10^{-4} +	0.027 –	0.151 ∙
20	0	0.005 –	1.4×10^{-6} –	3.6×10^{-11} +	5.3×10^{-8} +	0.001 +	0.001 –	0.424 ∙	4.9×10^{-4} +	0.027 –	0.151 ∙
25	0	0.005 –	1.4×10^{-6} –	3.6×10^{-11} +	3.9×10^{-9} +	0.001 +	0.001 –	0.424 ∙	4.9×10^{-4} +	0.027 –	0.151 ∙
30	0	0.001 –	3.7×10^{-7} –	3.6×10^{-11} +	9.9×10^{-9} +	0.001 +	0.001 –	0.424 ∙	4.9×10^{-4} +	0.027 –	0.151 ∙

Table S19: Cluster-level analysis. The 68 cases form 6 clusters (data set/radius in m) with nested N . $\overline{\Delta L}$: mean difference of the case-mean wake losses, first minus second method (pp; negative = first better), over the n cases in which both methods have at least 15 feasible runs; cluster columns: the same mean within each cluster. Fav.: clusters in which the first/second method has the lower mean. Tests over the 6 cluster means: exact two-sided sign test and exact Wilcoxon signed-rank test; flip: exact cluster sign-flip test of the case-weighted mean (all 2^6 sign patterns); CR: cluster-robust t test of the case-weighted mean (CR1 variance, 5 d.f.). With 6 clusters the smallest attainable exact two-sided p is 0.031. Case: the case-level Wilcoxon p of the paper (68 cases, imputation rule; unadjusted).

Pair (first vs. second)	n	$\overline{\Delta L}$	I/500	I/750	I/1000	II/500	II/750	II/1000	Fav.	p_{sign}	p_W	p_{flip}	p_{CR}	p_{case}
PSO-VNS vs. PSO	68	-0.018	+0.064	+0.043	-0.008	-0.063	-0.031	-0.090	4/2	0.688	0.688	0.562	0.486	0.296
PSO-VNS vs. SSA-VNS	68	-0.330	-0.468	-0.249	-0.156	-0.441	-0.379	-0.371	6/0	0.031	0.031	0.031	0.001	1.9×10^{-11}
PSO-VNS vs. SSA	66	-0.611	-0.670	-0.469	-0.371	-0.732	-0.718	-0.777	6/0	0.031	0.031	0.031	5.2×10^{-4}	7.6×10^{-12}
PSO-VNS vs. LX-SSA	66	-0.765	-0.913	-0.594	-0.478	-0.961	-0.890	-0.890	6/0	0.031	0.031	0.031	3.4×10^{-4}	7.6×10^{-12}
PSO-VNS vs. DE	50	-0.887	-0.431	-0.654	-0.750	-0.786	-1.110	-1.333	6/0	0.031	0.031	0.031	0.001	1.4×10^{-11}
PSO-VNS vs. VNS	68	-0.465	-0.513	-0.380	-0.252	-0.527	-0.568	-0.595	6/0	0.031	0.031	0.031	6.7×10^{-4}	6.3×10^{-11}
PSO-VNS vs. MS-SLSQP	68	-0.435	-0.144	-0.365	-0.391	-0.432	-0.503	-0.672	6/0	0.031	0.031	0.031	0.002	6.0×10^{-12}
SSA-VNS vs. SSA	66	-0.294	-0.218	-0.220	-0.215	-0.350	-0.339	-0.406	6/0	0.031	0.031	0.031	5.6×10^{-4}	5.7×10^{-11}
LX-SSA-VNS vs. LX-SSA	66	-0.394	-0.429	-0.320	-0.288	-0.539	-0.423	-0.433	6/0	0.031	0.031	0.031	1.1×10^{-4}	7.6×10^{-12}
PSO-VNS vs. RS-VNS	68	-0.399	-0.680	-0.304	-0.173	-0.559	-0.443	-0.380	6/0	0.031	0.031	0.031	0.003	2.6×10^{-11}
SSA-VNS vs. RS-VNS	68	-0.068	-0.212	-0.055	-0.018	-0.119	-0.064	-0.008	6/0	0.031	0.031	0.031	0.061	4.5×10^{-4}
LX-SSA-VNS vs. RS-VNS	68	-0.005	-0.126	-0.030	+0.017	-0.052	+0.024	+0.077	3/3	1.000	0.688	0.812	0.855	0.718
PSO-VNS vs. LX-SSA-VNS	68	-0.393	-0.554	-0.274	-0.190	-0.507	-0.467	-0.456	6/0	0.031	0.031	0.031	0.001	1.6×10^{-11}
SSA-VNS vs. LX-SSA-VNS	68	-0.063	-0.087	-0.025	-0.034	-0.067	-0.088	-0.085	6/0	0.031	0.031	0.031	0.003	5.4×10^{-6}
LX-SSA vs. SSA	66	+0.153	+0.244	+0.124	+0.108	+0.229	+0.172	+0.112	0/6	0.031	0.031	0.031	9.5×10^{-4}	1.9×10^{-7}
SSA-VNS vs. RSD-VNS	68	+0.024	-0.153	+0.016	+0.025	-0.013	+0.064	+0.137	2/4	0.688	0.562	0.562	0.546	0.005
LX-SSA-VNS vs. RSD-VNS	68	+0.087	-0.067	+0.041	+0.059	+0.053	+0.152	+0.222	1/5	0.219	0.219	0.125	0.087	1.4×10^{-6}
PSO-VNS vs. RSD-VNS	68	-0.306	-0.621	-0.234	-0.131	-0.454	-0.315	-0.235	6/0	0.031	0.031	0.031	0.006	3.6×10^{-11}
RSD-VNS vs. RS-VNS	68	-0.093	-0.059	-0.070	-0.043	-0.106	-0.128	-0.145	6/0	0.031	0.031	0.031	0.004	5.3×10^{-8}

Table S20: Dependence of the case-mean comparisons on the six clusters. 95% intervals of the mean case difference $\overline{\Delta L}$ (pp, first minus second): percentile bootstrap over the cases (as in the paper), over the clusters (block bootstrap: 10,000 resamples of the six clusters with replacement, case-weighted mean, fixed seed; with six clusters it tends to be too narrow) and cluster-robust t interval (CR1, 5 d.f.). Leave-one-cluster-out (LOCO): the case-mean Wilcoxon test (imputation rule, unadjusted) is repeated six times, each time without one cluster. Verdict: result on all 68 cases (first / second = that method significantly better at $\alpha = 0.05$; ns = not significant); p range over the six LOCO runs; Flips: LOCO runs in which the sign of $\overline{\Delta L}$ changes; Changed: clusters whose removal changes the verdict. The best-ranked method of the eight-method Friedman ranking is PSO-VNS in every LOCO run, and PSO is the only method not significantly different from it in rank.

Pair (first vs. second)	$\overline{\Delta L}$	Case boot. CI	Cluster boot. CI	CR1 CI	Verdict	LOCO p	Flips	Changed
PSO-VNS vs. PSO	-0.018	[-0.045, 0.008]	[-0.058, 0.025]	[-0.080, 0.044]	ns	0.091-0.658	1	-
PSO-VNS vs. SSA-VNS	-0.330	[-0.409, -0.255]	[-0.419, -0.242]	[-0.461, -0.200]	first	1.8×10^{-15} - 4.2×10^{-10}	0	-
PSO-VNS vs. SSA	-0.611	[-0.762, -0.471]	[-0.742, -0.476]	[-0.810, -0.412]	first	1.6×10^{-10} - 1.1×10^{-9}	0	-
PSO-VNS vs. LX-SSA	-0.765	[-0.964, -0.580]	[-0.916, -0.608]	[-0.992, -0.537]	first	1.6×10^{-10} - 1.1×10^{-9}	0	-
PSO-VNS vs. DE	-0.887	[-1.191, -0.604]	[-1.135, -0.636]	[-1.247, -0.527]	first	3.1×10^{-10} - 1.4×10^{-9}	0	-
PSO-VNS vs. VNS	-0.465	[-0.559, -0.371]	[-0.564, -0.353]	[-0.625, -0.306]	first	2.0×10^{-13} - 1.2×10^{-9}	0	-
PSO-VNS vs. MS-SLSQP	-0.435	[-0.519, -0.353]	[-0.556, -0.305]	[-0.618, -0.253]	first	2.4×10^{-11} - 1.5×10^{-9}	0	-
SSA-VNS vs. SSA	-0.294	[-0.383, -0.214]	[-0.361, -0.233]	[-0.391, -0.197]	first	1.6×10^{-10} - 1.0×10^{-8}	0	-
LX-SSA-VNS vs. LX-SSA	-0.394	[-0.528, -0.276]	[-0.460, -0.335]	[-0.487, -0.302]	first	1.6×10^{-10} - 1.1×10^{-9}	0	-
PSO-VNS vs. RS-VNS	-0.399	[-0.495, -0.308]	[-0.538, -0.281]	[-0.583, -0.214]	first	5.3×10^{-15} - 5.8×10^{-10}	0	-
SSA-VNS vs. RS-VNS	-0.068	[-0.108, -0.034]	[-0.127, -0.026]	[-0.142, 0.005]	first	1.5×10^{-4} -0.005	0	-
LX-SSA-VNS vs. RS-VNS	-0.005	[-0.041, 0.026]	[-0.061, 0.038]	[-0.076, 0.066]	ns	0.310-1.000	2	-
PSO-VNS vs. LX-SSA-VNS	-0.393	[-0.487, -0.303]	[-0.499, -0.288]	[-0.551, -0.236]	first	1.8×10^{-15} - 3.5×10^{-10}	0	-
SSA-VNS vs. LX-SSA-VNS	-0.063	[-0.091, -0.038]	[-0.084, -0.041]	[-0.094, -0.032]	first	1.7×10^{-6} - 1.8×10^{-4}	0	-
LX-SSA vs. SSA	+0.153	[0.096, 0.217]	[0.121, 0.202]	[0.097, 0.210]	second	8.4×10^{-8} - 1.3×10^{-6}	0	-
SSA-VNS vs. RSD-VNS	+0.024	[-0.033, 0.068]	[-0.053, 0.084]	[-0.072, 0.120]	second	0.001-0.080	1	II/1000
LX-SSA-VNS vs. RSD-VNS	+0.087	[0.039, 0.133]	[0.014, 0.158]	[-0.018, 0.193]	second	4.8×10^{-7} - 7.0×10^{-5}	0	-
PSO-VNS vs. RSD-VNS	-0.306	[-0.396, -0.227]	[-0.453, -0.204]	[-0.479, -0.133]	first	2.5×10^{-14} - 7.8×10^{-10}	0	-
RSD-VNS vs. RS-VNS	-0.093	[-0.134, -0.049]	[-0.124, -0.060]	[-0.141, -0.044]	first	1.4×10^{-8} - 4.4×10^{-7}	0	-

Table S21: Budget-split tests of PSO-VNS (Table S12; share ω of the 6,030 evaluations for PSO, $\omega = 1$: PSO alone) at the cluster level. The 12 split cases are two per (data set, radius) cluster. Columns as in Tables S19 and S20: mean case difference (pp, first minus second; negative = first better) overall and per cluster; Fav.: clusters favouring the first/second setting; p_W : exact Wilcoxon over the six cluster means; p_{CR} : cluster-robust t (CR1, 5 d.f.); cluster-bootstrap 95% CI; p_{case} : case-mean Wilcoxon over the 12 cases (unadjusted, as in the paper); Verdict and LOCO p range (10 cases each) as in Table S20; Changed: clusters whose removal changes the verdict.

Split test	n	$\overline{\Delta L}$	I/500	I/750	I/1000	II/500	II/750	II/1000	Fav.	p_W	p_{CR}	Cluster boot. CI	p_{case}	Verdict	LOCO p	Changed
$\omega = 0.5$ vs. $\omega = 0.25$	12	-0.153	-0.050	-0.226	-0.048	-0.150	-0.150	-0.297	6/0	0.031	0.012	[-0.224, -0.083]	0.001	first	0.002-0.006	-
$\omega = 0.5$ vs. $\omega = 0.75$	12	+0.100	+0.100	+0.148	+0.102	+0.054	+0.153	+0.043	0/6	0.031	0.003	[0.066, 0.134]	0.001	second	0.002-0.006	-
$\omega = 0.5$ vs. $\omega = 1$ (PSO)	12	-0.050	+0.088	+0.090	-0.025	-0.199	-0.059	-0.194	4/2	0.562	0.386	[-0.146, 0.042]	0.424	ns	0.160-0.770	-
$\omega = 0.75$ vs. $\omega = 1$ (PSO)	12	-0.150	-0.011	-0.058	-0.127	-0.253	-0.212	-0.237	6/0	0.031	0.015	[-0.218, -0.076]	4.9×10^{-4}	first	0.002-0.002	-
$\omega = 0.5$ vs. $\omega = 0.9$	12	+0.090	+0.199	+0.146	+0.074	+0.019	+0.096	+0.003	0/6	0.031	0.032	[0.037, 0.144]	0.027	second	0.014-0.084	I/500, I/750, I/1000, II/750
$\omega = 0.9$ vs. $\omega = 0.75$	12	+0.010	-0.100	+0.003	+0.028	+0.035	+0.057	+0.040	1/5	0.438	0.670	[-0.037, 0.043]	0.151	ns	0.020-0.375	I/500

Table S22: One multiplicity family over all 28 case-mean Wilcoxon tests quoted in the main text (tests that appear in two tables are counted once). n : cases entering the test (imputation rule); W/L: cases in which the first method has the lower/higher case-mean wake loss; $\overline{\Delta L}$: mean difference (pp, first minus second, both-qualified cases); p : unadjusted two-sided Wilcoxon p ; p_{Holm} : Holm-adjusted over all 28 tests (family-wise error); q_{BH} : Benjamini-Hochberg adjusted (false discovery rate). Holm sig.: significant after the all-family Holm correction ("—": not significant even unadjusted; **no**: significant only unadjusted). 23 of 28 tests are significant unadjusted, 21 after Holm and 22 after Benjamini-Hochberg.

Test (first vs. second)	n	W/L	$\overline{\Delta L}$	p	p_{Holm}	q_{BH}	Holm sig.
<i>Main comparison (Table II)</i>							
PSO-VNS vs. PSO	68	34/26	-0.018	0.296	1.000	0.331	—
PSO-VNS vs. SSA-VNS	68	59/1	-0.330	1.9×10^{-11}	4.2×10^{-10}	7.6×10^{-11}	yes
PSO-VNS vs. SSA	68	62/0	-0.611	7.6×10^{-12}	2.0×10^{-10}	5.3×10^{-11}	yes
PSO-VNS vs. LX-SSA	68	62/0	-0.765	7.6×10^{-12}	2.0×10^{-10}	5.3×10^{-11}	yes
PSO-VNS vs. DE	68	60/0	-0.887	1.4×10^{-11}	3.4×10^{-10}	7.6×10^{-11}	yes
PSO-VNS vs. VNS	68	55/4	-0.465	6.3×10^{-11}	1.1×10^{-9}	1.6×10^{-10}	yes
PSO-VNS vs. MS-SLSQP	68	66/2	-0.435	6.0×10^{-12}	1.7×10^{-10}	5.3×10^{-11}	yes
<i>Component analysis (Table IV; pairs not in the main comparison)</i>							
SSA-VNS vs. SSA	68	60/2	-0.294	5.7×10^{-11}	1.1×10^{-9}	1.6×10^{-10}	yes
LX-SSA-VNS vs. LX-SSA	68	62/0	-0.394	7.6×10^{-12}	2.0×10^{-10}	5.3×10^{-11}	yes
PSO-VNS vs. RS-VNS	68	58/1	-0.399	2.6×10^{-11}	5.6×10^{-10}	9.3×10^{-11}	yes
SSA-VNS vs. RS-VNS	68	42/18	-0.068	4.5×10^{-4}	0.006	7.8×10^{-4}	yes
LX-SSA-VNS vs. RS-VNS	68	27/33	-0.005	0.718	1.000	0.718	—
PSO-VNS vs. LX-SSA-VNS	68	60/0	-0.393	1.6×10^{-11}	3.7×10^{-10}	7.6×10^{-11}	yes
SSA-VNS vs. LX-SSA-VNS	68	47/13	-0.063	5.4×10^{-6}	7.5×10^{-5}	1.0×10^{-5}	yes
LX-SSA vs. SSA	66	9/51	+0.153	1.9×10^{-7}	3.0×10^{-6}	4.0×10^{-7}	yes
SSA-VNS vs. RSD-VNS	68	18/41	+0.024	0.005	0.037	0.006	yes
LX-SSA-VNS vs. RSD-VNS	68	10/50	+0.087	1.4×10^{-6}	2.1×10^{-5}	2.7×10^{-6}	yes
PSO-VNS vs. RSD-VNS	68	57/2	-0.306	3.6×10^{-11}	7.2×10^{-10}	1.1×10^{-10}	yes
RSD-VNS vs. RS-VNS	68	52/7	-0.093	5.3×10^{-8}	9.0×10^{-7}	1.2×10^{-7}	yes
<i>Post hoc subgroups of PSO-VNS vs. PSO</i>							
PSO-VNS vs. PSO, $N \geq 10$, both data sets	20	14/6	-0.079	0.048	0.291	0.059	no
PSO-VNS vs. PSO, $N \geq 10$, Data Set I	10	4/6	+0.052	0.322	1.000	0.347	—
PSO-VNS vs. PSO, $N \geq 10$, Data Set II	10	10/0	-0.210	0.002	0.018	0.003	yes
<i>Budget split (Table S12, 12 cases)</i>							
PSO-VNS ($\omega = 0.5$) vs. PSO-VNS ($\omega = 0.25$)	12	11/1	-0.153	0.001	0.016	0.002	yes
PSO-VNS ($\omega = 0.5$) vs. PSO-VNS ($\omega = 0.75$)	12	1/11	+0.100	0.001	0.016	0.002	yes
PSO-VNS ($\omega = 0.5$) vs. PSO	12	7/5	-0.050	0.424	1.000	0.440	—
PSO-VNS ($\omega = 0.75$) vs. PSO	12	12/0	-0.150	4.9×10^{-4}	0.006	8.0×10^{-4}	yes
PSO-VNS ($\omega = 0.5$) vs. PSO-VNS ($\omega = 0.9$)	12	2/10	+0.090	0.027	0.188	0.034	no
PSO-VNS ($\omega = 0.9$) vs. PSO-VNS ($\omega = 0.75$)	12	4/8	+0.010	0.151	0.757	0.177	—

Table S23: Tests of the post hoc $N \geq 10$ pattern of PSO-VNS vs. PSO. ΔL : case-mean wake loss of PSO-VNS minus that of PSO (pp; negative = PSO-VNS better). Interaction: mean ΔL of the $N \geq 10$ cases minus that of the $N < 10$ cases, two-sided permutation test (20,000 permutations, fixed seed) of the group labels over all cases (unrestr.) or within each (data set, radius) cluster (strat.). Spearman: correlation of ΔL with N (value: pooled / mean within-cluster ρ ; p : asymptotic / within-cluster permutation of ΔL , which uses N only within a farm and so is not confounded with the radius). Data Set II minus I: difference of ΔL between the two data sets over the identical (r, N) cases (in parentheses: pairs in which the PSO-VNS advantage is larger in Data Set II / in Data Set I), Wilcoxon signed-rank test.

Test	n	Estimate	p (unrestr.)	p (strat.)
Interaction $N \geq 10$ vs. $N < 10$, all cases	68	-0.086	0.003	0.002
Interaction $N \geq 10$ vs. $N < 10$, Data Set I	34	+0.035	0.235	0.289
Interaction $N \geq 10$ vs. $N < 10$, Data Set II	34	-0.207	5.0×10^{-5}	1.0×10^{-4}
Spearman $\rho(N, \Delta L)$, all cases	68	-0.25 / -0.16	0.038	0.228
Spearman $\rho(N, \Delta L)$, Data Set I	34	+0.09 / +0.34	0.626	0.065
Spearman $\rho(N, \Delta L)$, Data Set II	34	-0.62 / -0.66	1.0×10^{-4}	3.0×10^{-4}
Data Set II minus I, same (r, N) , $N \geq 10$ (9/1)	10	-0.262	0.004	-
Data Set II minus I, same (r, N) , $N < 10$ (8/12)	24	-0.020	0.294	-
Data Set II minus I, same (r, N) , all N (17/13)	34	-0.092	0.100	-

Table S24: Headline differences in benchmark energy. $\overline{\Delta L}$: mean difference of the case-mean wake losses (pp of the wake-free AEP, first minus second). Gain: mean over the cases of the first method's benchmark AEP minus the second's, in % of the second's and in MWh/yr of benchmark (model) energy per case, with 95% percentile bootstrap CIs over the cases (10,000 resamples, fixed seed); per turbine: gain divided by N ; largest N : mean gain over the cases with the largest N of the set. Benchmark AEP = objective/15 $\times$ 8.76 MWh/yr (Section II-B); the Kusiak-Song benchmark has a capacity factor far above real sites (62% wake-free for Data Set I), so these values indicate the order of magnitude of the differences, not the energy of a real farm.

Pair (first vs. second)	Cases	$\overline{\Delta L}$ (pp)	Gain (% AEP)	Gain (MWh/yr) per turbine	largest N	
PSO-VNS vs. PSO	68	-0.018	+0.020 [-0.008, 0.047]	+6 [-11, 22]	+0.2	+135 (15)
PSO-VNS vs. PSO (Data Set II, $N \geq 10$)	10	-0.210	+0.221 [0.141, 0.304]	+111 [68, 156]	+9.0	+245 (15)
SSA-VNS vs. RS-VNS	68	-0.068	+0.071 [0.035, 0.113]	+38 [15, 66]	+4.5	+162 (15)
PSO-VNS vs. RS-VNS	68	-0.399	+0.413 [0.318, 0.513]	+225 [168, 287]	+23.9	+439 (15)

Table S25: Wake-model uncertainty from the study's own layouts. Every feasible final layout of the eight main methods (68 cases, 6,030 evaluations) is re-evaluated (not re-optimized) with the benchmark Jensen model and with the Gaussian wake of Bastankhah and Porté-Agel ($k^* = 0.04$, Table S39); wake loss in pp of the wake-free power. Top: absolute difference $|L_{\text{Gauss}} - L_{\text{Jensen}}|$ of the same layout (mean, median, 10-90% quantiles) and mean signed difference, to be compared with the equivalence margin of ± 0.05 pp. Bottom: mean over the cases (both methods ≥ 15 feasible runs) of the case-mean difference under each model, the mean absolute change of the per-case difference between the models, and the cases in which its sign differs.

Layouts	n	mean	median	10-90%	signed
All feasible final layouts	15,614	0.378	0.274	[0.032, 0.864]	+0.343
Data Set I	7,808	0.295	0.225	-	-
Data Set II	7,806	0.461	0.341	-	-
<i>Case-mean difference PSO-VNS minus other method (pp)</i>					
Pair	Cases	Jensen	Gaussian	mean change	sign changes
PSO-VNS vs. PSO	68	-0.018	-0.009	0.031	18
PSO-VNS vs. SSA-VNS	68	-0.332	-0.354	0.064	3
PSO-VNS vs. SSA	66	-0.627	-0.586	0.080	2
PSO-VNS vs. LX-SSA	66	-0.774	-0.723	0.106	4
PSO-VNS vs. DE	50	-0.887	-0.649	0.249	4
PSO-VNS vs. VNS	68	-0.465	-0.509	0.088	4
PSO-VNS vs. MS-SLSQP	68	-0.438	-0.252	0.191	9

Table S26: Practical equivalence (margin $m = 0.05$ pp, set after the primary analysis) of the pairs of Table S15 at three levels of inference. $\overline{\Delta L}$: mean over the n cases of the case-mean wake-loss difference, first minus second method (pp; negative = first better; the same estimate at every level). *Seed level* (fixed benchmark: the 68 cases are fixed and only the run-to-run variability is random): 10,000 bootstrap resamples of the 30 seed-paired runs within every case (same seeds for both methods; case means over the feasible resampled runs), percentile 90% CI. *Case level* (generalization to exchangeable cases, as Table S15): percentile bootstrap over the cases; p_{TOST} as there. *Cluster level* (generalization to new farms; 6 (data set, radius) clusters): cluster-robust 90% CI with the CR2 (bias-reduced) variance and a t distribution with 5 d.f. (Bell–McCaffrey d.f. 4.8 for PSO-VNS vs. PSO), and a restricted wild-cluster bootstrap- t TOST (CR2-studentized, all 46,656 draws of the Webb six-point weights enumerated; $m_{\min}$ by test inversion). $m_{\min}$: smallest margin at which equivalence holds (rounded up). Eq.: equivalent at $\pm m$ (cluster level: CR2 CI inside $\pm m$ and wild $p_{\text{TOST}} \leq 0.05$).

Pair (first vs. second)	n	$\overline{\Delta L}$	Seed level				Case level				Cluster level					
			90% CI	$m_{\min}$	Eq.		90% CI	$m_{\min}$	p_{TOST}	Eq.	CR2 90% CI	$m_{\min}$	wild p_{TOST}	wild $m_{\min}$	Eq.	
<i>Component-analysis contrasts</i>																
SSA-VNS vs. RS-VNS	68	-0.068	[-0.092, -0.044]	0.093	no		[-0.101, -0.039]	0.102	0.839	no	[-0.126, -0.011]	0.126	0.716	0.130	no	
LX-SSA-VNS vs. RS-VNS	68	-0.005	[-0.030, 0.020]	0.031	yes		[-0.034, 0.021]	0.035	0.008	yes	[-0.061, 0.051]	0.062	0.082	0.068	no	
PSO-VNS vs. RS-VNS	68	-0.399	[-0.420, -0.378]	0.421	no		[-0.479, -0.324]	0.479	1.000	no	[-0.544, -0.254]	0.544	0.995	0.557	no	
SSA-VNS vs. LX-SSA-VNS	68	-0.063	[-0.086, -0.040]	0.086	no		[-0.086, -0.042]	0.087	0.841	no	[-0.087, -0.039]	0.088	0.843	0.087	no	
LX-SSA vs. SSA	66	+0.153	[0.123, 0.185]	0.185	no		[0.104, 0.206]	0.207	1.000	no	[0.109, 0.198]	0.198	0.995	0.213	no	
SSA-VNS vs. SSA	66	-0.294	[-0.319, -0.269]	0.319	no		[-0.368, -0.226]	0.368	1.000	no	[-0.371, -0.216]	0.372	0.995	0.376	no	
LX-SSA-VNS vs. LX-SSA	66	-0.394	[-0.422, -0.367]	0.423	no		[-0.505, -0.293]	0.506	1.000	no	[-0.467, -0.321]	0.468	0.997	0.467	no	
PSO-VNS vs. PSO	68	-0.018	[-0.026, -0.010]	0.026	yes		[-0.041, 0.004]	0.041	0.011	yes	[-0.067, 0.031]	0.067	0.105	0.067	no	
PSO-VNS vs. VNS	68	-0.465	[-0.489, -0.442]	0.489	no		[-0.545, -0.386]	0.546	1.000	no	[-0.593, -0.338]	0.593	0.996	0.575	no	
SSA-VNS vs. RSD-VNS	68	+0.024	[0.002, 0.046]	0.047	yes		[-0.022, 0.062]	0.062	0.141	no	[-0.051, 0.099]	0.100	0.263	0.089	no	
LX-SSA-VNS vs. RSD-VNS	68	+0.087	[0.065, 0.110]	0.111	no		[0.047, 0.126]	0.126	0.936	no	[0.004, 0.170]	0.171	0.788	0.165	no	
PSO-VNS vs. RSD-VNS	68	-0.306	[-0.325, -0.287]	0.326	no		[-0.380, -0.239]	0.381	1.000	no	[-0.442, -0.171]	0.442	0.994	0.454	no	
RSD-VNS vs. RS-VNS	68	-0.093	[-0.115, -0.070]	0.115	no		[-0.127, -0.055]	0.128	0.971	no	[-0.131, -0.054]	0.131	0.967	0.132	no	
<i>Main comparison, PSO-VNS vs. each method (PSO and VNS: see above)</i>																
PSO-VNS vs. SSA-VNS	68	-0.330	[-0.350, -0.310]	0.351	no		[-0.397, -0.266]	0.397	1.000	no	[-0.434, -0.227]	0.434	0.997	0.430	no	
PSO-VNS vs. SSA	66	-0.611	[-0.633, -0.589]	0.633	no		[-0.735, -0.493]	0.735	1.000	no	[-0.770, -0.452]	0.771	0.996	0.753	no	
PSO-VNS vs. LX-SSA	66	-0.765	[-0.791, -0.739]	0.792	no		[-0.930, -0.606]	0.930	1.000	no	[-0.946, -0.583]	0.946	0.997	0.921	no	
PSO-VNS vs. DE	50	-0.887	[-0.916, -0.858]	0.917	no		[-1.137, -0.648]	1.137	1.000	no	[-1.173, -0.600]	1.174	0.994	1.169	no	
PSO-VNS vs. MS-SLSQP	68	-0.435	[-0.452, -0.419]	0.452	no		[-0.505, -0.366]	0.505	1.000	no	[-0.579, -0.291]	0.580	0.994	0.580	no	

Table S27: Bayesian signed-rank test [15] on the case-mean differences (first minus second; region of practical equivalence $(-0.05, 0.05)$ pp; prior strength $s = 0.5$ at $z_0 = 0$; 50,000 posterior samples, fixed seed; as Table S15). $\theta_A, \theta_{\text{rope}}, \theta_B$: posterior shares of the Walsh averages $(d_i + d_j)/2$ below $-m$ (first better), inside and above $+m$ (second better) – the chance that a new pair of cases favours the first method by more than m , is within $\pm m$, or favours the second. $P_A, P_{\text{rope}}, P_B$: posterior probability that each region is the *most probable* of the three (the quantity of Table S15); this is not the probability that the mean difference lies within $\pm m$. Posterior means of θ with 95% credible intervals.

Pair (first vs. second)	n	P_A	P_{rope}	P_B	θ_A	θ_{rope}	θ_B			
SSA-VNS vs. RS-VNS	68	0.30	0.70	<0.01	0.42	[0.29, 0.57]	0.49	[0.37, 0.62]	0.08	[0.03, 0.16]
LX-SSA-VNS vs. RS-VNS	68	<0.01	>0.99	<0.01	0.23	[0.12, 0.36]	0.55	[0.43, 0.67]	0.22	[0.12, 0.35]
PSO-VNS vs. RS-VNS	68	>0.99	<0.01	<0.01	0.88	[0.79, 0.95]	0.12	[0.05, 0.21]	0.00	[0.00, 0.00]
SSA-VNS vs. LX-SSA-VNS	68	0.53	0.47	<0.01	0.49	[0.35, 0.63]	0.48	[0.35, 0.61]	0.03	[0.01, 0.08]
LX-SSA vs. SSA	66	<0.01	0.12	0.88	0.02	[0.00, 0.07]	0.40	[0.28, 0.54]	0.57	[0.43, 0.71]
SSA-VNS vs. SSA	66	>0.99	<0.01	<0.01	0.83	[0.72, 0.91]	0.16	[0.09, 0.26]	0.01	[0.00, 0.05]
LX-SSA-VNS vs. LX-SSA	66	>0.99	<0.01	<0.01	0.83	[0.73, 0.91]	0.17	[0.09, 0.27]	0.00	[0.00, 0.00]
PSO-VNS vs. PSO	68	<0.01	>0.99	<0.01	0.25	[0.14, 0.39]	0.62	[0.49, 0.75]	0.12	[0.05, 0.23]
PSO-VNS vs. VNS	68	>0.99	<0.01	<0.01	0.90	[0.82, 0.96]	0.10	[0.04, 0.17]	0.01	[0.00, 0.02]
SSA-VNS vs. RSD-VNS	68	<0.01	0.77	0.23	0.13	[0.05, 0.25]	0.48	[0.36, 0.61]	0.39	[0.25, 0.53]
LX-SSA-VNS vs. RSD-VNS	68	<0.01	0.10	0.90	0.05	[0.01, 0.13]	0.39	[0.27, 0.52]	0.56	[0.42, 0.69]
PSO-VNS vs. RSD-VNS	68	>0.99	<0.01	<0.01	0.86	[0.76, 0.93]	0.14	[0.07, 0.24]	0.00	[0.00, 0.00]
RSD-VNS vs. RS-VNS	68	0.95	0.05	<0.01	0.60	[0.46, 0.73]	0.37	[0.25, 0.51]	0.03	[0.00, 0.10]
PSO-VNS vs. SSA-VNS	68	>0.99	<0.01	<0.01	0.87	[0.78, 0.94]	0.13	[0.06, 0.22]	0.00	[0.00, 0.00]
PSO-VNS vs. SSA	66	>0.99	<0.01	<0.01	0.91	[0.84, 0.96]	0.09	[0.04, 0.16]	0.00	[0.00, 0.00]
PSO-VNS vs. LX-SSA	66	>0.99	<0.01	<0.01	0.92	[0.85, 0.97]	0.08	[0.03, 0.15]	0.00	[0.00, 0.00]
PSO-VNS vs. DE	50	>0.99	<0.01	<0.01	0.89	[0.78, 0.95]	0.11	[0.05, 0.22]	0.00	[0.00, 0.00]
PSO-VNS vs. MS-SLSQP	68	>0.99	<0.01	<0.01	0.93	[0.86, 0.97]	0.06	[0.02, 0.11]	0.01	[0.00, 0.05]
<i>PSO-VNS vs. PSO, subsets of the cases</i>										
$N < 10$	48	<0.01	>0.99	<0.01	0.06	[0.01, 0.17]	0.85	[0.71, 0.95]	0.08	[0.02, 0.21]
$N \geq 10$	20	0.93	0.05	0.02	0.57	[0.32, 0.80]	0.28	[0.15, 0.44]	0.15	[0.03, 0.34]
wake loss of PSO-VNS < 0.2 pp	24	<0.01	>0.99	<0.01	0.00	[0.00, 0.00]	1.00	[1.00, 1.00]	0.00	[0.00, 0.00]
wake loss of PSO-VNS ≥ 0.2 pp	44	0.14	0.85	<0.01	0.34	[0.18, 0.51]	0.49	[0.35, 0.64]	0.17	[0.07, 0.32]

Table S28: Heterogeneity of PSO-VNS vs. PSO (ΔL : case-mean wake loss of PSO-VNS minus that of PSO, pp; negative = PSO-VNS better). Beyond: cases with $\Delta L < -m$ / $\Delta L > +m$ ($m = 0.05$ pp); 90% CI, $m_{\min}$, Eq. as the case level of Table S26; Bayes: $P_A/P_{\text{rope}}/P_B$ (region most probable; A = PSO-VNS better) of the Bayesian signed-rank test (Table S27). Subsets: turbine number $N < 10$ / $N \geq 10$, and cases in which the case-mean wake loss of PSO-VNS is below / at least 0.2 pp (a case with a trivial wake loss cannot differ by m). Density: $\phi = N(\ell_{\min}/2r)^2$ with $\ell_{\min} = 308$ m, the nominal share of the farm disc covered by the N exclusion discs (range 0.05–0.95); stated after review, before this test was run. Interaction test: slope of the difference between the data sets over the identical (r, N) cases, HC3-studentized, 20,000 random sign flips of the pairs (valid when the data sets also differ in level); per data set: regression on ϕ with radius fixed effects, HC3-studentized, permutation of ΔL within radius. The cases beyond $\pm m$ are listed in Table S29.

Cases	n	Beyond	$\overline{\Delta L}$	90% CI	$m_{\min}$	Eq.	Bayes
All cases	68	11/8	−0.018	[−0.041, 0.004]	0.041	yes	<0.01 / >0.99 / <0.01
$N < 10$	48	2/4	+0.007	[−0.005, 0.021]	0.022	yes	<0.01 / >0.99 / <0.01
$N \geq 10$	20	9/4	−0.079	[−0.142, −0.017]	0.142	no	0.93 / 0.05 / 0.02
loss < 0.2 pp	24	0/0	−0.004	[−0.007, −0.002]	0.007	yes	<0.01 / >0.99 / <0.01
loss $\geq$ 0.2 pp	44	11/8	−0.026	[−0.060, 0.008]	0.060	no	0.14 / 0.85 / <0.01
Cluster means (data set/radius in m), pp: I/500 +0.064, I/750 +0.043, I/1000 −0.008, II/500 −0.063, II/750 −0.031, II/1000 −0.090; beyond $\pm m$: 2 favour PSO-VNS, 1 favour PSO							
Data set $\times$ density $\phi = N(\ell_{\min}/2r)^2$ (slope, pp per unit ϕ ; studentized permutation p)							
Interaction, $\Delta L_{II} - \Delta L_I$ on ϕ over the 34 identical (r, N) cases: slope −0.496, $p = 0.010$ (sign flips)							
Data Set I: slope +0.291 (PSO better at higher density), $p = 0.001$; Data Set II: slope −0.524 (PSO-VNS better), $p = 1.0 \times 10^{-4}$ (within-radius permutation)							

Table S29: The 19 of 68 cases in which PSO-VNS and PSO differ by more than the margin ($|\Delta L| > 0.05$ pp; $\Delta L = \text{PSO-VNS} - \text{PSO}$, pp): 11 favour PSO-VNS (up to 0.41 pp) and 8 favour PSO (up to 0.27 pp). $\phi = N(\ell_{\min}/2r)^2$.

Data set	r (m)	N	ϕ	ΔL	Better
I	500	8	0.76	+0.160	PSO
I	500	9	0.85	+0.274	PSO
I	500	10	0.95	+0.168	PSO
I	750	9	0.38	+0.068	PSO
I	750	10	0.42	+0.064	PSO
I	750	11	0.46	+0.183	PSO
I	750	12	0.51	+0.178	PSO
II	500	4	0.38	+0.062	PSO
II	500	8	0.76	−0.102	PSO-VNS
II	500	9	0.85	−0.168	PSO-VNS
II	500	10	0.95	−0.412	PSO-VNS
II	750	10	0.42	−0.101	PSO-VNS
II	750	11	0.46	−0.134	PSO-VNS
II	750	12	0.51	−0.108	PSO-VNS
II	1000	11	0.26	−0.192	PSO-VNS
II	1000	12	0.28	−0.211	PSO-VNS
II	1000	13	0.31	−0.230	PSO-VNS
II	1000	14	0.33	−0.326	PSO-VNS
II	1000	15	0.36	−0.382	PSO-VNS

Table S30: Estimates that match the tests. Mean: mean case-mean difference (first minus second, pp) with the 95% percentile bootstrap CI over the cases (as in the paper). HL: Hodges–Lehmann estimate (median of the $n(n+1)/2$ Walsh averages of the case-mean differences), the location estimate that belongs to the Wilcoxon signed-rank test, with the 95% CI from the exact inversion of that test (exact for untied differences; zero differences are kept). W/L and p_W : cases in which the first/second method has the lower case mean and the case-mean Wilcoxon p of the paper (zero differences dropped; imputation when only one method qualifies, so n of the test can exceed the n of the estimates). Bold: the pairs discussed in the text.

Pair (first vs. second)	n	Mean	95% CI	HL	95% CI	W/L	p_W
PSO-VNS vs. PSO	68	-0.018	[-0.045, 0.008]	-0.003	[-0.011, 0.003]	34/26	0.296
PSO-VNS vs. SSA-VNS	68	-0.330	[-0.409, -0.255]	-0.316	[-0.393, -0.226]	59/1	1.9×10^{-11}
PSO-VNS vs. SSA	66	-0.611	[-0.762, -0.471]	-0.581	[-0.735, -0.423]	62/0	7.6×10^{-12}
PSO-VNS vs. LX-SSA	66	-0.765	[-0.964, -0.580]	-0.698	[-0.918, -0.475]	62/0	7.6×10^{-12}
PSO-VNS vs. DE	50	-0.887	[-1.191, -0.604]	-0.726	[-1.163, -0.398]	60/0	1.4×10^{-11}
PSO-VNS vs. VNS	68	-0.465	[-0.559, -0.371]	-0.463	[-0.565, -0.354]	55/4	6.3×10^{-11}
PSO-VNS vs. MS-SLSQP	68	-0.435	[-0.519, -0.353]	-0.424	[-0.531, -0.344]	66/2	6.0×10^{-12}
SSA-VNS vs. SSA	66	-0.294	[-0.383, -0.214]	-0.249	[-0.338, -0.147]	60/2	5.7×10^{-11}
LX-SSA-VNS vs. LX-SSA	66	-0.394	[-0.528, -0.276]	-0.264	[-0.417, -0.170]	62/0	7.6×10^{-12}
PSO-VNS vs. RS-VNS	68	-0.399	[-0.495, -0.308]	-0.366	[-0.460, -0.278]	58/1	2.6×10^{-11}
SSA-VNS vs. RS-VNS	68	-0.068	[-0.108, -0.034]	-0.033	[-0.066, -0.010]	42/18	4.5×10^{-4}
LX-SSA-VNS vs. RS-VNS	68	-0.005	[-0.041, 0.026]	+0.003	[-0.017, 0.016]	27/33	0.718
PSO-VNS vs. LX-SSA-VNS	68	-0.393	[-0.487, -0.303]	-0.367	[-0.475, -0.259]	60/0	1.6×10^{-11}
SSA-VNS vs. LX-SSA-VNS	68	-0.063	[-0.091, -0.038]	-0.049	[-0.076, -0.021]	47/13	5.4×10^{-6}
LX-SSA vs. SSA	66	+0.153	[0.096, 0.217]	+0.077	[0.035, 0.179]	9/51	1.9×10^{-7}
SSA-VNS vs. RSD-VNS	68	+0.024	[-0.033, 0.068]	+0.025	[0.007, 0.058]	18/41	0.005
LX-SSA-VNS vs. RSD-VNS	68	+0.087	[0.039, 0.133]	+0.065	[0.037, 0.105]	10/50	1.4×10^{-6}
PSO-VNS vs. RSD-VNS	68	-0.306	[-0.396, -0.227]	-0.255	[-0.332, -0.181]	57/2	3.6×10^{-11}
RSD-VNS vs. RS-VNS	68	-0.093	[-0.134, -0.049]	-0.070	[-0.130, -0.041]	52/7	5.3×10^{-8}

Table S31: Feasibility of the final layouts on the Horns Rev 1 16-turbine block (6,030 evaluations, random starts), PSO-VNS vs. PSO, 30 runs paired by seed. Exact McNemar test on the discordant pairs: $p = 9.8 \times 10^{-4}$ (two-sided). One site and one initialization; on the 68 benchmark cases both methods are always feasible.

	PSO feasible	PSO infeasible
PSO-VNS feasible	19	11
PSO-VNS infeasible	0	0

S-IV Tests Beyond the Benchmark: Details

This section gives the per-case values of the feasibility-preserving initialization and the budget scaling (Section VIII-A), the full results of the Horns Rev 1 block (Section VIII-B) and of IEA37 Case Study 1 (Section VIII-C), and the computational cost (Section VIII-D).

S-IV-A Feasibility-Preserving Initialization and Budget Scaling

Tables S32 and S33 give the per-case values behind Table V for the six largest benchmark cases, together with the Horns Rev 1 block (the averages of Table V are over the six benchmark cases only). In the paired tests of Table S32, runs are scored as in Section S-I-C: an infeasible run ranks below every feasible run, so a positive r_{rb} can come from more feasible runs even when the mean of the feasible runs is not better. With feasible starts, PSO and DE do not move away from the best initial layout (Section VIII-A), so their feasible-start values coincide in every case; Section S-VII-C analyzes why.

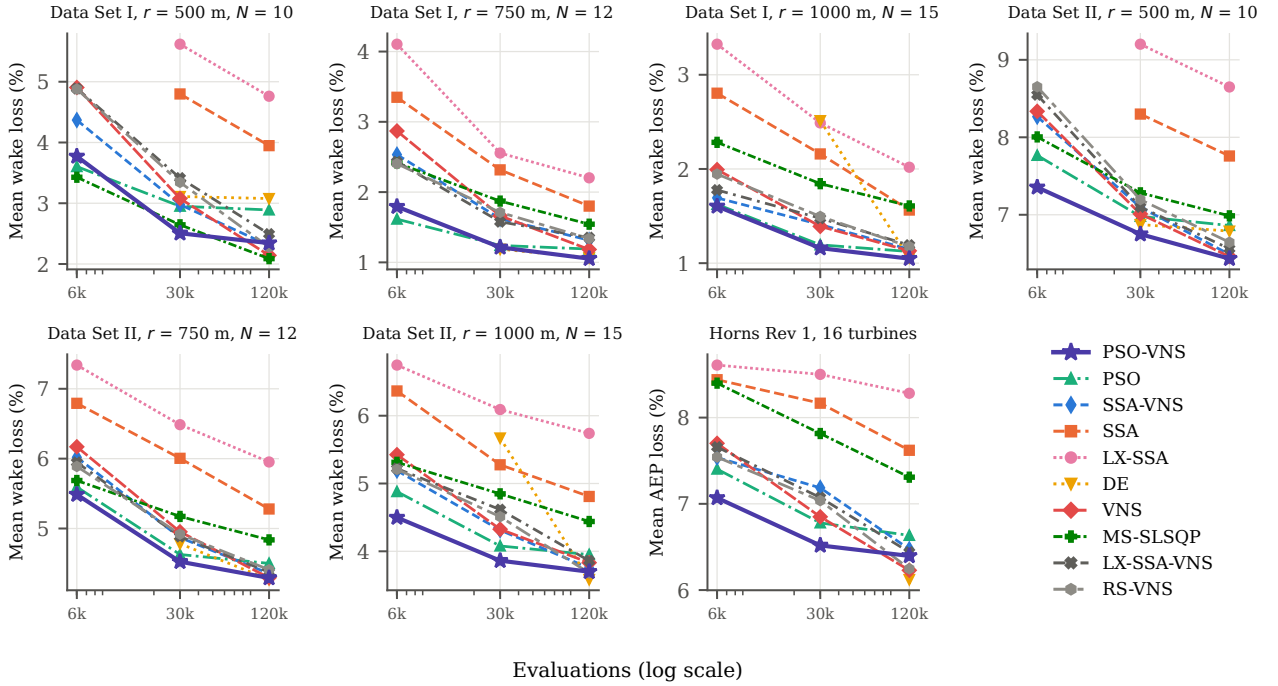


Figure S11: Budget scaling: mean wake loss of the six largest benchmark cases and mean AEP loss of the Horns Rev 1 block (feasible runs; methods with at least half of their runs feasible) versus the budget (6,030, 30,030 and 120,030 evaluations, random initialization; 10 seeds for Horns Rev 1 at 120,030 evaluations).

Table S32: Feasible versus random initialization per case (6,030 evaluations): mean wake loss (%; Horns Rev: mean AEP in GWh/yr) of the feasible runs, feasible runs, Holm-adjusted Wilcoxon signed-rank p (over the methods of the case) and rank-biserial r_{rb} (positive = feasible initialization better).

Method	Random	Feasible	Feas. (R)	Feas. (F)	p_{Holm}	r_{rb}
<i>Data Set I, $r = 500$ m, $N = 10$</i>						
PSO-VNS	3.771	2.871	30/30	30/30	0.008	+0.59
PSO	3.604	3.698	30/30	30/30	0.477	-0.15
SSA-VNS	4.369	2.932	26/30	30/30	2.76×10^{-6}	+0.91
SSA	5.582	3.611	12/30	30/30	9.31×10^{-8}	+0.98
LX-SSA	6.756	3.649	13/30	30/30	2.98×10^{-8}	+1.00
DE	6.737	3.698	1/30	30/30	1.68×10^{-8}	+1.00
VNS	4.905	2.782	30/30	30/30	2.98×10^{-8}	+1.00
MS-SLSQP	3.428	2.894	30/30	30/30	0.001	+0.70
LX-SSA-VNS	4.888	2.860	29/30	30/30	2.98×10^{-8}	+1.00
<i>Data Set I, $r = 750$ m, $N = 12$</i>						
PSO-VNS	1.793	1.909	30/30	30/30	0.339	-0.20
PSO	1.615	3.252	30/30	30/30	1.68×10^{-8}	-1.00
SSA-VNS	2.533	1.947	30/30	30/30	0.004	+0.66
SSA	3.349	2.623	30/30	30/30	0.006	+0.63
LX-SSA	4.105	2.695	28/30	30/30	8.34×10^{-6}	+0.90
DE	–	3.252	0/30	30/30	1.68×10^{-8}	+1.00
VNS	2.869	1.778	30/30	30/30	9.59×10^{-5}	+0.83
MS-SLSQP	2.415	2.169	30/30	30/30	0.110	+0.40
LX-SSA-VNS	2.431	1.911	30/30	30/30	0.001	+0.72
<i>Data Set I, $r = 1000$ m, $N = 15$</i>						
PSO-VNS	1.603	1.586	30/30	30/30	0.598	-0.11
PSO	1.624	3.162	30/30	30/30	1.68×10^{-8}	-1.00
SSA-VNS	1.693	1.561	30/30	30/30	0.328	+0.29
SSA	2.804	2.493	30/30	30/30	0.213	+0.42
LX-SSA	3.325	2.619	29/30	30/30	0.004	+0.68
DE	–	3.162	0/30	30/30	1.68×10^{-8}	+1.00
VNS	1.993	1.541	30/30	30/30	0.001	+0.74
MS-SLSQP	2.283	2.092	30/30	30/30	0.213	+0.42
LX-SSA-VNS	1.777	1.585	30/30	30/30	0.220	+0.38
<i>Data Set II, $r = 500$ m, $N = 10$</i>						
PSO-VNS	7.355	7.411	30/30	30/30	0.761	-0.07
PSO	7.767	9.147	30/30	30/30	1.89×10^{-6}	-0.93
SSA-VNS	8.266	7.401	25/30	30/30	1.49×10^{-4}	+0.79
SSA	9.942	8.831	12/30	30/30	3.26×10^{-7}	+0.97
LX-SSA	9.959	8.924	13/30	30/30	1.04×10^{-7}	+0.98
DE	12.185	9.147	1/30	30/30	1.68×10^{-8}	+1.00
VNS	8.336	7.263	30/30	30/30	2.76×10^{-6}	+0.91
MS-SLSQP	8.005	7.819	29/30	30/30	0.248	+0.32
LX-SSA-VNS	8.548	7.430	29/30	30/30	2.36×10^{-6}	+0.92
<i>Data Set II, $r = 750$ m, $N = 12$</i>						
PSO-VNS	5.486	5.304	30/30	30/30	0.192	+0.35
PSO	5.594	7.395	30/30	30/30	1.68×10^{-8}	-1.00
SSA-VNS	6.019	5.385	30/30	30/30	7.15×10^{-6}	+0.90
SSA	6.792	6.475	30/30	30/30	0.165	+0.40
LX-SSA	7.339	6.624	28/30	30/30	0.003	+0.68
DE	–	7.395	0/30	30/30	1.68×10^{-8}	+1.00
VNS	6.170	5.059	30/30	30/30	1.79×10^{-6}	+0.94
MS-SLSQP	5.687	5.646	30/30	30/30	0.919	-0.02
LX-SSA-VNS	5.946	5.365	30/30	30/30	1.73×10^{-4}	+0.80
<i>Data Set II, $r = 1000$ m, $N = 15$</i>						
PSO-VNS	4.498	4.549	30/30	30/30	0.556	-0.13
PSO	4.880	6.639	30/30	30/30	1.68×10^{-8}	-1.00
SSA-VNS	5.190	4.504	30/30	30/30	4.99×10^{-6}	+0.91
SSA	6.365	5.648	30/30	30/30	9.38×10^{-4}	+0.72
LX-SSA	6.747	5.775	29/30	30/30	1.53×10^{-6}	+0.94
DE	–	6.639	0/30	30/30	1.68×10^{-8}	+1.00
VNS	5.425	4.439	30/30	30/30	1.30×10^{-7}	+0.98
MS-SLSQP	5.313	5.145	30/30	30/30	0.192	+0.35
LX-SSA-VNS	5.220	4.559	30/30	30/30	1.99×10^{-4}	+0.79
<i>Horns Rev 1, 16 turbines</i>						
PSO-VNS	138.29	138.43	30/30	30/30	0.617	+0.22
PSO	137.79	136.39	19/30	30/30	0.617	+0.20
SSA-VNS	137.59	138.42	30/30	30/30	1.46×10^{-4}	+0.82
SSA	136.25	137.16	30/30	30/30	9.27×10^{-4}	+0.73
LX-SSA	136.00	137.07	28/30	30/30	1.66×10^{-4}	+0.82
DE	–	136.39	0/30	30/30	1.68×10^{-8}	+1.00
VNS	137.35	138.53	30/30	30/30	1.66×10^{-4}	+0.82
MS-SLSQP	136.31	136.90	30/30	30/30	0.007	+0.62
LX-SSA-VNS	137.41	138.36	30/30	30/30	1.66×10^{-4}	+0.81

Note: with feasible starts, PSO and DE do not move away from the best (packed) initial layout (Section VIII-A), so their feasible-start values coincide in every case. Infeasible runs enter the paired test ranked below every feasible run (Section S-I-C).

Table S33: Budget scaling per case: mean wake loss (%; Horns Rev: mean AEP in GWh/yr) of the feasible runs and feasible runs at each budget (random initialization).

Method	6,030	Feas.	30,030	Feas.	120,030	Feas.
<i>Data Set I, $r = 500$ m, $N = 10$ (wake loss, %)</i>						
PSO-VNS	3.771	30/30	2.506	30/30	2.343	30/30
PSO	3.604	30/30	2.951	30/30	2.890	30/30
SSA-VNS	4.369	26/30	2.978	30/30	2.305	30/30
SSA	5.582	12/30	4.799	28/30	3.948	30/30
LX-SSA	6.756	13/30	5.618	25/30	4.761	29/30
DE	6.737	1/30	3.111	30/30	3.077	30/30
VNS	4.905	30/30	3.078	30/30	2.142	30/30
MS-SLSQP	3.428	30/30	2.645	30/30	2.088	30/30
LX-SSA-VNS	4.888	29/30	3.416	30/30	2.494	30/30
RS-VNS	4.879	24/30	3.344	30/30	2.278	30/30
<i>Data Set I, $r = 750$ m, $N = 12$ (wake loss, %)</i>						
PSO-VNS	1.793	30/30	1.214	30/30	1.053	30/30
PSO	1.615	30/30	1.241	30/30	1.190	30/30
SSA-VNS	2.533	30/30	1.621	30/30	1.316	30/30
SSA	3.349	30/30	2.316	30/30	1.801	30/30
LX-SSA	4.105	28/30	2.556	30/30	2.203	30/30
DE	–	0/30	1.188	30/30	1.066	30/30
VNS	2.869	30/30	1.663	30/30	1.184	30/30
MS-SLSQP	2.415	30/30	1.872	30/30	1.543	30/30
LX-SSA-VNS	2.431	30/30	1.578	30/30	1.352	30/30
RS-VNS	2.404	30/30	1.711	30/30	1.339	30/30
<i>Data Set I, $r = 1000$ m, $N = 15$ (wake loss, %)</i>						
PSO-VNS	1.603	30/30	1.160	30/30	1.048	30/30
PSO	1.624	30/30	1.195	30/30	1.124	30/30
SSA-VNS	1.693	30/30	1.410	30/30	1.146	30/30
SSA	2.804	30/30	2.160	30/30	1.564	30/30
LX-SSA	3.325	29/30	2.489	30/30	2.018	30/30
DE	–	0/30	2.513	21/30	1.057	30/30
VNS	1.993	30/30	1.390	30/30	1.131	30/30
MS-SLSQP	2.283	30/30	1.844	30/30	1.608	30/30
LX-SSA-VNS	1.777	30/30	1.477	30/30	1.192	30/30
RS-VNS	1.946	30/30	1.496	30/30	1.175	30/30
<i>Data Set II, $r = 500$ m, $N = 10$ (wake loss, %)</i>						
PSO-VNS	7.355	30/30	6.750	30/30	6.434	30/30
PSO	7.767	30/30	6.979	30/30	6.861	30/30
SSA-VNS	8.266	25/30	7.097	30/30	6.488	30/30
SSA	9.942	12/30	8.300	28/30	7.759	30/30
LX-SSA	9.959	13/30	9.202	25/30	8.650	29/30
DE	12.185	1/30	6.874	30/30	6.789	30/30
VNS	8.336	30/30	7.011	30/30	6.461	30/30
MS-SLSQP	8.005	29/30	7.283	30/30	6.986	30/30
LX-SSA-VNS	8.548	29/30	7.101	30/30	6.561	30/30
RS-VNS	8.652	24/30	7.196	30/30	6.645	30/30
<i>Data Set II, $r = 750$ m, $N = 12$ (wake loss, %)</i>						
PSO-VNS	5.486	30/30	4.525	30/30	4.292	30/30
PSO	5.594	30/30	4.628	30/30	4.500	30/30
SSA-VNS	6.019	30/30	4.870	30/30	4.356	30/30
SSA	6.792	30/30	6.005	30/30	5.279	30/30
LX-SSA	7.339	28/30	6.486	30/30	5.952	30/30
DE	–	0/30	4.783	30/30	4.269	30/30
VNS	6.170	30/30	4.953	30/30	4.295	30/30
MS-SLSQP	5.687	30/30	5.173	30/30	4.837	30/30
LX-SSA-VNS	5.946	30/30	4.885	30/30	4.375	30/30
RS-VNS	5.888	30/30	4.919	30/30	4.410	30/30
<i>Data Set II, $r = 1000$ m, $N = 15$ (wake loss, %)</i>						
PSO-VNS	4.498	30/30	3.862	30/30	3.699	30/30
PSO	4.880	30/30	4.079	30/30	3.951	30/30
SSA-VNS	5.190	30/30	4.312	30/30	3.754	30/30
SSA	6.365	30/30	5.276	30/30	4.809	30/30
LX-SSA	6.747	29/30	6.091	30/30	5.740	30/30
DE	–	0/30	5.671	21/30	3.584	30/30
VNS	5.425	30/30	4.326	30/30	3.832	30/30
MS-SLSQP	5.313	30/30	4.848	30/30	4.440	30/30
LX-SSA-VNS	5.220	30/30	4.616	30/30	3.860	30/30
RS-VNS	5.210	30/30	4.510	30/30	3.681	30/30
<i>Horns Rev 1, 16 turbines (AEP, GWh/yr)</i>						
PSO-VNS	138.29	30/30	139.11	30/30	139.29	10/10
PSO	137.79	19/30	138.72	28/30	138.94	10/10
SSA-VNS	137.59	30/30	138.11	30/30	139.17	10/10
SSA	136.25	30/30	136.65	30/30	137.47	10/10
LX-SSA	136.00	28/30	136.16	30/30	136.48	10/10
DE	–	0/30	137.79	8/30	139.70	10/10
VNS	137.35	30/30	138.61	30/30	139.54	10/10
MS-SLSQP	136.31	30/30	137.18	30/30	137.93	10/10
LX-SSA-VNS	137.41	30/30	138.28	30/30	139.22	10/10
RS-VNS	137.58	28/30	138.34	30/30	139.51	10/10

S-IV-B Horns Rev 1 16-Turbine Block

Table S34 gives the full results for the Horns Rev 1 block, and Fig. S12 its convergence and layouts. All values come from the runs with the corrected direction binning (Section S-I-G). The 1° re-evaluation of every Horns Rev 1 run is reported in Section S-V-C.

Check against PyWake. Our model was checked on the installed 80-turbine Horns Rev 1 farm against the Jensen (NOJ) model of PyWake 2.6.20 [9] ($k = 0.04$), with the same turbine curves and wind climate and at identical direction and speed bins (the bin probabilities agree to 7×10^{-18} , and the wake-free AEP is identical, 744.04 GWh/yr; Table S47). PyWake gives 661.39 GWh/yr (wake loss 11.11%) and our model 666.75 GWh/yr (10.39%): our AEP is 0.81% higher, and the difference of the wake losses (ours minus PyWake’s) is -0.72 pp, -6% of PyWake’s wake loss. Rotor averaging does not explain the difference: PyWake’s hub-centre variant (659.16 GWh/yr) lies further from our value than its default. The difference comes from the thrust coefficient, which our model takes at the free-stream speed: with C_T at the local (waked) speed, our model gives 660.24 GWh/yr and reproduces PyWake’s NOJ with hub-centre test and momentum-theory induction exactly; the remaining small gap to PyWake’s default NOJ is its rotor averaging and induction polynomial. Our model therefore slightly underestimates the wake loss, in the same way for all methods. For the installed 16-turbine block, PyWake gives 139.08 GWh/yr and our model 139.82 GWh/yr (0.53% higher).

Table S34: Horns Rev 1 site case, 16-turbine block (Vestas V80 power and thrust curves, measured 12-sector wind climate, installed outline; Jensen wake $k = 0.04$; minimum spacing $4D = 320$ m; wake-free AEP 148.81 GWh/yr): AEP in GWh/yr, mean (SD) over the feasible runs of 30 seed-paired runs at 6,030 evaluations, mean wake loss (%), feasible runs, Wilcoxon signed-rank p of PSO-VNS vs. each method (Holm-adjusted over these 9 comparisons) and rank-biserial r_{rb} (positive = PSO-VNS better; zero differences dropped); run-level Friedman $p = 1.64 \times 10^{-29}$ (over the seeds common to all methods).

Layout / method	AEP	Loss (%)	Feas.	p_{Holm}	r_{rb}
Installed layout	139.82	6.04	–	–	–
PSO-VNS	138.29 (0.75)	7.07	30/30	–	–
SSA-VNS	137.59 (0.72)	7.54	30/30	2.56×10^{-4}	+0.72
LX-SSA-VNS	137.41 (0.85)	7.66	30/30	1.58×10^{-4}	+0.77
RS-VNS	137.58 (0.67)	7.54	28/30	1.08×10^{-4}	+0.81
VNS	137.35 (1.04)	7.70	30/30	1.08×10^{-4}	+0.81
PSO	137.79 (0.88)	7.40	19/30	4.61×10^{-5}	+0.85
SSA	136.25 (1.01)	8.44	30/30	2.12×10^{-7}	+0.97
LX-SSA	136.00 (0.83)	8.61	28/30	6.52×10^{-8}	+0.99
DE	–	–	0/30	1.68×10^{-8}	+1.00
MS-SLSQP	136.31 (0.83)	8.40	30/30	1.68×10^{-8}	+1.00

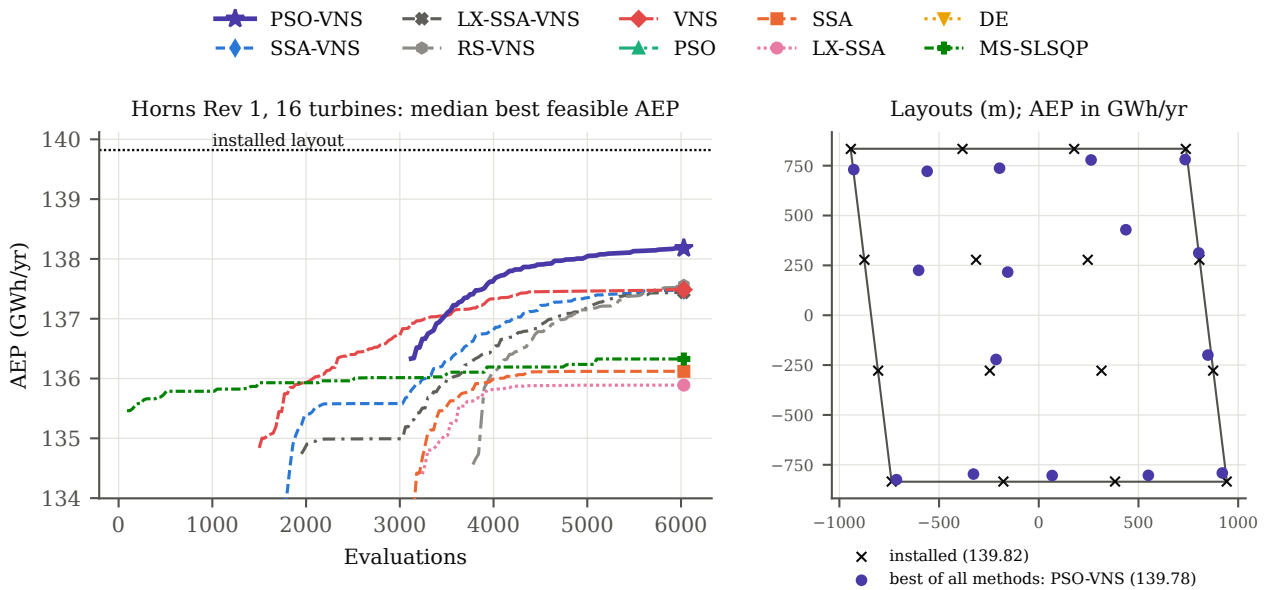


Figure S12: Horns Rev 1 16-turbine block (corrected direction binning): median best feasible AEP versus evaluations (left) and layouts (right).

S-IV-C IEA37 Case Study 1

Table S35 gives the results of all methods at both budgets, Table S36 compares our best layouts with the submitted ones [12, 13] under the published 1-mm feasibility tolerance, Fig. S13 shows the convergence at the largest budget, and Fig. S14 the best layouts of IEA37 Case Study 1 and of the Horns Rev 1 block. Some published layouts lie slightly outside the boundary circle (by a few millimetres, and participant 12 with 16 turbines by 3.518 m); after the outside turbines are projected radially onto the circle, 5 of these layouts become feasible, and the best feasible published layout is that of participant 12 in both scenarios (Table S48). With this projection, the gap of the best PSO-VNS layout at 30,030 evaluations to the best feasible published layout is -1.93% for 16 turbines and -6.23% for 36 turbines (strict 1-mm convention: -1.33% and -4.20%); Table S49 gives these gaps also in wake-loss percentage points.

Table S35: IEA Wind Task 37 Case Study 1 (16- and 36-turbine scenarios; official AEP model): AEP (MWh) mean, best and SD over the feasible runs, mean wake loss relative to the wake-free AEP (%), feasible runs, rank (ranking rule of Table II) and Holm-adjusted Wilcoxon signed-rank p of PSO-VNS vs. each method, per scenario and budget.

Method	Mean	Best	SD	Loss (%)	Feas.	Rank	p_{Holm}
<i>IEA37 CS1, 16 turbines, 6,030 evaluations</i>							
PSO-VNS	395,032	402,782	4,587	15.87	30/30	1	–
PSO	390,540	403,753	7,192	16.82	30/30	2	7.15×10^{-6}
SSA-VNS	388,563	403,251	8,564	17.25	30/30	3	3.80×10^{-4}
VNS	387,637	400,169	5,591	17.44	30/30	4	4.61×10^{-5}
LX-SSA-VNS	386,930	397,429	6,911	17.59	30/30	5	4.61×10^{-5}
RS-VNS	386,206	398,682	5,817	17.75	30/30	6	4.61×10^{-5}
MS-SLSQP	382,605	405,434	20,279	18.51	18/30	7	6.91×10^{-5}
SSA	378,073	395,150	9,076	19.48	30/30	8	6.52×10^{-8}
LX-SSA	373,977	393,285	7,867	20.35	30/30	9	1.68×10^{-8}
DE	338,611	357,391	8,837	27.88	27/30	10	1.68×10^{-8}
<i>IEA37 CS1, 16 turbines, 30,030 evaluations</i>							
PSO-VNS	402,583	413,332	4,678	14.26	30/30	1	–
PSO	401,347	412,702	4,764	14.52	30/30	2	0.006
LX-SSA-VNS	399,197	407,330	4,330	14.98	30/30	3	0.006
SSA-VNS	396,921	406,925	6,267	15.47	30/30	4	4.42×10^{-4}
VNS	396,192	404,076	4,928	15.62	30/30	5	7.99×10^{-5}
RS-VNS	396,163	408,205	4,977	15.63	30/30	6	8.52×10^{-6}
MS-SLSQP	394,567	410,350	15,316	15.97	29/30	7	0.006
SSA	392,876	401,459	4,394	16.33	30/30	8	4.30×10^{-7}
LX-SSA	390,267	398,770	5,052	16.88	30/30	9	1.68×10^{-8}
DE	359,952	397,403	12,352	23.34	30/30	10	2.98×10^{-8}
<i>IEA37 CS1, 36 turbines, 6,030 evaluations</i>							
VNS	741,566	785,791	24,100	29.81	30/30	1	0.404
PSO-VNS	736,832	787,709	19,268	30.25	30/30	2	–
PSO	729,189	787,989	21,150	30.98	27/30	3	8.49×10^{-4}
SSA-VNS	728,870	766,803	22,859	31.01	30/30	4	0.248
SSA	717,258	750,707	19,300	32.11	30/30	5	0.001
LX-SSA-VNS	715,952	757,838	30,626	32.23	29/30	6	0.002
LX-SSA	709,661	746,301	19,845	32.83	30/30	7	4.75×10^{-4}
RS-VNS	709,276	750,170	21,667	32.86	28/30	8	3.60×10^{-5}
MS-SLSQP	733,342	748,571	13,798	30.58	4/30	9	1.49×10^{-7}
DE	–	–	–	–	0/30	10	1.68×10^{-8}
<i>IEA37 CS1, 36 turbines, 30,030 evaluations</i>							
PSO-VNS	807,446	827,385	11,697	23.57	30/30	1	–
PSO	802,252	830,256	13,260	24.06	30/30	2	4.87×10^{-5}
VNS	801,797	824,230	11,876	24.10	30/30	3	0.119
SSA-VNS	791,404	819,931	14,356	25.09	30/30	4	4.18×10^{-4}
RS-VNS	787,892	807,208	9,532	25.42	30/30	5	4.16×10^{-6}
LX-SSA-VNS	785,677	810,989	13,216	25.63	30/30	6	1.12×10^{-7}
SSA	771,298	802,538	14,756	26.99	30/30	7	2.98×10^{-8}
LX-SSA	752,030	789,311	18,838	28.82	30/30	8	2.98×10^{-8}
MS-SLSQP	798,913	833,365	41,511	24.38	14/30	9	2.37×10^{-4}
DE	–	–	–	–	0/30	10	1.68×10^{-8}

Table S36: IEA Wind Task 37 Case Study 1: AEP (MWh, official calculator) of the baseline (example) layout, of the best feasible and the best overall participant layout, our best feasible layout over all methods and budgets, the best PSO-VNS layout, mean PSO-VNS AEP (budgets 6,030 / 30,030 evaluations), and difference (%) of the best PSO-VNS layout from the best feasible participant layout and from the baseline. “infeas.”: violates the boundary or spacing constraint by more than 1 mm.

Scenario	Baseline	Best feasible publ.	Best publ.	Our best	PSO-VNS best	PSO-VNS mean	Δ_{publ}	Δ_{base}
IEA37 CS1, 16 turbines	366,941.6	418,924.4 (par4)	421,561.9 (par12, infeas.)	413,331.9 (PSO-VNS)	413,331.9	395,031.8 / 402,583.1	-1.33	+12.64
IEA37 CS1, 36 turbines	737,883.1	863,676.3 (par4)	882,383.3 (par12, infeas.)	833,365.0 (MS-SLSQP)	827,385.0	736,832.4 / 807,446.0	-4.20	+12.13

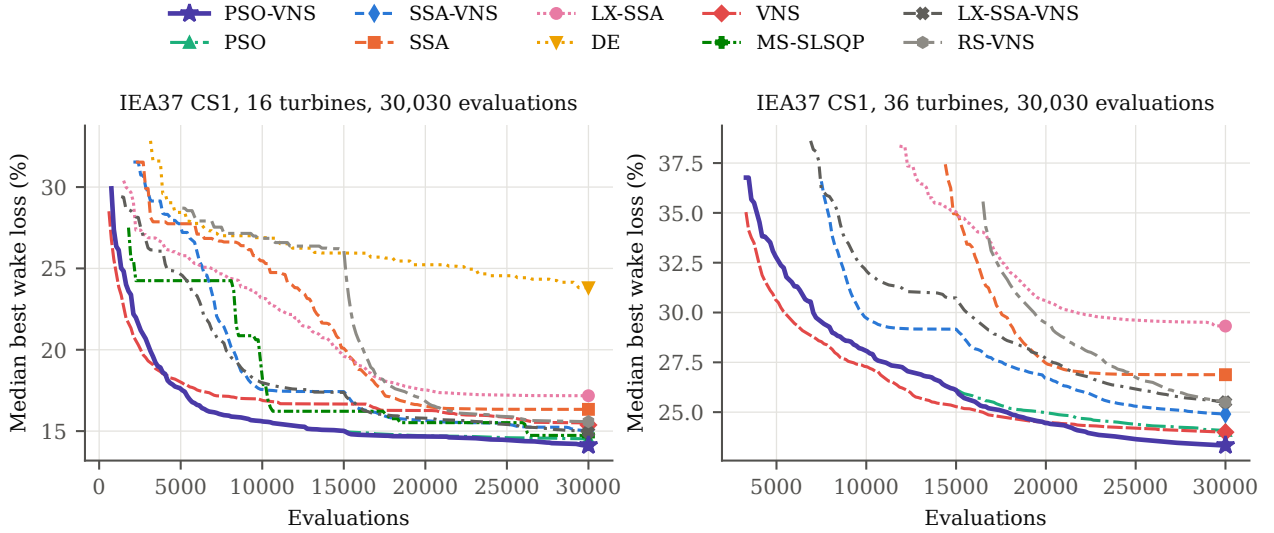


Figure S13: IEA37 Case Study 1: median best feasible wake loss versus evaluations at the largest budget.

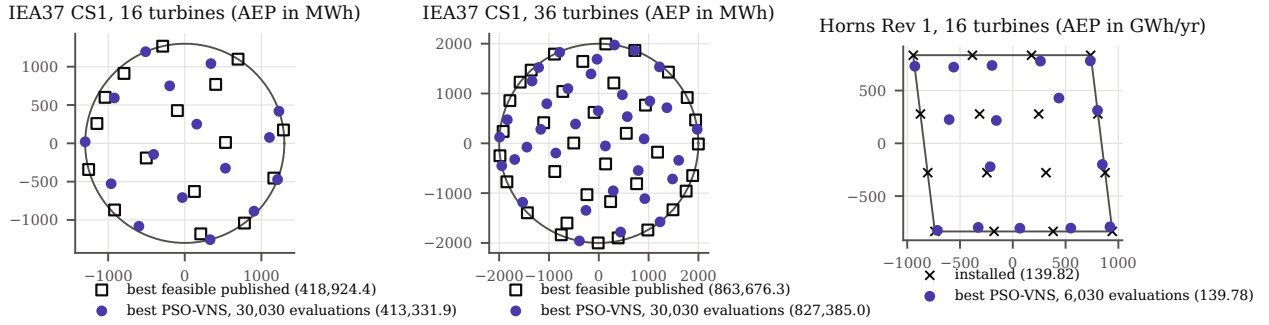


Figure S14: Best layouts. Left and middle: IEA37 Case Study 1, 16 and 36 turbines, best feasible published layout and best PSO-VNS layout at 30,030 evaluations. Right: Horns Rev 1 16-turbine block (corrected direction binning), installed layout (7D grid) and best PSO-VNS layout at 6,030 evaluations (minimum spacing 4D).

S-IV-D Computational Cost

Table S37 gives the time per evaluation and per run. The metaheuristics need 0.23–0.45 ms per evaluation and MS-SLSQP 0.53 ms, about 2.3 times as much as PSO. These timings are indicative only: they come from the machines that produced the respective result files, so small differences between methods are not meaningful.

Table S37: Computational cost at 6,030 evaluations: mean wall-clock time per evaluation (including the optimizer overhead) and range over N of the mean time per run (single core). Times come from the machine that produced each result file.

Method	ms per evaluation	s per run
PSO-VNS	0.349	1.5–3.2
PSO	0.244	1.0–2.2
SSA-VNS	0.388	1.7–3.4
SSA	0.399	1.7–3.6
LX-SSA	0.396	1.7–3.6
DE	0.467	2.0–4.3
VNS	0.371	1.6–3.4
MS-SLSQP	1.820	2.6–21.7

S-V Sensitivity to the Modeling Assumptions: Details

This section defines the Gaussian wake model of the re-evaluation in Section IX and gives the full re-evaluation results and the details of the spacing and boundary checks of Section IX.

Table S38: Robustness to the Benchmark Model: Final Layouts Re-Evaluated, Not Re-Optimized

Model	Δ (%)		Rank	Best	$\bar{\tau}$	Same
	DS I	DS II				
Benchmark	+0.0	+0.0	1.82	PSO-VNS	—	—
Cubic curve	-17.7	-36.9	1.82	PSO-VNS	0.95	94
Cubic, 25 m/s cut-out	-21.5	-37.0	1.80	PSO-VNS	0.95	93
Gaussian wake	-0.3	-0.4	1.97	PSO	0.66	54

All feasible final layouts of the 68 cases. Δ : mean relative change of the objective (%); Rank: average rank of PSO-VNS; Best: best-ranked method; $\bar{\tau}$: mean Kendall correlation between the benchmark and the alternative ordering within a case; Same: cases (%) with unchanged best method. Ranks of all methods: supplementary material.

S-V-A Gaussian Wake Model Used for the Re-Evaluation

For the re-evaluation of Section IX, the deficit of the Jensen model is replaced by the Gaussian wake model of Bastankhah and Porté-Agel [10],

$$\Delta_{ij} = \left(1 - \sqrt{1 - C_T/(8\sigma^2/D^2)}\right) \exp(-v_{ij}^2/(2\sigma^2)), \quad \sigma/D = K_G d_{ij}/D + \epsilon, \quad (S2)$$

$$\epsilon = 0.2\sqrt{\frac{1}{2}(1 + \sqrt{1 - C_T})/\sqrt{1 - C_T}},$$

for turbines i downstream of j , with $K_G = 0.04$, the lateral offset v_{ij} and downstream distance d_{ij} of Section II, and the same root-sum-square superposition and Weibull scaling as the benchmark. The Gaussian model differs from the Jensen cone both in its smooth lateral profile and in its expansion rate ($K_G = 0.04$ vs. $K = 0.075$, not calibrated to each other), and, unlike the cone test of the benchmark, it gives no deficit to turbines upstream of j . It is not valid in the near wake (below about 2–3 rotor diameters), which does not matter at $\ell_{\min} = 4D$.

S-V-B Re-Evaluation, Spacing and Boundary Handling

Table S39 gives the full re-evaluation of the final layouts under the alternative power-curve and wake models, and Table S40 gives the constructive capacity of the three farms at $4D$, $5D$ and $6D$ spacing. Table S41 is an archived check from the previous version of this study: six cases were re-optimized at $5D$ and $6D$ with the original SSA and LX-SSA code only, and PSO-VNS and PSO were not re-run at larger spacing. It shows how the attainable benchmark objective changes when the spacing is enlarged; it is not a comparison of the methods of this paper. Table S38 (moved here from the main text) summarizes the re-evaluation: PSO-VNS keeps the best average rank under the cubic power curve, with and without a 25-m/s cut-out, whereas under the Gaussian wake with the benchmark’s 15° bins PSO is narrowly first; with 1° bins, PSO-VNS is first under the Gaussian wake as well (Section S-V-C).

Table S39: Robustness of the results to the benchmark model. All feasible final layouts of the 8 methods are re-evaluated (not re-optimized) with alternative power-curve and wake models. Δ : mean relative change of the objective (expected power) with respect to the benchmark model (both evaluated on the stored coordinates, rounded to 1 mm); average rank of each method over the 68 cases (ranking rule of Table II: methods with fewer than half of their runs feasible are ranked last, by their number of feasible runs); $\bar{\tau}$: mean Kendall rank correlation between the benchmark and the alternative ordering of the methods with at least 15 feasible runs within a case; “Same best”: percentage of cases in which the best method is unchanged.

Model	Δ (%)		Average rank								$\bar{\tau}$	Same best (%)
	DS I	DS II	PSO-VNS	PSO	SSA-VNS	SSA	LX-SSA	DE	VNS	MS-SLSQP		
Benchmark (linear curve, Jensen)	+0.0	+0.0	1.82	1.97	3.45	5.38	6.06	7.04	4.40	5.88	–	–
Cubic power curve	-17.7	-36.9	1.82	1.90	3.53	5.68	5.97	6.90	4.42	5.79	0.95	94
Cubic curve with 25 m/s cut-out	-21.5	-37.0	1.80	1.91	3.53	5.68	5.97	6.90	4.42	5.79	0.95	93
Gaussian wake ($k^* = 0.04$)	-0.3	-0.4	1.97	1.91	3.93	5.43	6.04	6.94	5.12	4.66	0.66	54

Table S40: Largest turbine count for which a layout satisfying the circular boundary and the minimum spacing was constructed (multi-start packing search; constructive lower bound), compared with the largest tested N .

r (m)	$4D$ (308 m)	$5D$ (385 m)	$6D$ (462 m)	Largest tested N
500	13	8	7	10
750	26	19	13	12
1000	42	29	21	15

Table S41: Minimum-spacing re-optimization with the original LX-SSA and SSA code (6,030 evaluations, 30 seed-paired runs per cell): mean benchmark objective, Wilcoxon signed-rank p for LX-SSA vs. SSA, and the mean LX-SSA wake loss as a percentage of the ideal objective at $4D$ and $6D$. A superscript gives the number of feasible runs when fewer than 30.

DS	r (m)	N	$4D$ (308 m)			$5D$ (385 m)			$6D$ (462 m)			LX-SSA loss (%)
			LX-SSA	SSA	p	LX-SSA	SSA	p	LX-SSA	SSA	p	
1	500	4	56078.8	56086.8	0.237	56082.4	56085.2	0.700	56075.5	56083.9	0.839	0.19 $\rightarrow$ 0.19
1	750	8	111374.0	111506.5	0.245	111154.1	111163.4	0.871	110314.4 ²⁸	110847.5	0.021	0.88 $\rightarrow$ 1.85
1	1000	8	111924.6	111947.1	0.887	111953.5	111946.4	0.777	111781.5	111964.7	0.005	0.39 $\rightarrow$ 0.52
2	500	4	29004.2	29028.7	0.146	28999.5	29019.6	0.229	28981.8	28971.1	0.612	0.88 $\rightarrow$ 0.96
2	750	8	56659.2	56806.2	0.061	56621.0	56673.3	0.584	56533.9 ²⁸	56690.6	0.038	3.18 $\rightarrow$ 3.47
2	1000	8	57645.1	57739.3	0.084	57580.7	57655.7	0.140	57475.5	57587.2	0.114	1.50 $\rightarrow$ 1.79

S-V-C Direction Resolution

The benchmark evaluates the wake only at the centres of 24 direction bins of 15° , and the Horns Rev 1 model at the centres of 5° bins. Between neighbouring bin centres a top-hat wake leaves direction gaps in which a turbine is not waked, and optimized layouts can exploit them. All feasible final layouts were therefore re-evaluated, not re-optimized, with finer bins (`analysis/mpce_direction.py`). On the benchmark (21,716 layouts of the 68 cases), every 15° bin is split into 3 (5°) or 15 (1°) sub-bins that keep its Weibull parameters and share its frequency, so the wake-free power is unchanged (Table S42). With 1° sub-bins, the mean wake loss of every method rises, by 0.40 to 1.04 pp; for PSO-VNS from 1.322% to 2.335%, about 20 times the equivalence margin. The optimized layouts thus exploit the gaps between the 15° bins, the VNS-based methods and PSO most. PSO-VNS has the best average rank with 15° , 5° and 1° bins (at 1° : 2.09 against 2.37 for PSO), also under the Gaussian wake at 1° (2.22 against 2.35); PSO is narrowly first only under the Gaussian wake at 15° (Table S38), a property of the coarse rose. The order beyond the top two changes: MS-SLSQP moves from sixth to third (average rank 5.88 to 4.03), and the within-case orderings agree only moderately with those of the benchmark (mean Kendall $\bar{\tau} = 0.52$; same best method in 37% of the cases). For PSO-VNS vs. PSO (Table S43), the 15° result is case-level equivalence at ± 0.05 pp. With 1° sub-bins, the mean difference is -0.043 pp (90% CI $[-0.062, -0.025]$ pp): PSO-VNS is slightly but significantly better (case-mean Wilcoxon $p = 2.5 \times 10^{-4}$), and equivalence at ± 0.05 pp is no longer shown (minimal margin 0.062 pp). The sign of the per-case difference changes in 26 of the 60 cases with a nonzero difference; on the 44 non-trivial cases, equivalence is shown at neither resolution; and the Data Set II advantage for $N \geq 10$ survives (lower mean wake loss in 10 of 10 cases, -0.160 pp). The component-analysis conclusions hold with 1° sub-bins (Table S44): SSA-VNS and LX-SSA-VNS have a higher case-mean wake loss than the disc-sampling control RSD-VNS ($p = 1.7 \times 10^{-4}$ and 3.2×10^{-6}), and PSO-VNS beats every control and keeps the best average rank of the nine variants; the small gain of SSA-VNS over the square-sampling control RS-VNS is no longer significant ($p = 0.057$).

Horns Rev 1 (Tables S45 and S46). Every run of the Horns Rev 1 study was re-evaluated with 1° bins and with PyWake’s NOJ model at the same bins. With 1° bins, no feasible run of any method at any budget exceeds the installed block (0 of 901, against 16 with the 5° bins of the paper), and under PyWake only 1 run does, a DE run at 120,030 evaluations. The installed block loses 0.32 GWh/yr under the finer rose, the optimized layouts 1.08 GWh/yr on average, the VNS-based methods most. At 6,030 evaluations with random starts, the AEP advantage of PSO-VNS over PSO on the 19 jointly feasible seeds holds only with the 5° bins ($+0.54$ GWh/yr, $p = 0.016$); with 1° bins (-0.05 GWh/yr, $p = 0.65$) and under PyWake (-0.12 GWh/yr, $p = 0.37$) PSO has the nonsignificantly higher mean. The robust result at 6,030 evaluations is feasibility (30/30 vs. 19/30 feasible runs). At 30,030 evaluations, PSO-VNS keeps the highest mean AEP with 1° bins (137.71 GWh/yr) and under PyWake (137.85 GWh/yr), and its advantage over PSO on the 28 jointly feasible seeds is significant under both ($p = 6.5 \times 10^{-4}$ and 3.8×10^{-4}). The results of this paper are therefore results for the discretized wind roses of the benchmark and of Horns Rev 1; the 1° re-evaluation is a robustness check of the final layouts, not a re-optimization with a finer rose.

Table S42: Direction resolution of the benchmark: all feasible final layouts of the 68 cases (6,030 evaluations, random initialization, 30 runs) re-evaluated, not re-optimized, with every 15° direction bin split into 3 (5°) or 15 (1°) sub-bins. Each sub-bin keeps the Weibull parameters of its bin and receives the bin frequency divided by the number of sub-bins (piecewise-constant wind rose; wake-free power unchanged). Mean wake loss (%) over the cases in which the method qualifies (at least 15 feasible runs); average rank over the 68 cases (ranking rule of Table II) for the Jensen benchmark model and the Gaussian wake ($k^* = 0.04$) at 15° and 1° ; ΔL : mean over the cases of the PSO-VNS minus method case-mean wake loss (pp; negative = PSO-VNS better) and Holm-adjusted case-mean Wilcoxon p . $\bar{\tau}$: mean within-case Kendall correlation with the benchmark ordering; same best: cases with unchanged best method.

Method	Mean wake loss (%)			Average rank				ΔL , 15°		ΔL , 1°	
	15°	5°	1°	Jensen 15°	Jensen 1°	Gauss 15°	Gauss 1°	ΔL	p_{Holm}	ΔL	p_{Holm}
PSO-VNS	1.322	2.126	2.335	1.82	2.09	1.97	2.22	–	–	–	–
PSO	1.341	2.179	2.378	1.97	2.37	1.91	2.35	–0.018	0.296	–0.043	2.5×10^{-4}
SSA-VNS	1.653	2.436	2.646	3.45	4.24	3.93	4.35	–0.330	5.7×10^{-11}	–0.311	9.1×10^{-11}
SSA	1.805	2.489	2.650	5.38	5.31	5.43	5.24	–0.611	4.5×10^{-11}	–0.468	5.3×10^{-10}
LX-SSA	1.959	2.628	2.784	6.06	5.81	6.04	5.74	–0.765	4.5×10^{-11}	–0.603	5.3×10^{-10}
DE	1.493	1.815	1.888	7.04	6.51	6.94	6.50	–0.887	5.6×10^{-11}	–0.482	5.9×10^{-10}
VNS	1.788	2.589	2.803	4.40	5.65	5.12	5.72	–0.465	1.3×10^{-10}	–0.468	5.0×10^{-11}
MS-SLSQP	1.758	2.330	2.470	5.88	4.03	4.66	3.88	–0.435	4.2×10^{-11}	–0.135	5.9×10^{-7}
Best-ranked $\bar{\tau}$ / same best (%)				PSO-VNS –	PSO-VNS 0.52 / 37	PSO 0.66 / 54	PSO-VNS 0.50 / 37				

Table S43: PSO-VNS vs. PSO under the benchmark direction bins and finer sub-bins (re-evaluation): mean over the cases of the per-case difference of the mean wake losses (PSO-VNS minus PSO, pp; negative = PSO-VNS better), 90% percentile case-bootstrap CI (10,000 resamples), smallest equivalence margin, equivalence at ± 0.05 pp (90% CI inside the margin; TOST at $\alpha = 0.05$), case-mean Wilcoxon p , cases with lower/equal/higher PSO-VNS loss, cases with $|\Delta L| > 0.05$ pp, and cases whose sign differs from the benchmark (zero differences not counted). Non-trivial: the 44 cases with a PSO-VNS wake loss of at least 0.2% in the benchmark model. ^b95% CI.

Direction bins	Cases	ΔL (pp)	90% CI	Min. margin	Equiv.	p	W/T/L	>margin	Sign changes
15° (benchmark)	all (68)	–0.018	[–0.041, 0.004]	0.041	yes	0.296	34/8/26	19	–
15° (benchmark)	non-trivial (44)	–0.026	[–0.060, 0.008]	0.060	no	0.623			
15° (benchmark)	DS II, $N \geq 10$ (10)	–0.210	[–0.288, –0.135] ^b			0.002	10/0/0		
15° (re-evaluated)	all (68)	–0.018	[–0.041, 0.004]	0.041	yes	0.296	34/8/26	19	0
15° (re-evaluated)	non-trivial (44)	–0.025	[–0.060, 0.008]	0.060	no	0.623			
15° (re-evaluated)	DS II, $N \geq 10$ (10)	–0.209	[–0.288, –0.134] ^b			0.002	10/0/0		
5°	all (68)	–0.054	[–0.074, –0.035]	0.074	no	2.0×10^{-5}	44/8/16	26	24
5°	non-trivial (44)	–0.087	[–0.114, –0.062]	0.114	no	5.0×10^{-8}			
5°	DS II, $N \geq 10$ (10)	–0.189	[–0.255, –0.130] ^b			0.002	10/0/0		
1°	all (68)	–0.043	[–0.062, –0.025]	0.062	no	2.5×10^{-4}	42/8/18	32	26
1°	non-trivial (44)	–0.073	[–0.098, –0.050]	0.098	no	1.2×10^{-6}			
1°	DS II, $N \geq 10$ (10)	–0.160	[–0.227, –0.107] ^b			0.002	10/0/0		

Table S44: Component analysis under 15° (benchmark) and 1° direction bins (re-evaluation of the same final layouts). Top: mean wake loss (%), qualified cases) and average rank among the nine variants. Bottom: case-mean difference of the wake losses (first minus second method, pp) and unadjusted case-mean Wilcoxon p .

Variant / contrast	Loss (%)		Avg. rank	
	15°	1°	15°	1°
PSO-VNS	1.322	2.335	1.96	1.97
SSA-VNS	1.653	2.646	4.46	4.91
LX-SSA-VNS	1.716	2.707	5.41	5.84
RS-VNS	1.721	2.690	5.29	5.62
RSD-VNS	1.629	2.572	3.77	4.28
VNS	1.788	2.803	6.12	6.75
PSO	1.341	2.378	2.24	2.19
SSA	1.805	2.650	7.50	6.46
LX-SSA	1.959	2.784	8.25	6.99
PSO-VNS – PSO	−0.018 (0.296)		−0.043 (2.5×10^{-4})	
SSA-VNS – RSD-VNS	+0.024 (0.005)		+0.074 (1.7×10^{-4})	
LX-SSA-VNS – RSD-VNS	+0.087 (1.4×10^{-6})		+0.135 (3.2×10^{-6})	
PSO-VNS – RSD-VNS	−0.306 (3.6×10^{-11})		−0.236 (2.0×10^{-11})	
RSD-VNS – RS-VNS	−0.093 (5.3×10^{-8})		−0.119 (2.7×10^{-7})	
SSA-VNS – RS-VNS	−0.068 (4.5×10^{-4})		−0.044 (0.057)	
LX-SSA-VNS – RS-VNS	−0.005 (0.718)		+0.016 (0.167)	
PSO-VNS – RS-VNS	−0.399 (2.6×10^{-11})		−0.355 (4.1×10^{-12})	
PSO-VNS – VNS	−0.465 (6.3×10^{-11})		−0.468 (7.1×10^{-12})	
SSA-VNS – LX-SSA-VNS	−0.063 (5.4×10^{-6})		−0.061 (2.4×10^{-4})	

Table S45: Horns Rev 1, 16-turbine block: every run of the study re-evaluated (not re-optimized) with 1° direction bins (centres 0.5°, 1.5°, ...; each 30° sector keeps its Weibull A , k and spreads its frequency uniformly over its 30 bins; 1 m/s speed bins unchanged) and with PyWake’s NOJ model ($k = 0.04$) at the same 1° bins. Mean AEP (GWh/yr) of the feasible runs and number of feasible runs whose AEP exceeds that of the installed block under the same model. R/F: random / feasibility-preserving initialization; 6k, 30k, 120k: evaluations (10 seeds at 120k). 5°: the paper’s model re-evaluated on the stored coordinates (rounded to 1 cm; this lowers the AEP of 99 of the 901 feasible runs by more than 0.01 GWh/yr, at most 0.53, so a few 5° means differ slightly from Table VI).

Setting	Method	Feas.	Mean AEP (GWh/yr)			Runs above installed		
			5°	1°	PyWake NOJ 1°	5°	1°	PyWake
Installed		–	139.82	139.50	139.16	–	–	–
6k R	PSO-VNS	30/30	138.29	136.99	137.22	0	0	0
	SSA-VNS	30/30	137.59	136.13	136.37	0	0	0
	LX-SSA-VNS	30/30	137.41	135.98	136.23	0	0	0
	RS-VNS	28/30	137.58	136.25	136.50	0	0	0
	VNS	30/30	137.35	135.93	136.20	0	0	0
	PSO	19/30	137.79	137.04	137.31	0	0	0
	SSA	30/30	136.25	135.38	135.74	0	0	0
	LX-SSA	28/30	135.98	135.25	135.62	0	0	0
	DE	0/30	–	–	–	0	0	0
	MS-SLSQP	30/30	136.31	135.84	136.21	0	0	0
6k F	PSO-VNS	30/30	138.43	137.03	137.25	0	0	0
	SSA-VNS	30/30	138.42	137.03	137.25	0	0	0
	LX-SSA-VNS	30/30	138.36	136.94	137.18	0	0	0
	VNS	30/30	138.52	137.12	137.32	1	0	0
	PSO	30/30	136.39	136.25	136.65	0	0	0
	SSA	30/30	137.15	136.33	136.66	0	0	0
	LX-SSA	30/30	137.04	136.34	136.69	0	0	0
	DE	30/30	136.39	136.25	136.65	0	0	0
	MS-SLSQP	30/30	136.90	136.42	136.77	0	0	0
30k R	PSO-VNS	30/30	139.11	137.71	137.85	3	0	0
	SSA-VNS	30/30	138.11	136.66	136.84	0	0	0
	LX-SSA-VNS	30/30	138.28	136.88	137.03	0	0	0
	RS-VNS	30/30	138.33	136.88	137.07	0	0	0
	VNS	30/30	138.61	137.19	137.37	0	0	0
	PSO	28/30	138.72	137.48	137.65	1	0	0
	SSA	30/30	136.64	135.49	135.81	0	0	0
	LX-SSA	30/30	136.13	135.16	135.49	0	0	0
	DE	8/30	137.79	137.29	137.55	0	0	0
	MS-SLSQP	30/30	137.18	136.64	136.96	0	0	0
120k R	PSO-VNS	10/10	139.24	137.88	137.96	2	0	0
	SSA-VNS	10/10	139.17	137.79	137.94	0	0	0
	LX-SSA-VNS	10/10	139.22	137.83	137.95	1	0	0
	RS-VNS	10/10	139.51	138.07	138.14	3	0	0
	VNS	10/10	139.54	138.06	138.14	1	0	0
	PSO	10/10	138.60	137.58	137.76	0	0	0
	SSA	10/10	137.32	136.17	136.44	0	0	0
	LX-SSA	10/10	136.40	135.32	135.67	0	0	0
	DE	10/10	139.59	138.23	138.33	4	0	1
	MS-SLSQP	10/10	137.93	137.39	137.66	0	0	0

Table S46: Horns Rev 1, 16 turbines, random initialization: PSO-VNS minus each method, mean AEP difference (GWh/yr; positive = PSO-VNS better) on the seeds on which both runs are feasible, with the unadjusted Wilcoxon signed-rank p , under the paper’s 5° bins, 1° bins and PyWake NOJ (1°). n : jointly feasible seeds.

Budget	vs.	n	5°	1°	PyWake
6,030	SSA-VNS	30	+0.70 (2.6×10^{-4})	+0.86 (7.9×10^{-5})	+0.84 (6.3×10^{-5})
6,030	LX-SSA-VNS	30	+0.88 (6.3×10^{-5})	+1.01 (1.1×10^{-5})	+0.99 (4.4×10^{-6})
6,030	RS-VNS	28	+0.74 (1.2×10^{-4})	+0.76 (3.7×10^{-5})	+0.73 (4.0×10^{-6})
6,030	VNS	30	+0.94 (2.7×10^{-5})	+1.06 (3.8×10^{-6})	+1.02 (3.8×10^{-6})
6,030	PSO	19	+0.54 (0.016)	−0.05 (0.651)	−0.12 (0.374)
6,030	SSA	30	+2.04 (3.5×10^{-8})	+1.60 (1.3×10^{-7})	+1.48 (1.0×10^{-7})
6,030	LX-SSA	28	+2.34 (3.7×10^{-8})	+1.77 (3.7×10^{-8})	+1.64 (5.2×10^{-8})
6,030	DE	0	—	—	—
6,030	MS-SLSQP	30	+1.98 (1.9×10^{-9})	+1.15 (3.1×10^{-7})	+1.01 (1.0×10^{-6})
30,030	SSA-VNS	30	+1.00 (4.7×10^{-7})	+1.05 (3.5×10^{-8})	+1.01 (2.6×10^{-8})
30,030	LX-SSA-VNS	30	+0.83 (1.4×10^{-5})	+0.83 (2.1×10^{-5})	+0.82 (8.0×10^{-6})
30,030	RS-VNS	30	+0.78 (6.0×10^{-6})	+0.83 (3.2×10^{-6})	+0.78 (2.0×10^{-6})
30,030	VNS	30	+0.49 (0.009)	+0.52 (0.005)	+0.48 (0.006)
30,030	PSO	28	+0.45 (1.5×10^{-6})	+0.28 (6.5×10^{-4})	+0.25 (3.8×10^{-4})
30,030	SSA	30	+2.47 (1.9×10^{-9})	+2.22 (1.9×10^{-9})	+2.03 (1.9×10^{-9})
30,030	LX-SSA	30	+2.98 (1.9×10^{-9})	+2.55 (1.9×10^{-9})	+2.36 (1.9×10^{-9})
30,030	DE	8	+1.28 (0.016)	+0.35 (0.195)	+0.21 (0.383)
30,030	MS-SLSQP	30	+1.93 (1.9×10^{-9})	+1.06 (1.3×10^{-8})	+0.89 (3.9×10^{-7})

Table S47: Horns Rev 1 wake model against PyWake 2.6.20 (NOJ, $k = 0.04$, V80 curves and Weibull climate of PyWake’s Hornsrev1Site) at identical direction and speed bins (the bin probabilities agree to 7×10^{-18}). AEP and wake-free AEP in GWh/yr, wake loss in %. PyWake’s NOJ uses rotor-area overlap, C_T at the effective speed of the upstream turbine and the Madsen $a(C_T)$ polynomial by default. ^aDiagnostic variant of our model only.

N	Direction bins	Model	AEP	Wake-free	Loss (%)
80	5° at 2.5° (paper)	This paper (hornsrev_model)	666.75	744.04	10.39
80	5° at 2.5° (paper)	This paper, C_T at local speed ^a	660.24	744.04	11.26
80	5° at 2.5° (paper)	PyWake NOJ	661.39	744.04	11.11
80	5° at 2.5° (paper)	PyWake NOJ, rotor centre	659.16	744.04	11.41
80	5° at 2.5° (paper)	PyWake NOJ, rotor centre, momentum $a(C_T)$	660.24	744.04	11.26
80	1° at 0.5°	This paper (hornsrev_model)	664.64	744.04	10.67
80	1° at 0.5°	This paper, C_T at local speed ^a	657.88	744.04	11.58
80	1° at 0.5°	PyWake NOJ	661.87	744.04	11.04
80	1° at 0.5°	PyWake NOJ, rotor centre	656.77	744.04	11.73
80	1° at 0.5°	PyWake NOJ, rotor centre, momentum $a(C_T)$	657.88	744.04	11.58
80	1° at 0° (PyWake default)	This paper (hornsrev_model)	663.33	744.04	10.85
80	1° at 0° (PyWake default)	This paper, C_T at local speed ^a	656.25	744.04	11.80
80	1° at 0° (PyWake default)	PyWake NOJ	661.93	744.04	11.04
80	1° at 0° (PyWake default)	PyWake NOJ, rotor centre	655.12	744.04	11.95
80	1° at 0° (PyWake default)	PyWake NOJ, rotor centre, momentum $a(C_T)$	656.25	744.04	11.80
16	5° at 2.5° (paper)	This paper (hornsrev_model)	139.82	148.81	6.04
16	5° at 2.5° (paper)	This paper, C_T at local speed ^a	139.34	148.81	6.36
16	5° at 2.5° (paper)	PyWake NOJ	139.08	148.81	6.53
16	5° at 2.5° (paper)	PyWake NOJ, rotor centre	139.21	148.81	6.45
16	5° at 2.5° (paper)	PyWake NOJ, rotor centre, momentum $a(C_T)$	139.34	148.81	6.36
16	1° at 0.5°	This paper (hornsrev_model)	139.50	148.81	6.25
16	1° at 0.5°	This paper, C_T at local speed ^a	139.00	148.81	6.59
16	1° at 0.5°	PyWake NOJ	139.16	148.81	6.48
16	1° at 0.5°	PyWake NOJ, rotor centre	138.87	148.81	6.68
16	1° at 0.5°	PyWake NOJ, rotor centre, momentum $a(C_T)$	139.00	148.81	6.59
16	1° at 0° (PyWake default)	This paper (hornsrev_model)	139.24	148.81	6.43
16	1° at 0° (PyWake default)	This paper, C_T at local speed ^a	138.70	148.81	6.79
16	1° at 0° (PyWake default)	PyWake NOJ	139.17	148.81	6.47
16	1° at 0° (PyWake default)	PyWake NOJ, rotor centre	138.57	148.81	6.88
16	1° at 0° (PyWake default)	PyWake NOJ, rotor centre, momentum $a(C_T)$	138.70	148.81	6.79

Table S48: IEA37 Case Study 1 published layouts: largest distance outside the boundary circle, minimum spacing and feasibility at the published 1 mm tolerance (strict), and after projecting every turbine outside the circle radially onto it (AEP recomputed with our calculator, which reproduces the official one; spacing re-checked against $2D = 260$ m with a 10^{-6} m tolerance). Layouts violating the spacing cannot be repaired by the projection.

N	Part.	AEP publ. (MWh)	Outside (m)	Min. spacing (m)	Feas. strict	AEP projected	Spacing proj. (m)	Feas. projected
16	1	411,182.2	0.0010	439.1	yes	411,182.2	439.1	yes
16	2	409,689.4	0.0000	522.0	yes	409,689.4	522.0	yes
16	3	402,318.8	0.0000	376.2	yes	402,318.8	376.2	yes
16	4	418,924.4	0.0000	357.6	yes	418,924.4	357.6	yes
16	5	414,141.3	0.0000	419.0	yes	414,141.3	419.0	yes
16	6	388,758.4	0.0000	387.2	yes	388,758.4	387.2	yes
16	7	392,587.9	0.0000	366.1	yes	392,587.9	366.1	yes
16	8	412,251.2	0.0010	260.0	no	412,251.2	260.0	yes
16	9	388,342.7	0.0000	263.3	yes	388,342.7	263.3	yes
16	10	408,360.8	0.0000	308.3	yes	408,360.8	308.3	yes
16	11	409,850.3	0.0053	464.1	no	409,849.9	464.1	yes
16	12	421,561.9	3.5182	563.3	no	421,451.1	562.9	yes
36	1	844,281.2	0.0000	260.0	yes	844,281.2	260.0	yes
36	2	849,369.8	0.0000	286.1	yes	849,369.8	286.1	yes
36	3	828,745.6	0.0000	268.6	yes	828,745.6	268.6	yes
36	4	863,676.3	0.0000	262.2	yes	863,676.3	262.2	yes
36	5	820,394.2	0.0000	166.3	no	820,394.2	166.3	no
36	6	776,000.1	0.0000	282.7	yes	776,000.1	282.7	yes
36	7	777,475.8	0.0000	238.3	no	777,475.8	238.3	no
36	8	846,357.8	0.0065	465.6	no	846,357.6	465.6	yes
36	9	813,544.2	0.0000	260.0	yes	813,544.2	260.0	yes
36	10	851,631.9	0.0000	260.0	yes	851,631.9	260.0	yes
36	11	846,255.2	0.0002	326.3	yes	846,255.1	326.3	yes
36	12	882,383.3	0.0049	596.2	no	882,382.8	596.2	yes

Table S49: IEA37 Case Study 1: gap of PSO-VNS (best and mean of 30 runs) and of the best run of all our methods to the best feasible published layout, under the strict 1 mm tolerance and with boundary-projected published layouts; AEP gap in % and wake-loss gap in percentage points (pp, positive = more wake loss than the reference). Rank: position of the best PSO-VNS run among the feasible published layouts.

N	Budget	Convention	Ref.	Ref. AEP	Ref. loss (%)	Best gap (%)	Best gap (pp)	Rank	Mean gap (%)	Mean gap (pp)	Best of ours (%)
16	6,030	strict	4	418,924.4	10.78	-3.85	+3.44	sixth	-5.70	+5.09	-3.22 (MS-SLSQP)
16	6,030	projected	12	421,451.1	10.24	-4.43	+3.98	ninth	-6.27	+5.63	-3.80 (MS-SLSQP)
16	30,030	strict	4	418,924.4	10.78	-1.33	+1.19	third	-3.90	+3.48	-1.33 (PSO-VNS)
16	30,030	projected	12	421,451.1	10.24	-1.93	+1.73	fourth	-4.48	+4.02	-1.93 (PSO-VNS)
36	6,030	strict	4	863,676.3	18.25	-8.80	+7.19	eighth	-14.69	+12.01	-8.76 (PSO)
36	6,030	projected	12	882,382.8	16.48	-10.73	+8.96	tenth	-16.50	+13.78	-10.70 (PSO)
36	30,030	strict	4	863,676.3	18.25	-4.20	+3.44	seventh	-6.51	+5.32	-3.51 (MS-SLSQP)
36	30,030	projected	12	882,382.8	16.48	-6.23	+5.21	ninth	-8.49	+7.09	-5.56 (MS-SLSQP)

S-VI Theoretical Details

This section states and proves the theoretical facts used in the main text: the stability conditions of the implemented PSO update and where the two PSO settings lie (Section S-VI-A), the budget accounting of the two-phase hybrids and the strength of the random-sampling control (Section S-VI-B), what can and cannot be proven about PSO and DE from feasible starts (Section S-VI-C), the practical-equivalence statements (Section S-VI-D) and the property of the VNS local search (Section S-VI-E). All statements refer to the code in `analysis/`; all numbers of this section are recomputed and checked by `analysis/make_theory_figures.py`.

S-VI-A Stability of the Implemented PSO Update

In the code, a move of particle i updates every coordinate m by

$$v \leftarrow wv + c_1\rho_1(p - x) + c_2\rho_2(g - x), \quad x \leftarrow \min(\max(x + v, -r), r), \quad (\text{S3})$$

where $x = X_m^i$, $v = V_m^i$, $p = P_m^i$, $g = G_m$, and ρ_1, ρ_2 are independent $U(0, 1)$ numbers drawn anew for every coordinate and every move; v is not changed when x is clipped. We use two assumptions: (A1) *stagnation*: p and g are fixed; (A2) *no active bound*: the clipping never acts, so $v_t = x_t - x_{t-1}$. Under (A1)–(A2), with $\varphi_{k,t} = c_k\rho_{k,t}$ and $\varphi_t = \varphi_{1,t} + \varphi_{2,t}$, each coordinate follows, independently of the others,

$$x_{t+1} = (1 + w - \varphi_t)x_t - wx_{t-1} + \varphi_{1,t}p + \varphi_{2,t}g. \quad (\text{S4})$$

The coefficient φ_t is random, with mean $\mu = (c_1 + c_2)/2$ and variance $\sigma^2 = (c_1^2 + c_2^2)/12$. We call the update *order-1 stable* if, for all initial values (x_0, x_1) , $E x_t$ converges to a limit that does not depend on them, and *order-2 stable* if, in addition, $E x_t^2$ does [7].

Proposition S1 (Stability of the implemented update) *Let $c_1, c_2 \geq 0$, $c_1 + c_2 > 0$, and assume (A1)–(A2).*

(a) $E x_t \rightarrow x^* = (c_1p + c_2g)/(c_1 + c_2)$ for all initial values if and only if $|w| < 1$ and $0 < \mu < 2(1 + w)$, i.e.,

$$|w| < 1 \quad \text{and} \quad c_1 + c_2 < 4(1 + w). \quad (\text{S5})$$

(b) The update is order-2 stable if and only if $|w| < 1$ and

$$\chi(1) := 2\mu(1 - w^2) - (1 - w)\mu^2 - (1 + w)\sigma^2 > 0. \quad (\text{S6})$$

For $c_1 = c_2$, (S6) is Poli's condition (8), $c_1 + c_2 < 24(1 - w^2)/(7 - 5w)$ [7]; for $c_1 \neq c_2$ it is stricter than (8), since σ^2 grows when $c_1 + c_2$ is kept and $c_1 \neq c_2$.

(c) Under (b), $\text{Var } x_t \rightarrow V_\infty = (1 + w)q/\chi(1)$ with $q = c_1^2c_2^2(p - g)^2/[6(c_1 + c_2)^2]$; for $c_1 = c_2 = c$,

$$V_\infty = \frac{c(1 + w)}{4[12(1 - w^2) - c(7 - 5w)]} (p - g)^2, \quad (\text{S7})$$

as in [7]. The limit is zero if and only if $p = g$.

Proof. (a) Since $\rho_{1,t}, \rho_{2,t}$ are independent of (x_t, x_{t-1}) , the deviation $e_t = E x_t - x^*$ satisfies $e_{t+1} = m e_t - w e_{t-1}$ with $m = 1 + w - \mu$ (the constant terms cancel because $E\varphi_{1,t}(p - x^*) + E\varphi_{2,t}(g - x^*) = 0$). Its characteristic polynomial $\lambda^2 - m\lambda + w$ has both roots strictly inside the unit circle if and only if $|w| < 1$ and $|m| < 1 + w$ (Schur–Cohn conditions for a real monic quadratic), i.e., $0 < \mu < 2(1 + w)$; otherwise an initial value exciting a root of modulus ≥ 1 keeps e_t away from zero.

(b) Let $y_t = x_t - x^*$. Then $y_{t+1} = (1 + w - \varphi_t)y_t - wy_{t-1} + \eta_t$ with $\eta_t = \varphi_{1,t}(p - x^*) + \varphi_{2,t}(g - x^*)$, $E\eta_t = 0$, and (φ_t, η_t) is independent of (y_t, y_{t-1}) . Hence $z_t = (Ey_t^2, Ey_t y_{t-1}, Ey_{t-1}^2)^\top$ satisfies $z_{t+1} = M z_t + b_t$ with

$$M = \begin{pmatrix} a & -2wm & w^2 \\ m & -w & 0 \\ 1 & 0 & 0 \end{pmatrix}, \quad a = E(1 + w - \varphi_t)^2 = m^2 + \sigma^2,$$

and $b_t = (q - 2\text{Cov}(\varphi_t, \eta_t)Ey_t, 0, 0)^\top$, $q = E\eta_t^2$. Order-2 stability is equivalent to a spectral radius of M below one (and $|w| < 1$). Sufficiency: Ey_t obeys the mean recursion of (a), which does not involve η_t , so it equals the mean of the process with $p = g$ (where $\eta_t = 0$ and $z_t = M^t z_0 \rightarrow 0$) and $(Ey_t)^2 \leq Ey_t^2 \rightarrow 0$ there; hence $b_t \rightarrow (q, 0, 0)^\top$ and $z_t \rightarrow (I - M)^{-1}(q, 0, 0)^\top$, independent of the initial values. Necessity: order-2 stability includes order-1 stability, so $|w| < 1$ by (a) and $Ey_t \rightarrow 0$; hence $Ey_t^2 = Ex_t^2 - 2x^*Ex_t + x^{*2}$ converges to a limit

that does not depend on the initial values. The second row of M is the recursion $Ey_{t+1}y_t = mEy_t^2 - wEy_t y_{t-1}$ (the term $E\eta_t Ey_t$ vanishes), which, with $|w| < 1$, makes $Ey_t y_{t-1}$ converge whenever Ey_t^2 does (also if $wm = 0$), and the third component of z_t is the first one delayed by one step; so the whole vector z_t converges to a limit that does not depend on the initial values. Now, for the initial values $(\lambda y_0, \lambda y_1)$, Ey_t is proportional to λ , so $z_t = \lambda^2 M^t z_0 + \lambda u_t + v_t$ with u_t, v_t independent of λ ; if z_t has the same limit for three values of λ , then $M^t z_0 \rightarrow 0$, and since the vectors $z_0 = (y_1^2, y_1 y_0, y_0^2)$ span $\mathbb{R}^3$, $M^t \rightarrow 0$. The characteristic polynomial of M is $\chi(\lambda) = \lambda^3 + a_2 \lambda^2 + a_1 \lambda + a_0$ with $a_2 = w - a$, $a_1 = 2wm^2 - aw - w^2$, $a_0 = -w^3$. By Jury's criterion, all roots lie strictly inside the unit circle if and only if (i) $\chi(1) > 0$, (ii) $\chi(-1) < 0$, (iii) $|a_0| < 1$ and (iv) $|a_0^2 - 1| > |a_0 a_2 - a_1|$. Direct expansion gives $\chi(1) = (1+w)(1-w^2) - (1+w)\sigma^2 - (1-w)m^2$, which equals (S6), and $\chi(-1) = -(1-w)(1-w^2 + \sigma^2) - (1+w)m^2 < 0$ for $|w| < 1$; (iii) is $|w| < 1$. For (iv), $a_0 a_2 - a_1 = wX$ with $X = \sigma^2(1+w^2) - (1-w^2)m^2 + w(1-w^2)$. If (i) holds, then $(1-w)m^2 < (1+w)(1-w^2)$, i.e., $m^2 < (1+w)^2$, and $\sigma^2 < 1-w^2$; therefore $-(1-w^2)(1+w+w^2) < X < (1-w^2)(1+w+w^2)$, and $|wX| < |w|(1+w+w^2)(1-w^2) \leq (1+w^2+w^4)(1-w^2) = 1-w^6$, because $1+w^2+w^4 - |w|(1+w+w^2)$ equals $(1-w)(1-w^3)$ for $w \geq 0$ and $1-u+2u^2-u^3+u^4 > 0$ for $w = -u < 0$. So (iv) follows from (i), and order-2 stability holds if and only if $|w| < 1$ and $\chi(1) > 0$. For $c_1 = c_2 = c$: $\mu = c$, $\sigma^2 = c^2/6$ and $\chi(1) = c[12(1-w^2) - c(7-5w)]/6$, which is positive if and only if $2c < 24(1-w^2)/(7-5w)$. For fixed $c_1 + c_2$, σ^2 is smallest at $c_1 = c_2$, which gives the last statement of (b). (The general condition restates, in our notation, the second-moment analysis of [7] for $c_1 \neq c_2$; we do not claim it as new.)

(c) In the limit $Ey_t \rightarrow 0$, so $b_t \rightarrow (q, 0, 0)^\top$ and $z_t \rightarrow z_\infty$ with $z_\infty = Mz_\infty + (q, 0, 0)^\top$. The second row gives $Ey_t y_{t-1} \rightarrow mV_\infty/(1+w)$, and the first row then gives $V_\infty \chi(1)/(1+w) = q$. With $p - x^* = c_2(p-g)/(c_1+c_2)$ and $g - x^* = -c_1(p-g)/(c_1+c_2)$, $q = [c_2^2 \text{Var } \varphi_1 + c_1^2 \text{Var } \varphi_2](p-g)^2/(c_1+c_2)^2 = c_1^2 c_2^2 (p-g)^2/[6(c_1+c_2)^2]$, which is $c^2(p-g)^2/24$ for $c_1 = c_2 = c$ and gives (S7). $\square$

Which φ enters the order-1 condition. For constant coefficients, the classical condition is $0 < \varphi < 2(1+w)$ with $\varphi = c_1 + c_2$. In the implemented update the coefficients are $c_k \rho_k$ with $\rho_k \sim U(0, 1)$, and the order-1 condition involves their mean, $E\varphi_t = (c_1 + c_2)/2$; hence $c_1 + c_2 < 4(1+w)$. Inserting $c_1 + c_2$ into the constant-coefficient condition would give $c_1 + c_2 < 3.4$ at $w = 0.7$ and wrongly classify the old setting as order-1 unstable. The analytic order-2 region of Proposition S1(b) agrees with the numerically computed spectral radius of M at 200,000 random points $(w, c_1, c_2) \in (-1, 1) \times [0, 6]^2$ and on a 601×601 grid for $c_1 = c_2$ (points within 10^{-3} of the boundary excluded; no disagreement).

Table S50: Stability of the two PSO settings under stagnation and without active bounds (Proposition S1). ρ_1 : spectral radius of the mean recursion; ρ_2 : spectral radius of the second-moment matrix M (order-2 stable iff $\rho_2 < 1$); $V_\infty/(p-g)^2$: limit variance factor (S7).

Setting	w	$c_1 + c_2$	$4(1+w)$	$\frac{24(1-w^2)}{7-5w}$	margin	ρ_1	ρ_2	$V_\infty/(p-g)^2$
Old	0.7	4	6.8	3.50	-0.50	0.84	1.12	unbounded
Constriction	0.7298	2.99	6.92	3.35	+0.36	0.85	0.94	1.09

Margin: order-2 bound minus $c_1 + c_2$. Constriction: $c_1 + c_2 = 2.99236$ is 10.6% below the bound 3.3475; old: $c_1 + c_2 = 4$ is 14.4% above the bound 3.4971.

Proposition S2 (The two settings) *Under (A1)–(A2), both settings are order-1 stable. The constriction setting is order-2 stable ($c_1 + c_2 = 2.99 < 3.35$, $\rho_2 = 0.94$); each coordinate of a particle with $p \neq g$ has, in the limit, mean $(p+g)/2$ and standard deviation $1.04|p-g|$ ($V_\infty = 1.09(p-g)^2$). The old setting is not order-2 stable ($4 > 3.50$): Ex_t^2 is unbounded for every initial value if $p \neq g$, and for every initial value with $(x_0, x_1) \neq (x^*, x^*)$ if $p = g$; a positive weighted sum of the second moments grows at least by the factor $\rho_2 = 1.12$ per iteration.*

Proof. The numbers follow from Proposition S1 (Table S50). For the old setting, the dominant eigenvalue $\rho_2 = 1.12$ of M is real and has a left eigenvector u with positive components ($u \approx (0.90, 0.21, 0.39)$). Since $c_1 = c_2$, $\text{Cov}(\varphi_t, \eta_t) = 0$ and $b_t = (q, 0, 0)^\top$, so $u^\top z_{t+1} = \rho_2 u^\top z_t + u_1 q \geq \rho_2 u^\top z_t$. For a nonzero $z_0 = (\alpha, \beta, \gamma)$ with $\alpha, \gamma \geq 0$, $|\beta| \leq \sqrt{\alpha\gamma} \leq (\alpha + \gamma)/2$ gives $u^\top z_0 > 0$; if $z_0 = 0$ and $p \neq g$, then $u^\top z_1 = u_1 q > 0$. Hence $u^\top z_{t+1} \geq \rho_2^t u^\top z_1 \rightarrow \infty$ (with $u^\top z_1 > 0$ in both cases). Since $|Ey_t y_{t-1}| \leq (Ey_t^2 + Ey_{t-1}^2)/2$, $u^\top z_t \leq (u_1 + u_2/2)Ey_t^2 + (u_3 + u_2/2)Ey_{t-1}^2$, so the sequence Ey_t^2 is unbounded, and so is $Ex_t^2 = Ey_t^2 + 2x^*Ey_t + x^{*2}$, because $Ey_t \rightarrow 0$ (order-1 stability). We show unboundedness only; that the second moments themselves grow asymptotically by the factor ρ_2 per iteration is not needed here. $\square$

What the conditions do not cover. (i) *Stagnation.* In a run, $\mathbf{P}^i$ and $\mathbf{G}$ change. For $c_1 = c_2$, Cleghorn and Engelbrecht [14] obtain the same region under the weaker non-stagnate distribution assumption; we do not claim that their region extends to $c_1 \neq c_2$, where the condition under stagnation is stricter (Proposition S1(b)). No stability result says how good $\mathbf{G}$ is: order-2 stability means that each particle samples a distribution of bounded

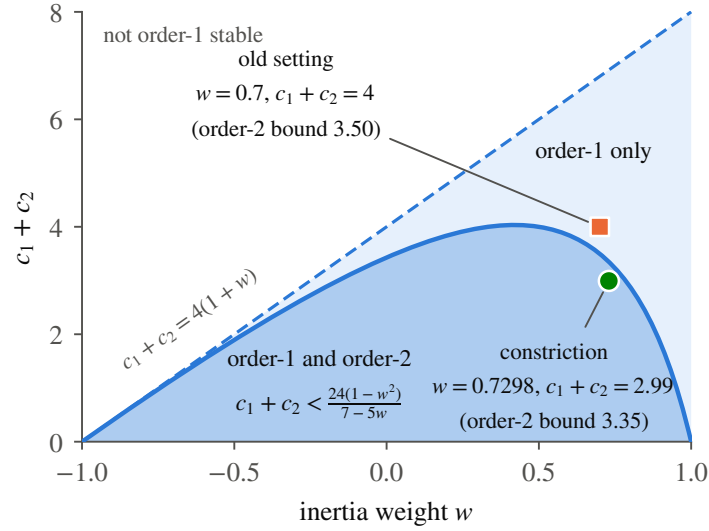


Figure S15: Stability regions of the implemented PSO update under stagnation and without active bounds ($c_1 = c_2$; Proposition S1). Light: order-1 stable only, $c_1 + c_2 < 4(1 + w)$; dark: order-1 and order-2 stable, $c_1 + c_2 < 24(1 - w^2)/(7 - 5w)$. The old setting ($w = 0.7$, $c_1 = c_2 = 2$) is order-1 but not order-2 stable; the constriction setting (7) is order-2 stable.

variance around $(c_1 \mathbf{P}^i + c_2 \mathbf{G}) / (c_1 + c_2)$, whose variance vanishes only if $\mathbf{P}^i = \mathbf{G}$ (Proposition S1(c)); it implies neither convergence to a point nor local optimality. (ii) *Bounds*. The code clips the position to $[-r, r]$ and keeps the velocity, so after a clip $v_t \neq x_t - x_{t-1}$, and (S4) and both results no longer describe the trajectory. (iii) *Causality*. That the old setting violates the order-2 condition does not imply that this causes its poor results; the coefficient sweep of Section S-II-B shows that the loss of feasibility begins near the boundary and grows progressively beyond it, and that velocity clamping restores much of it. What the code implies at a bound is the following.

Lemma S3 (Boundary handling of the PSO code) (a) *If a move leaves a coordinate of turbine k at $\pm r$, then $x_k^2 + y_k^2 \geq r^2$, with equality only if its other coordinate is exactly zero; so the evaluated layout violates the boundary constraint of the circular farm unless that coordinate is zero.* (b) *If a coordinate is at r after a move and its kept velocity is $v \geq 0$, the next velocity is $v' = wv + \varphi_1(p - r) + \varphi_2(g - r) \leq wv$, and the coordinate is clipped to r again as long as $v' \geq 0$; the same holds at $-r$ with the signs reversed.*

Proof. (a) $x_k^2 + y_k^2 = r^2 + y_k^2$ if $x_k = \pm r$ (and likewise for y_k). (b) p and g are evaluated, hence clipped, positions, so $p - r \leq 0$ and $g - r \leq 0$; the position update gives $\min(r + v', r) = r$ for $v' \geq 0$. $\square$

Thus, in the implementation, the unbounded second moments of the old setting cannot materialize as growing positions; a coordinate that reaches the bound stays there while its velocity points outward, and by (a) every such evaluation is infeasible (almost surely). Lemma S3 does not say how often clipping occurs; that is an empirical question, answered by the diagnostics of Section S-VII-A (Fig. S16 and Table S52).

S-VI-B Budget Accounting and the Random-Sampling Control

An *evaluation* is one call of the objective, counted by a wrapper that stops the run at the budget B (for MS-SLSQP also the finite-difference calls). A population method with N_p members uses N_p evaluations for the initial population and c per iteration, $c = N_p$ for PSO, SSA and DE and $c = 2N_p$ for LX-SSA, so T iterations cost $N_p + cT$ evaluations; stand-alone, $T = \lfloor (B - N_p)/c \rfloor$ (Section S-I-D). A two-phase hybrid with split ω uses $B_1 = N_p + cT_1$ evaluations in Phase 1, $T_1 = \text{round}((\omega B - N_p)/c)$ with Python's rounding half to even (for PSO and SSA at $\omega = 0.5$, 99.5 is rounded to $T_1 = 100$), and $B_2 = B - B_1$ in Phase 2; RS-VNS and RSD-VNS use $B_1 = \text{round}(\omega B)$, including the seeded initial population. At $B = 6,030$ and $\omega = 0.5$, $B_1 = 3,030$ for the swarms and 3,015 for the two random-sampling controls, a difference of 15 evaluations (0.25% of B).

Proposition S4 (Prefix and monotonicity) *For every seed: (a) Phase 1 of PSO-VNS evaluates exactly the first B_1 layouts of the stand-alone PSO run; (b) the final F_p of PSO-VNS is at most the best F_p of these B_1 evaluations; (c) (a) does not hold for SSA-VNS and LX-SSA-VNS.*

Proof. (a) Both runs seed the global generator once and then draw, in the same order, the initial population and the vectors $\boldsymbol{\rho}_1, \boldsymbol{\rho}_2$ of every move; the PSO update does not depend on the number of iterations, so the first

T_1 iterations coincide. (b) Phase 2 starts with $\mathbf{G}$ as its best evaluated point, replaces it only by strictly better evaluated points and returns the best one. (c) The SSA coefficient $\xi_1 = 2 \exp[-(4t/T)^2]$ depends on t/T , and the hybrid uses $T = T_1$ instead of the stand-alone iteration count. $\square$

Why equal budgets matter. Let $L_A(b)$ be the best F_p found by method A within b evaluations. For a fixed seed, $L_A(b)$ is nonincreasing in b , so any comparison at unequal budgets mixes the effect of the method with that of the budget: counting only Phase-2 evaluations, or counting an LX-SSA iteration as N_p instead of $2N_p$ evaluations, would give one method more evaluations and an advantage by monotonicity alone. At the same B , Proposition S4 makes PSO-VNS versus PSO a clean contrast: both continue from the identical state after B_1 evaluations, PSO with its whole swarm and VNS from $\mathbf{G}$ alone, for the same B_2 evaluations. It guarantees $L_{\text{PSO-VNS}}(B) \leq L_{\text{PSO}}(B_1)$ but not $L_{\text{PSO-VNS}}(B) \leq L_{\text{PSO}}(B)$; which of the two continuations is better is an empirical question. Likewise, the contrast between a hybrid and RS-VNS isolates Phase 1 (a search method versus uniform sampling with about the same number of evaluations, followed by the same Phase 2). How weak this control is follows from the next result.

Proposition S5 (Uniform layouts in the bounding square) *Let the $2N$ coordinates be independent and uniform in $[-r, r]$, and let P_N be the probability that the layout is feasible (all turbines in the disc D of radius r and all pairwise distances $\geq s$). Then*

(a) *all N turbines lie in D with probability $(\pi/4)^N$;*

(b) $P_N \leq (\pi/4)^N \prod_{k=1}^{N-1} \max\left\{0, 1 - \frac{\max\{A(s), kA(s/2)\}}{\pi r^2}\right\}$, *where, for $0 < \rho \leq 2r$,*

$$A(\rho) = r^2 \arccos\left(1 - \frac{\rho^2}{2r^2}\right) + \rho^2 \arccos \frac{\rho}{2r} - \frac{\rho}{2} \sqrt{4r^2 - \rho^2} \quad (\text{S8})$$

is the area of the part of D covered by a disc of radius ρ centred on the boundary of D ;

(c) *among M such layouts, the expected number of feasible ones is MP_N .*

Proof. (a) The N turbine positions are independent and uniform in the square, and each lies in D with probability $\pi r^2/(2r)^2$. (b) Conditional on all turbines in D , the positions $\mathbf{p}_1, \dots, \mathbf{p}_N$ are independent and uniform in D . Let E_k be the event that $\mathbf{p}_{k+1}$ is at least s from $\mathbf{p}_1, \dots, \mathbf{p}_k$. Given $E_1 \cap \dots \cap E_{k-1}$ and the first k positions, $\mathbf{p}_{k+1}$ must avoid $U_k = D \cap \bigcup_{j \leq k} B(\mathbf{p}_j, s)$. The area of $B(\mathbf{c}, \rho) \cap D$ for $\mathbf{c} \in D$ is nonincreasing in $\|\mathbf{c}\|$ (its derivative with respect to the distance of the centres is minus the length of the common chord), so it is at least $A(\rho)$, the value at $\|\mathbf{c}\| = r$. Hence $|U_k| \geq A(s)$; and since the discs $B(\mathbf{p}_j, s/2)$ are pairwise disjoint (centres at least s apart) and contained in $B(\mathbf{p}_j, s)$, also $|U_k| \geq kA(s/2)$. So $\Pr(E_k \mid E_1, \dots, E_{k-1}) \leq 1 - \max\{A(s), kA(s/2)\}/(\pi r^2)$, and the chain rule gives (b). (c) Linearity of expectation. $\square$

Table S51: Probability P_N that a layout drawn uniformly in the bounding square $[-r, r]^{2N}$ is feasible (Proposition S5): all turbines in the farm, $(\pi/4)^N$; rigorous upper bound (b); Monte Carlo estimate from 10^7 layouts per row (fixed seed; turbines drawn uniformly in the disc, multiplied by $(\pi/4)^N$; 95% Clopper–Pearson interval); expected number of feasible layouts among the 3,015 uniform layouts of the Phase 1 of RS-VNS at $B = 6,030$ (including the initial population). Rows $N = 10, 12, 15$: largest benchmark cases ($s = 8R = 308$ m); $N = 16, 36$: IEA37 Case Study 1 ($s = 2D = 260$ m); $N = 20$: hypothetical.

N	r (m)	s (m)	$(\pi/4)^N$	bound on P_N	P_N (Monte Carlo) [95% CI]	expected feasible of 3,015
2	500	308	0.62	0.52	0.44 [0.44, 0.44]	1,337
10	500	308	0.089	$6.5 \cdot 10^{-3}$	0 [0, $3.3 \cdot 10^{-8}$]	$< 9.9 \cdot 10^{-5}$
12	750	308	0.055	0.012	$9.4 \cdot 10^{-8}$ [$5.5 \cdot 10^{-8}$, $1.5 \cdot 10^{-7}$]	$2.8 \cdot 10^{-4}$
15	1,000	308	0.027	$7.0 \cdot 10^{-3}$	$3.5 \cdot 10^{-7}$ [$2.9 \cdot 10^{-7}$, $4.1 \cdot 10^{-7}$]	$1.0 \cdot 10^{-3}$
16	1,300	260	0.021	0.011	$1.5 \cdot 10^{-4}$ [$1.5 \cdot 10^{-4}$, $1.5 \cdot 10^{-4}$]	0.45
20	1,000	308	$8.0 \cdot 10^{-3}$	$7.0 \cdot 10^{-4}$	0 [0, $2.9 \cdot 10^{-9}$]	$< 8.9 \cdot 10^{-6}$
36	2,000	260	$1.7 \cdot 10^{-4}$	$4.3 \cdot 10^{-5}$	$1.8 \cdot 10^{-9}$ [$1.5 \cdot 10^{-9}$, $2.2 \cdot 10^{-9}$]	$5.4 \cdot 10^{-6}$

For $N = 2$ the exact value, from the distance distribution of two uniform points in a disc, is $P_2 = 0.443$. “<”: upper end of the 95% interval (no feasible layout among the 10^7).

The bound (b) is rigorous but loose; the Monte Carlo estimates show that for the largest benchmark cases a uniform layout in the square is feasible with probability below 10^{-6} , so the expected number of feasible layouts in the Phase 1 of RS-VNS is about 10^{-3} or less (Table S51). Since any violation above $4.6 \cdot 10^{-8}$ makes F_p exceed every wake loss (Section S-I-H and Lemma S6), and $F_p = L + \text{Pen}$ with $0 \leq L \leq NE_{\text{ideal}} + \varepsilon$ (ε : the small contribution of the negative part of the power curve just above cut-in), the Phase-1 result of RS-VNS in these cases is almost surely the sampled layout with the smallest F_p , which is decided by the penalty: $F_p(\mathbf{X}) < F_p(\mathbf{Y})$

whenever $\text{Pen}(\mathbf{Y}) - \text{Pen}(\mathbf{X}) > NE_{\text{ideal}}$. RS-VNS thus hands VNS the least-violating of its samples, not a layout selected for low wake loss, and “better than RS-VNS” is a minimum bar. The factor $(\pi/4)^N$ explains only a small part of this weakness: the Monte Carlo column of Table S51 lies orders of magnitude below $(\pi/4)^N$, so the spacing constraint dominates, and in the runs, disc sampling gives a feasible sample to only a minority of the RS-VNS runs without one (Section S-III).

S-VI-C Feasible Starts: What Is Proven and What Is Observed

With the feasibility-preserving initialization (Algorithm S2), all N_p initial layouts are feasible, and $\mathbf{G}$ is the best of them. From feasible starts, in all runs of the study (six largest cases and Horns Rev 1, 30 seeds each), $\mathbf{G}$ never moved (Section VIII-A). The following facts follow from the code; the stall itself does not, and is an empirical finding.

Lemma S6 (Improvement needs a feasible trial) *If $\mathbf{G}$ is feasible, a trial $\mathbf{X}$ replaces it only if $L(\mathbf{X}) + \text{Pen}(\mathbf{X}) < L(\mathbf{G})$; since every violated constraint adds more than 1 and a violation g adds $(1 + \mu g)^2$, every violation of $\mathbf{X}$ must be below $(\sqrt{L(\mathbf{G})} - 1)/\mu$, i.e., below $4.6 \cdot 10^{-8}$ (m or m^2) in the benchmark cases, and $\mathbf{X}$ must have a lower wake loss than $\mathbf{G}$. So $\mathbf{X}$ need not be exactly feasible, but its violations must be far below the reporting tolerance of 10^{-6} m .*

Proof. Definition of F_p (4), $\text{Pen} \geq 0$ and $L \geq 0$; for the benchmark cases $L(\mathbf{G}) \leq NE_{\text{ideal}} + \varepsilon$ with $N \leq 15$ and $E_{\text{ideal}} \leq 14,045.74$ (Data Set I), so $L(\mathbf{G}) < 2.11 \cdot 10^5$, $\sqrt{L(\mathbf{G})} < 459.1$, and with $\mu = 10^{10}$ the bound is below $4.6 \cdot 10^{-8}$. $\square$

Proposition S7 (PSO from feasible starts) *Let all initial layouts be feasible and let i^* be the particle with the best initial layout, so that $\mathbf{X}^{i^*} = \mathbf{P}^{i^*} = \mathbf{G}$ and $\mathbf{V}^{i^*} = \mathbf{0}$.*

- (a) *As long as $\mathbf{G}$ does not change, particle i^* keeps $\mathbf{V}^{i^*} = \mathbf{0}$ and $\mathbf{X}^{i^*} = \mathbf{G}$: it re-evaluates $\mathbf{G}$ once per iteration (200 of the 6,030 evaluations if $\mathbf{G}$ never changes) and evaluates no other point.*
- (b) *The first move of every other particle is $\mathbf{X}^i \leftarrow \mathbf{X}^i + c_2 \mathbf{p}_2 \circ (\mathbf{G} - \mathbf{X}^i)$, clipped to the box: every coordinate moves a random fraction in $[0, c_2] = [0, 1.50]$ of the way to the same coordinate of $\mathbf{G}$, i.e., turbine k of layout i is moved toward turbine k of $\mathbf{G}$, although the numbering of the turbines in two independently packed layouts is arbitrary.*
- (c) *If no evaluation improves any $\mathbf{P}^i$, the swarm is in exact stagnation, and, as long as no coordinate is clipped, Propositions S1(c) and S2 apply: every coordinate of particle $i \neq i^*$ is sampled, in the limit, with mean $(P_m^i + G_m)/2$ and standard deviation $1.04 |P_m^i - G_m|$, so the samples do not concentrate at $\mathbf{G}$.*

Proof. (a) By induction: with $\mathbf{V} = \mathbf{0}$ and $\mathbf{P}^{i^*} = \mathbf{X}^{i^*} = \mathbf{G}$, (5) gives $\mathbf{V} = \mathbf{0}$, the position stays $\mathbf{G}$ (inside the box), and its value equals $F_p(\mathbf{G})$, which is not a strict improvement, so neither $\mathbf{P}^{i^*}$ nor $\mathbf{G}$ changes. (b) $\mathbf{V} = \mathbf{0}$ and $\mathbf{P}^i = \mathbf{X}^i$ at the start. (c) Stagnation is (A1), and no clipping is (A2). $\square$

Part (c) is illustrative only: with the typical distance of the feasible starts to $\mathbf{G}$ (median matched RMS turbine distance 291 m, Table S54), a standard deviation of $1.04 |P_m^i - G_m|$ would place many samples outside the box, so (A2) fails and this limit is not reached in the runs.

Proposition S8 (Move structure of DE/rand/1/bin) *In the DE of the code, the trial of member i takes coordinate j from the mutant $\mathbf{x}_{r_1} + F(\mathbf{x}_{r_2} - \mathbf{x}_{r_3})$ ($r_1, r_2, r_3 \neq i$ distinct; clipped to the box) if $j = j_{\text{rand}}$ or $U_j < \text{CR}$, and from $\mathbf{x}_i$ otherwise. The number K of mutant coordinates is $1 + \text{Bin}(n - 1, \text{CR})$, with mean $1 + \text{CR}(n - 1)$, e.g., 27.1 of the $n = 30$ coordinates for $N = 15$ ($\text{CR} = 0.9$). A turbine whose two coordinates both come from the mutant is placed at $\mathbf{x}_{r_1, k} + F(\mathbf{x}_{r_2, k} - \mathbf{x}_{r_3, k})$, a combination of the k th turbines of three other layouts. If no trial is accepted, the population keeps its initial layouts and every mutant comes from the fixed set of $N_p(N_p - 1)(N_p - 2)$ vectors $\mathbf{x}_a + F(\mathbf{x}_b - \mathbf{x}_c)$.*

Proof. The binomial crossover loop and the one-to-one replacement of the code. $\square$

For comparison, the SSA leaders sample $H_m + U(-\xi_1 r, \xi_1 r)$ independently for every coordinate (the code’s step $\xi_1((ub - lb)\xi_2 + lb) = \xi_1 r(2\xi_2 - 1)$ with a random sign), with $\xi_1(t) = 2 \exp[-(4t/T)^2] \leq 0.037$ for $t \geq T/2$ (0.0366 at $t = T/2$); the VNS local search changes one coordinate of the incumbent by $\pm h$, and its shaking perturbs all coordinates of the incumbent. These moves perturb one layout and keep the numbering of its turbines, whereas the moves of Propositions S7(b) and S8 combine coordinates of different, independently numbered layouts. That these combined moves almost never yield a feasible layout with lower wake loss, which by Lemma S6 is the only way to move $\mathbf{G}$, is the plausible mechanism of the stall, but it is not proven here; the diagnostics measure it (share of feasible trials, violation type, personal-best updates, accepted DE trials; Section S-VII-C and Table S54).

S-VI-D Practical Equivalence

Definition S9 (Practical equivalence of two methods) For methods A and B , let $d_c = \bar{L}_{A,c} - \bar{L}_{B,c}$ (pp of wake loss) be the difference of the case means in case c , over the n cases in which both methods have at least 15 of 30 feasible runs, and let Δ be their mean. A and B are practically equivalent at margin δ if the hypothesis $H_0: |\Delta| \geq \delta$ is rejected in favour of $H_1: |\Delta| < \delta$ by the two one-sided tests (TOST) at $\alpha = 0.05$: $H_0^-: \Delta \leq -\delta$ and $H_0^+: \Delta \geq \delta$ are both rejected at level α . The margin is $\delta = 0.05$ pp, a convention fixed after the primary analysis.

Because $H_0 = H_0^- \cup H_0^+$ is rejected only if both parts are, TOST has size at most α without a multiplicity correction (intersection–union principle).

Lemma S10 (Confidence-interval form) With the t statistic, TOST at level α rejects H_0 if and only if the $(1 - 2\alpha)$ confidence interval $\bar{d} \pm t_{1-\alpha, n-1}$ se lies inside $(-\delta, \delta)$; for $\alpha = 0.05$ this is the 90% interval. Equivalence then holds for every margin larger than $\delta_{\min} = \max(|\text{lower end}|, |\text{upper end}|)$ and for none smaller.

Proof. H_0^- is rejected iff $(\bar{d} + \delta)/\text{se} > t_{1-\alpha, n-1}$, i.e., iff the lower end exceeds $-\delta$; H_0^+ is rejected iff the upper end is below δ . Both ends do not depend on δ , which gives the statement on $\delta_{\min}$. $\square$

The pipeline uses the percentile-bootstrap analogue (`analysis/mpce_results.py`, function `tost`): equivalence holds if the 90% percentile interval of the 10,000 bootstrap means (cases resampled with replacement) lies inside $(-\delta, \delta)$, the reported p value is the larger of the two one-sided bootstrap tail shares (with the add-one correction; an inversion of the percentile interval rather than a p value calibrated at the null boundary), and $\delta_{\min}$ is reported for every pair, so readers can apply their own margin. With 10,000 resamples, the interval rule and the rule $p \leq 0.05$ can differ only in the boundary case in which exactly 500 bootstrap means lie at or beyond one of the margins (interpolation of the 5% quantile). As the cases are not independent (Section S-I-C), the interval describes this benchmark, not a population of farms.

Levels of inference. The same interval rule is applied at three levels, which answer different questions (Table S26): at the *seed level*, the 68 cases are fixed and only the run-to-run variability is resampled (the runs within every case), which describes the fixed benchmark; at the *case level*, the cases are resampled as if exchangeable (Table S15); at the *cluster level*, a cluster-robust interval over the six (data set, radius) clusters describes generalization to new farms. The minimal margin grows from level to level, so an equivalence statement must name its level; for PSO-VNS vs. PSO equivalence at ± 0.05 pp holds at the seed and case levels but is not shown at the cluster level (Section S-III).

One-sided statements. “ A gives no gain larger than δ over B ” (non-superiority of A) needs only one of the two tests of TOST: it holds at level α if the lower end of the $(1 - 2\alpha)$ interval of $\Delta = \bar{L}_A - \bar{L}_B$ lies above $-\delta$, whatever its upper end. This is the form in which we state that the SSA phase adds nothing of practical size beyond sampling in the disc (SSA-VNS vs. RSD-VNS), where two-sided equivalence is not shown.

Non-significance is not equivalence. A test of $\Delta = 0$ that does not reject only says that zero lies in its confidence interval; the interval can still extend far beyond $\pm\delta$, for example when the differences vary much between cases or there are few of them. Conversely, a difference can be significant and practically negligible. The two questions give four outcomes:

	90% CI inside $(-\delta, \delta)$	90% CI not inside $(-\delta, \delta)$
95% CI contains 0	equivalent, no significant difference	inconclusive
95% CI excludes 0	significant but practically equivalent	significant, not equivalent

Only the left column supports statements such as “as good as”; a non-significant result in the right column (“inconclusive”) does not, and “no gain larger than δ ” needs the one-sided form below. (The case-mean Wilcoxon test concerns a median-type location of the d_c , the TOST interval their mean; the table applies to each test with its own interval.)

Bayesian signed-rank test with a region of practical equivalence (ROPE). The Bayesian test of the pipeline [15] (function `bayes_signrank`) places a Dirichlet-process prior (strength 0.5, pseudo-observation at 0) on the distribution of the d_c and, for every posterior sample, computes the probabilities $\theta_A, \theta_{\text{rope}}, \theta_B$ that the average of two case differences, $(d_c + d_{c'})/2$, lies below $-\delta$, inside $[-\delta, \delta]$ or above δ . The reported $P(\text{rope})$ is the posterior probability that θ_{rope} is the largest of the three (likewise $P(A \text{ better})$, $P(B \text{ better})$). It is thus (i) a posterior probability given the prior and the exchangeability of the cases, not an error rate; (ii) a statement about the typical pairwise-averaged case difference, not about the mean Δ of TOST, so the two can disagree, e.g., when most cases differ little but a few differ much; and (iii) a Monte Carlo share, so a reported value of 1.00 means that θ_{rope} was largest in (almost) every posterior sample, not certainty. We therefore report it as the posterior probability that practical equivalence is the most probable of the three outcomes, together with the posterior means of θ_A , θ_{rope} and θ_B (Table S27); like the case-level TOST, it treats the cases as exchangeable.

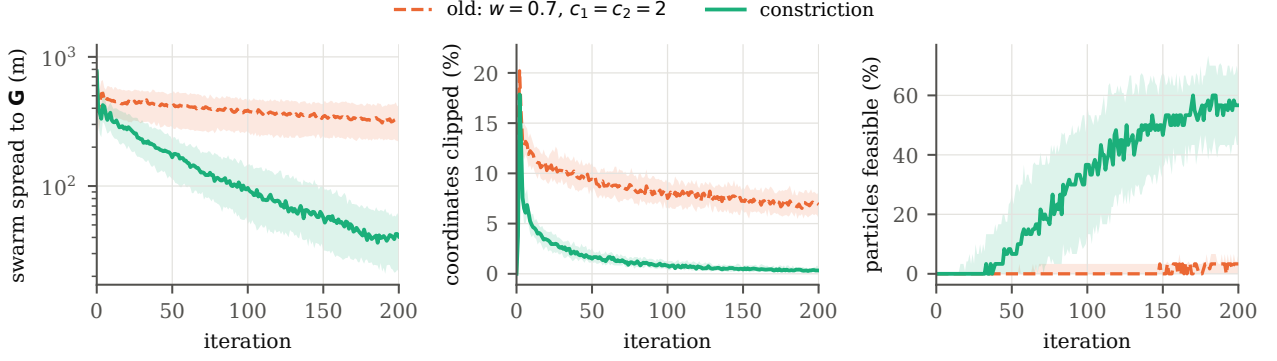


Figure S16: PSO dynamics with the old setting ($w = 0.7$, $c_1 = c_2 = 2$) and the constriction setting on 6 cases $\times$ 10 seeds at 6,030 evaluations (200 iterations): median (line) and interquartile range (band) over the runs. Left: swarm spread, the mean over particles of the RMS distance of the turbines to the global best $\mathbf{G}$ (log scale). Middle: share of the coordinates that leave the bounding square in an iteration and are clipped (velocities are kept). Right: share of particle evaluations whose layout is feasible. Per-case values: Table S52.

S-VI-E The Local Search of VNS

The local search of VNS (Section III-C) is a compass search: from $\mathbf{y}$ with step h it evaluates the $2n$ points $\text{clip}(\mathbf{y} \pm h\mathbf{e}_m)$, moves to the best one if it strictly improves F_p , and halves h otherwise, starting at $h_0 = 0.05r$ and stopping when $h < h_{\min} = 10^{-3}r$. Shaking draws a point uniformly in the ℓ_∞ shell $\delta_{k-1} < \|\mathbf{X}' - \mathbf{X}\|_\infty \leq \delta_k$, $\delta_k = 0.1kr$ (by rejection), and then clips it to the box; the clipped point is therefore neither uniform on the shell nor necessarily in it.

Proposition S11 (Coordinate-wise local minimality) *Unless the budget ends first, the local search stops after finitely many evaluations at a point $\mathbf{y}$ with $F_p(\mathbf{y}) \leq F_p(\mathbf{y}_0)$ and*

$$F_p(\text{clip}(\mathbf{y} \pm \bar{h}\mathbf{e}_m)) \geq F_p(\mathbf{y}) \quad \text{for all } m = 1, \dots, n, \quad \bar{h} = h_0/2^5 = 0.0015625r. \quad (\text{S9})$$

In PSO-VNS, if B_2 suffices for the first descent, the Phase-2 incumbent therefore satisfies (S9) and $F_p \leq F_p(\mathbf{G})$, and so does every later incumbent, since each results from a completed descent.

Proof. For a fixed h , every accepted move strictly decreases F_p , so no point is visited twice, and every coordinate of a visited point lies in the finite set $\{a + kh : k \in \mathbb{Z}\} \cap [-r, r]$ with a the value of that coordinate at the start of this h , $-r$ or r ; hence each h ends after finitely many moves, and there are six values of h . The loop ends only by halving the step to below $h_{\min}$, which happens only after a complete sweep with no strict improvement at the current step $\bar{h}$; $\bar{h} = h_0/2^5$ because $0.05/2^5 = 0.0015625 \geq 10^{-3} > 0.05/2^6$. $\square$

Condition (S9) is a coordinate-wise optimality at one step length: $\bar{h}$ is 0.78 m for $r = 500$ m and 1.56 m for $r = 1000$ m. It does not imply optimality for smaller or larger steps, for moves of several coordinates, or a local minimum in the usual sense; F_p is discontinuous (top-hat wake), so no stationarity statement is available either. It does not require a converged PSO start; the start only sets $F_p(\mathbf{y}_0)$. If the budget ends during a descent, the returned layout (the best one evaluated) has F_p no larger than the incumbent but no such certificate. At $B = 6,030$ this is the rule, not the exception, for the larger cases: for $N \geq 12$ the first descent of PSO-VNS is unfinished in 39 of the instrumented runs (Table S53), so Proposition S11 gives no certificate for the largest layouts; with best-improvement polling, one move costs $2n$ evaluations.

S-VII Algorithm Diagnostics

This section looks inside the runs: the dynamics of the two PSO settings (Section S-VII-A), what happens in the VNS phase of the hybrids (Section S-VII-B), why PSO and DE do not move from feasible starts (Section S-VII-C) and the Phase 1 of RS-VNS (Section S-VII-D). The data come from instrumented reruns of subsets of the runs of the paper (`analysis/mpce_diagnostics.py`). The instrumented optimizers are verbatim copies of the code of the paper with logging lines that only read the state and draw no random numbers. All 522 instrumented runs reproduce the stored runs of the paper (same wake loss, coordinates, convergence curve and number of evaluations; 522 of 522), and seed 1 of every case and method, rerun with the original, uninstrumented code in the same process, gives bit-identical results (62 of 62 runs).

Table S52: PSO dynamics per case, old setting / constriction setting (10 seeds each, 6,030 evaluations). Spread: median over the runs of the swarm spread at the last iteration (m); $|V|$: median mean absolute velocity per coordinate at the last iteration (m); clipped: share of coordinates clipped per iteration (iterations 1–200); feasible, outside, spacing: share of particle evaluations in iterations 101–200 with a feasible layout, with a turbine outside the circle, with two turbines closer than $8R$; **G** feas.: runs whose final global best is feasible; L : median wake loss of the feasible final global bests (%). Note below the table: violations of the infeasible evaluations and of the final layouts of all stored runs of both settings (68 cases $\times$ 30 seeds).

Case (DS, r , N)	Spread (m)	$ V $ (m)	Clipped (%)	Feasible (%)	Outside (%)	Spacing (%)	G feas.	L (%)
I, 500, 6	188.9 / 21.7	127.7 / 9.5	8.8 / 1.9	5.8 / 54.8	81.0 / 40.6	79.3 / 10.6	10 / 10	0.52 / 0.43
I, 1000, 10	436.1 / 58.3	268.9 / 21.8	8.4 / 1.3	3.6 / 66.6	87.6 / 23.4	85.1 / 17.8	10 / 10	0.78 / 0.45
I, 750, 12	332.1 / 51.4	190.5 / 18.6	9.1 / 1.6	0.7 / 41.9	92.9 / 31.1	98.8 / 49.0	4 / 10	3.79 / 1.44
II, 750, 8	311.3 / 39.4	177.8 / 16.6	8.7 / 2.0	5.4 / 55.7	86.6 / 35.2	78.8 / 20.2	10 / 10	2.67 / 1.82
II, 500, 10	197.8 / 14.5	125.4 / 4.8	8.4 / 1.5	0.0 / 27.2	89.9 / 36.2	100.0 / 64.4	0 / 10	– / 7.75
II, 1000, 15	488.5 / 53.8	283.8 / 19.3	8.7 / 1.6	0.5 / 44.1	95.5 / 27.6	99.3 / 48.9	3 / 10	7.13 / 4.94
All 6 cases	319.3 / 40.7	187.3 / 15.2	8.7 / 1.6	2.6 / 48.4	88.9 / 32.3	90.2 / 35.2	37 / 60	

Infeasible particle evaluations (iterations 1–200; old / constriction): 98.3 / 68.4% of all; of these, outside the circle only 5.8 / 18.4%, spacing only 7.9 / 34.3%, both 86.3 / 47.3%. Final layouts of all stored runs (old / constriction; feasibility rule of the paper: a turbine more than 10^{-6} m outside the circle, a pair more than 10^{-6} m closer than $8R$): infeasible 286 / 0 of 2,040; of these, outside the circle only 10 / 0, spacing only 224 / 0, both 52 / 0; with a turbine outside the circle that sits on the box bound at a tangent point ($|x| = r$ or $|y| = r$): 60 / 0. (Spacing from the stored minimum spacing; boundary from the coordinates, which are stored to 1 mm; the 46 old-setting finals whose boundary status is not decidable at 1 mm, or that are among the instrumented runs, are classified with the unrounded positions of the reproduced runs. With a 1-mm tolerance instead, 282 finals are infeasible and 246 violate only the spacing.) Final layouts with a coordinate on the box bound: 68 / 2 (infeasible ones: 60 / 0).

Table S53: Inside the VNS phase (Phase 2) of the hybrids at 6,030 evaluations: 12 split cases (both data sets; $r = 500, 750, 1000$ m with $N = 6, 8, 10$ and $N = 10, 12, 15$) $\times$ 10 seeds. A cycle is one shaking step followed by the complete local search (compass search); the first descent is the local search from the switch point, before any shaking; every evaluation after it belongs to a cycle. Evaluation shares are pooled over the runs (sum over runs of the evaluations of a stage divided by the sum of the Phase-2 evaluations); medians of the per-run shares are given separately. Improvement shares refer to the reduction of the wake loss in Phase 2 in the runs whose switch point is feasible (pooled); the gain of the cycles includes the local search of a cycle cut off by the budget.

	PSO-VNS	SSA-VNS	RS-VNS
Runs with feasible switch point	116 of 120	100 of 120	48 of 120
Phase-2 evaluations (median)	3,000	3,000	3,015
First descent: evaluations (median)	2,860	2,920	3,015
share of Phase-2 evaluations (pooled, %)	78.5	75.9	84.8
share of Phase-2 evaluations (median over runs, %)	95.3	97.3	100.0
Shake-plus-search cycles: share of Phase-2 evaluations (pooled, %)	21.5	24.1	15.2
in the runs with a feasible switch point (pooled, %)	22.3	25.8	28.1
runs with at least one cycle evaluation	62	62	42
Shaking steps per run (median)	1.0	1.0	0.0
range over runs	0–3	0–3	0–3
median for $N = 6$	2.0	2.0	2.0
median for $N = 8$	1.0	1.0	1.0
median for $N = 10$	1.0	1.0	0.0
median for $N = 12$	0.0	0.0	0.0
median for $N = 15$	0.0	0.0	0.0
Shaking evaluations (% of Phase 2; rest: local search)	0.02	0.03	0.02
Accepted cycles per run (mean)	0.1	0.1	0.1
acceptance rate of cycles (%)	22.2	25.8	80.0
share accepted with $k = 1$ (%)	100.0	100.0	100.0
accepted with $k = 1, \dots, 5$	6/0/0/0/0	8/0/0/0/0	16/0/0/0/0
runs without an accepted cycle	114	112	104
shaken point already better (% of accepted)	0.0	0.0	0.0
Improvement from the first descent (pooled, %)	97.5	90.3	88.6
Improvement from accepted cycles (%)	0.5	3.4	3.5
Improvement from all cycles, incl. the one cut off (%)	2.5	9.7	11.4
Infeasible at switch, feasible after descent	1	6	11
Infeasible at switch, feasible after a cycle	0	0	0
Feasible incumbent made infeasible	0	0	0
Start is not the best feasible Phase-1 layout ^a	0	0	0

^a runs in which Phase 1 evaluated at least one feasible layout but handed an infeasible or worse layout to VNS (the penalized comparison picks the best feasible layout whenever one exists).

S-VII-A PSO Dynamics of the Two Settings

Fig. S16 and Table S52 compare the old and the constriction setting on 6 cases $\times$ 10 seeds. At the last iteration, the swarm of the old setting is 8.9 times as wide and its mean absolute velocity 13 times as large as with the constriction setting: it contracts much less, consistent with its missing order-2 stability (Proposition S2; Section S-II-B). It clips 8.7% of the coordinates per iteration (constriction: 1.6%), and only 2.6% of its particle evaluations in iterations 101–200 are feasible (constriction: 48.4%). Most of its infeasible evaluations violate both constraints. The infeasible *final* layouts, however, mostly violate only the spacing constraint: with the

Table S54: Why PSO (constriction) and DE do not move from feasible starts: the six largest cases and the Horns Rev 1 16-turbine block $\times$ 3 seeds with feasibility-preserving initialization, 6,030 evaluations; every candidate (trial) is logged; feasibility with the 1e-6 m tolerance of the paper. Mixture test: first-update PSO candidates $\mathbf{P}^i + c_2 \mathbf{r} \odot (\mathbf{G} - \mathbf{P}^i)$ between a particle's feasible start and the global best (200 draws per particle, separate random stream; $c_2 = 1.496$, independent $\mathbf{r} \sim U(0,1)$ per coordinate, clipped to the box), with the original turbine labels and after relabelling the turbines of $\mathbf{P}^i$ to their nearest turbines of $\mathbf{G}$ (assignment problem); then, relabelled, without overshoot (weights in $[0,1]$) and with one common weight (a convex combination of the two layouts).

	PSO	DE
Runs with an improved global best	0 of 21	0 of 21
Candidates evaluated after the initial population	126,000	126,000
feasible (%)	3.33	0.00
feasible, other than re-evaluations of $\mathbf{G}$	0	0
Infeasible candidates: outside the site only / spacing only / both (%)	0.0 / 3.6 / 96.4	0.0 / 0.6 / 99.4
Feasible at the first update (%)	3.3	0.0
Median min. spacing at the first update (m; limit 308, Horns Rev 320)	95	84
Personal-best updates / accepted DE trials	0	0
Feasible candidates better than $\mathbf{G}$	0	0
Runs in which the $\mathbf{G}$ -particle keeps $\mathbf{V} = \mathbf{0}$	21 of 21	–
Iterations with exactly one feasible candidate ($\mathbf{G}$ re-evaluated)	4,200 of 4,200	–
Mixture test (feasible % / turbine outside the site % / median min. spacing, m):		
first PSO move, original labels	0.0 / 93.1 / 91	
first PSO move, relabelled	0.1 / 80.3 / 189	
relabelled, weights in $[0,1]$	0.8 / 39.4 / 222	
relabelled, one weight (convex comb.)	16.6 / 0.0 / 286	
Initial layouts: RMS turbine distance to $\mathbf{G}$ after relabelling (m, median / min)	291 / 111	
All stored runs of the paper (30 seeds): final $\neq$ best initial layout	0 of 210	0 of 210

Table S55: Phase 1 of RS-VNS replayed (geometry only, the seeded random stream of the runs): feasible samples among the 3,015 uniform samples in the bounding square, over 30 seeds (90,450 samples per row; the geometry is the same for both data sets). Inside: all N turbines inside the circle, observed rate and $(\pi/4)^N$. Disc: Monte Carlo feasibility rate of the same number of samples uniform in the disc (the Phase-1 distribution of the disc-sampling control RSD-VNS). Rows with $N \leq 5$ at 750 and 1000 m omitted (all 30 seeds have feasible samples).

r (m)	N	feasible	seeds with one	inside (%)	$(\pi/4)^N$ (%)	disc feasible (%)
500	2	39,952	30	61.45	61.69	71.9
500	3	15,525	30	48.32	48.45	35.4
500	4	3,740	30	37.94	38.05	11.1
500	5	546	30	29.76	29.88	2.11
500	6	53	23	23.12	23.47	0.186
500	7	1	1	18.19	18.43	0.0166
500	8	0	0	14.22	14.48	0
500	9	0	0	11.19	11.37	0
500	10	0	0	8.70	8.93	0
750	6	1,698	30	23.12	23.47	8.28
750	7	458	30	18.19	18.43	2.71
750	8	75	25	14.22	14.48	0.711
750	9	15	13	11.19	11.37	0.129
750	10	2	2	8.70	8.93	0.0155
750	11	0	0	6.89	7.01	0.00221
750	12	0	0	5.44	5.51	0
1000	6	5,275	30	23.12	23.47	25.1
1000	7	2,264	30	18.19	18.43	14.1
1000	8	906	30	14.22	14.48	7.03
1000	9	280	30	11.19	11.37	3.11
1000	10	101	27	8.70	8.93	1.25
1000	11	23	15	6.89	7.01	0.385
1000	12	8	8	5.44	5.51	0.138
1000	13	1	1	4.21	4.33	0.0376
1000	14	0	0	3.26	3.40	0.00663
1000	15	0	0	2.64	2.67	0.00111

10^{-6} m rule of the paper, 224 of the 286 infeasible final layouts of the old setting violate only the spacing, 10 only the boundary and 52 both. In F_p (4), the boundary violation $g_i^b = x_i^2 + y_i^2 - r^2$ is measured in m^2 and the spacing violation g_{ij}^s in m, so a turbine clearly outside the circle is penalized much more than two turbines slightly too close, and the best infeasible layout tends to lie inside the circle with turbines too close. The remaining boundary violations are mostly a penalty-scaling effect as well: in 60 of the 62 final layouts that violate the boundary, an outside turbine is pinned at the box, near a point where the box touches the circle (one coordinate at $\pm r$), where $g_i^b = y_i^2$ is of second order in the other coordinate and hence cheap in F_p . A penalty with normalized violations, e.g., $(\|\mathbf{x}_i\| - r)/r$ and $(\ell_{\min} - \ell_{ij})/\ell_{\min}$, would avoid both effects. The infeasibility of the final layouts of the old setting is therefore mainly a spacing problem, not one of turbines left outside the

circle.

S-VII-B Inside the VNS Phase

Table S53 follows Phase 2 of PSO-VNS, SSA-VNS and RS-VNS on the 12 cases of the split study $\times$ 10 seeds. At 6,030 evaluations, Phase 2 is essentially one compass-search descent: for PSO-VNS, the first descent from the switch point takes a median of 2,860 evaluations (pooled share 78.5% of the Phase-2 evaluations; median share per run 95.3%) and yields 97.5% of the Phase-2 improvement, and 114 of 120 runs have no accepted shaking cycle. A shake costs a single evaluation; the relevant cost is that of the cycles (shake plus the local search that follows it), which take 21.5% of the Phase-2 evaluations (pooled; median run 4.7%) and give 0.5% of the Phase-2 gain (2.5% including the cycle cut off by the budget). For $N \geq 12$ the median run has no shaking step at all, because the first descent almost never finishes within the budget, and all accepted cycles use the smallest neighbourhood $k = 1$. VNS never makes a feasible incumbent infeasible, and its start is always the best feasible Phase-1 layout when Phase 1 evaluated one. The component analysis (Section VII) therefore measures the value of a local-search phase after Phase 1, a truncated best-improvement compass search at this budget; the neighbourhood structure of VNS can matter only for small N , and the local-minimality certificate of Proposition S11 does not apply to the runs in which the first descent is unfinished.

S-VII-C Feasible Starts

Table S54 logs every candidate of PSO and DE from feasible starts (7 cases $\times$ 3 seeds). Neither method improves on the best initial layout, in these runs or in any of the 210 stored runs of each. Of the 126,000 PSO candidates, 3.33% are feasible, and every one of them is the particle holding $\mathbf{G}$ (with zero velocity) re-evaluating $\mathbf{G}$, as in Proposition S7(a); no personal best is ever updated, and none of the 126,000 DE trials is feasible or accepted. Label symmetry does not explain the stall: after the turbines of each start are relabelled to their nearest turbines of $\mathbf{G}$, the first PSO move is still feasible in only 0.1% of the draws, and only a common weight for all coordinates (a convex combination of two layouts) gives 16.6%. The feasible starts are distinct dense layouts (median RMS turbine distance to $\mathbf{G}$ after relabelling 291 m), and the independent per-coordinate weights, with overshoot beyond $\mathbf{G}$, place turbines too close to each other or outside the site.

S-VII-D Phase 1 of RS-VNS

Table S55 replays the random stream of Phase 1 of RS-VNS. The replay agrees with the stored runs, and the share of samples with all turbines inside the circle matches $(\pi/4)^N$ (Proposition S5). In 730 RS-VNS runs, among them all runs of 14 cases, Phase 1 draws no feasible layout at all, and VNS starts from the least-violating sample (Section S-VI-B); in all of these runs the spacing constraint is binding (Section S-III). The last column of Table S55 gives the feasibility rate of samples drawn uniformly in the disc, the Phase-1 distribution of the control RSD-VNS (Section S-III).

S-VIII Model Derivations

Figs. S17–S19 illustrate the farm geometry and the wake model of Section II, and Section S-VIII-A gives the discrete form of the expected power used by the benchmark objective.

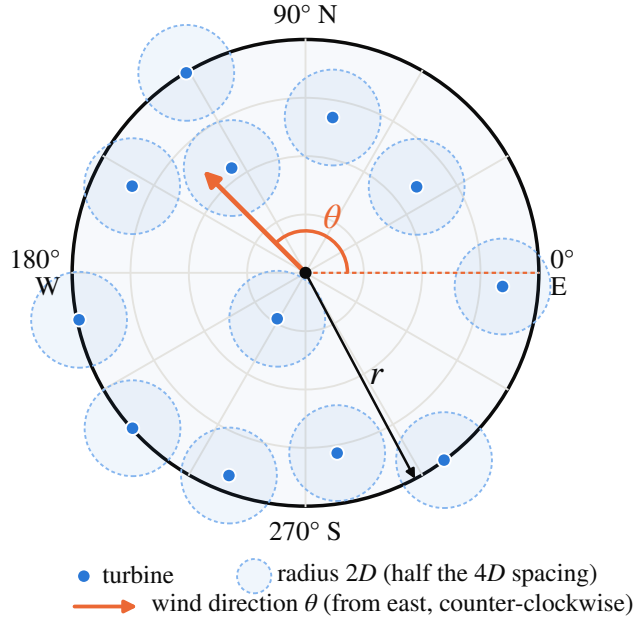


Figure S17: Circular wind farm of radius r . The wind direction θ (direction the wind blows towards) is measured counter-clockwise from east. The points show, for illustration only, the best LX-SSA-VNS layout for 12 turbines in the 750-m farm (Wind Data Set I) from the previous version of this study; the shaded discs have radius $2D$, so two discs touch when their turbines are exactly $4D$ apart.

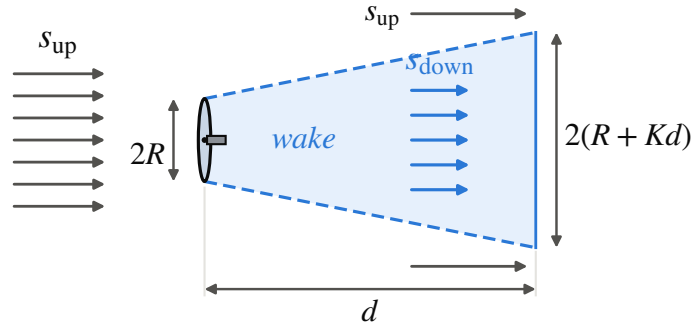


Figure S18: Jensen wake behind a rotor of radius R (schematic, not to scale). The wake widens linearly with the spreading constant K , so that its width at the downstream distance d is $2(R + Kd)$, and the wind speed inside it drops from s_{up} to s_{down} .

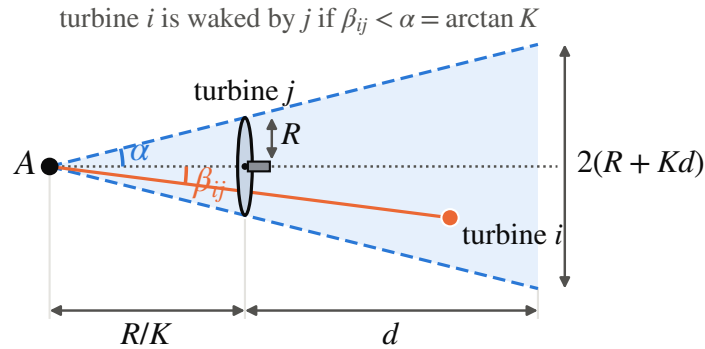


Figure S19: Wake half cone of turbine j (schematic, K exaggerated). The virtual vertex A lies a distance R/K upstream of the rotor, the half-cone angle is $\alpha = \arctan K$, and a turbine i lies in the wake of j when $\beta_{ij} < \alpha$. As in [1], the test also admits a turbine that lies less than R/K upstream of j and close to its axis (between A and the rotor); this non-physical upstream deficit is kept for comparability with the published results.

S-VIII-A Discrete Form of the Expected Power

Following [1], the integrals of (3) are evaluated with a Riemann sum. The wind direction is divided into bins $\theta_0 = 0^\circ, \theta_1, \dots, \theta_{N_\theta} = 360^\circ$ with probabilities p_l , and the speed range between $s_{\text{cut-in}}$ and s_{rated} into bins $s_0 = s_{\text{cut-in}}, s_1, \dots, s_{N_s} = s_{\text{rated}}$. With $\bar{\theta}_l = (\theta_{l-1} + \theta_l)/2$ and $W_i(s, \theta) = \exp\{-[s/\psi_i(\theta)]^{\zeta(\theta)}\}$,

$$E(P_i) = \sum_{l=1}^{N_\theta} (\theta_l - \theta_{l-1}) p_l \left\{ P_{\text{rated}} W_i(s_{\text{rated}}, \bar{\theta}_l) + \lambda_1 \sum_{q=1}^{N_s} \frac{s_{q-1} + s_q}{2} [W_i(s_{q-1}, \bar{\theta}_l) - W_i(s_q, \bar{\theta}_l)] \right. \\ \left. + \lambda_0 [W_i(s_{\text{cut-in}}, \bar{\theta}_l) - W_i(s_{\text{rated}}, \bar{\theta}_l)] \right\}. \quad (\text{S10})$$

Like the earlier studies, we keep the bin-width factor $(\theta_l - \theta_{l-1})$. Because all 24 direction bins are 15° wide and $\sum_l p_l = 1$, the objective equals 15 times the expected farm power in kW, and the gross (wake-only) AEP of the benchmark is $\text{AEP [MWh]} = (\text{objective}/15) \times 8.76$. The linear part of the power curve is -7.0 kW at 3.5 m/s and zero at 3.55 m/s.

S-IX Published Kusiak–Song Results

The Kusiak–Song benchmark has been solved with an evolution strategy [1], ant colony optimization [2], particle filtering [3] and biogeography-based optimization [4]. We do not compare our results numerically with these studies. The published values available to us are transcriptions whose source tables could not be checked against the original papers, several of them are internally inconsistent (e.g., objective and wake loss that do not add up to the wake-free value), and the budgets, the numbers of runs and the handling of infeasible layouts of these studies differ from ours or are not reported. The transcribed values and the status of each are listed in `analysis/literature_values.csv` and `analysis/literature_values.md`.

S-X Reproducibility Map

Table S56 lists which script produces which part of the results; `analysis/README_reproduce.md` gives the commands, the run files and the order in which to call them. The per-run results of the main comparison come from two generations of experiment scripts with the same objective and optimizer code: SSA, LX-SSA, DE, the old-setting PSO, VNS, SSA-VNS and LX-SSA-VNS from the grid scripts of the previous version (`fresh_*.csv`), and PSO, PSO-VNS, RS-VNS, RSD-VNS and MS-SLSQP, as well as the split, initialization, budget, Horns Rev 1 and IEA37 studies, from `analysis/mpce_experiments.py` and `analysis/iea37_experiments.py` (`mpce_*.csv`); `mpce_results.py` records the source file of every run. Later code changes were behavior-preserving: re-runs of archived runs with the current code reproduce their objectives and coordinates exactly. Convergence curves are stored at every $\lfloor (B - 30)/200 \rfloor$ evaluations. Table S57 gives the source of the runs of every method and study.

Table S56: Reproducibility map: script (in `analysis/`) that produces each part of the results. Commands and run files: `analysis/README_reproduce.md`.

Results	Script
Runs of PSO, PSO-VNS, RS-VNS, RSD-VNS, MS-SLSQP; split, feasible-initialization, budget and Horns Rev 1 studies	<code>mpce_experiments.py</code> (one experiment id per study)
Runs of IEA37 Case Study 1	<code>iea37_experiments.py</code>
Runs of SSA, LX-SSA, DE, old-setting PSO, VNS, SSA-VNS, LX-SSA-VNS	<code>full_grid_experiments.py</code> of the previous version, experiments <code>grid</code> (SSA, LX-SSA, DE, old-setting PSO), <code>vgrid</code> (VNS) and <code>bgrid</code> (SSA-VNS, LX-SSA-VNS)
Main-text tables, generated tables of this document (<code>mpce_supplementary.tex</code>), figures in <code>figures_mpce/</code> , <code>mpce_summary.json</code>	<code>mpce_results.py</code> (called by <code>build_mpce_paper.sh</code>)
Number macros <code>mpce_numbers.tex</code> ; checks of outcome-dependent statements	<code>mpce_numbers.py</code> ; <code>mpce_check_final.py</code> (both called by <code>mpce_results.py</code>)
Re-evaluation with the cubic power curve and the Gaussian wake	<code>mpce_robustness.py</code> (cached in <code>mpce_reevaluation.cache.csv</code>)
Spacing re-optimization with SSA and LX-SSA (Table S41)	<code>analyze_authors_runs.py</code>
Packing capacity (Table S40)	<code>packing_capacity.py</code>
Model schematics (Figs. S17–S19)	<code>make_model_figures.py</code>
Verification of the evaluator against the original code and archived runs; IEA37 calculator	<code>validate_evaluator.py</code> , <code>calibrate_authors_code.py</code> ; <code>iea37_model.py</code> (run as a script)
Robustness of the statistical conclusions, equivalence at three levels, equivalence curve (Section S-III-A, Fig. S10); <code>mpce_numbers_extra.tex</code>	<code>mpce_inference_extra.py</code> and <code>mpce_check_extra.py</code> (checks X01–X57), after <code>mpce_results.py</code>
Algorithm diagnostics (Section S-VII); <code>mpce_numbers_diag.tex</code> , <code>diag*.pdf</code>	<code>mpce_diagnostics.py</code> (instrumented reruns, about 18 min; run separately) and <code>mpce_check_diag.py</code> (checks D01–D22)
PSO coefficient sweep and bound handling (Section S-II-B); <code>mpce_numbers_csweep.tex</code> , <code>csweep.pdf</code>	runs: <code>mpce_experiments.py</code> ; analysis: <code>mpce_csweep.py</code> and <code>mpce_check_csweep.py</code> (checks S01–S15)
Direction resolution, Horns Rev 1 re-evaluation and PyWake check, projected IEA37 layouts (Sections S-IV-B, S-IV-C and S-V-C); <code>mpce_numbers_dir.tex</code>	<code>mpce_direction.py</code> , <code>pywake_check.py</code> (PyWake 2.6.20), <code>iea37_projected.py</code> ; <code>mpce_check_dir.py</code> (checks F01–F46)
Theoretical details (Section S-VI), <code>theory*.pdf</code>	<code>make_theory_figures.py</code> ; with <code>--check</code> (about 3 min) it verifies every number of Section S-VI (checks T01–T13)

Table S57: Provenance of the per-run results used in this paper: study, budget (evaluations), initialization, methods, source file (mpce_<exp>: all shards mpce_<exp>_s<i>of<k>.csv of experiment <exp> of mpce_experiments.py / iea37_experiments.py; fresh_*.csv: earlier grid runs, see the repository README) and number of runs.

Study	Budget	Init.	Methods	Source	Runs
Benchmark	6,030	feasible	VNS, DE, LX-SSA-VNS, LX-SSA, PSO, MS-SLSQP, SSA, SSA-VNS	mpce_feas	1440
Benchmark	6,030	feasible	PSO-VNS	mpce_feasp	180
Benchmark	6,030	random	LX-SSA-VNS, SSA-VNS	fresh_bgrid.csv	4080
Benchmark	6,030	random	DE, LX-SSA, PSO (old), SSA	fresh_grid.csv	8160
Benchmark	6,030	random	VNS	fresh_vgrid.csv	2040
Benchmark	6,030	random	PSO-VNS ($\omega = 0.9$)	mpce_omega90	360
Benchmark	6,030	random	PSO-VNS	mpce_psobv	2040
Benchmark	6,030	random	PSO	mpce_psoc	2040
Benchmark	6,030	random	PSO-VNS ($\omega = 0.25$), PSO-VNS ($\omega = 0.75$)	mpce_psosplit	720
Benchmark	6,030	random	RSD-VNS	mpce_rsdisc	2040
Benchmark	6,030	random	RS-VNS	mpce_rsvns	2040
Benchmark	6,030	random	MS-SLSQP	mpce_slsqp	2040
Benchmark	30,030	random	VNS, DE, LX-SSA-VNS, LX-SSA, PSO, RS-VNS, MS-SLSQP, SSA, SSA-VNS	mpce_b30k	1620
Benchmark	30,030	random	PSO-VNS	mpce_b30kp	180
Benchmark	120,030	random	VNS, DE, LX-SSA-VNS, LX-SSA, PSO, RS-VNS, MS-SLSQP, SSA, SSA-VNS	mpce_b120k	1620
Benchmark	120,030	random	PSO-VNS	mpce_b120kp	180
Horns Rev 1	6,030	feasible	VNS, DE, LX-SSA-VNS, LX-SSA, PSO-VNS, PSO, MS-SLSQP, SSA, SSA-VNS	mpce_hrfix	270
Horns Rev 1	6,030	random	VNS, DE, LX-SSA-VNS, LX-SSA, PSO-VNS, PSO, RS-VNS, MS-SLSQP, SSA, SSA-VNS	mpce_hrfix	300
Horns Rev 1	30,030	random	VNS, DE, LX-SSA-VNS, LX-SSA, PSO-VNS, PSO, RS-VNS, MS-SLSQP, SSA, SSA-VNS	mpce_hrfix	300
Horns Rev 1	120,030	random	VNS, DE, LX-SSA-VNS, LX-SSA, PSO-VNS, PSO, RS-VNS, MS-SLSQP, SSA, SSA-VNS	mpce_hrfix	100
IEA37	6,030	random	VNS, DE, LX-SSA-VNS, LX-SSA, PSO, RS-VNS, MS-SLSQP, SSA, SSA-VNS	mpce_iea16	270
IEA37	6,030	random	PSO-VNS	mpce_iea16p	30
IEA37	6,030	random	VNS, DE, LX-SSA-VNS, LX-SSA, PSO, RS-VNS, MS-SLSQP, SSA, SSA-VNS	mpce_iea36	270
IEA37	6,030	random	PSO-VNS	mpce_iea36p	30
IEA37	30,030	random	VNS, DE, LX-SSA-VNS, LX-SSA, PSO, RS-VNS, MS-SLSQP, SSA, SSA-VNS	mpce_iea16	270
IEA37	30,030	random	PSO-VNS	mpce_iea16p	30
IEA37	30,030	random	VNS, DE, LX-SSA-VNS, LX-SSA, PSO, RS-VNS, MS-SLSQP, SSA, SSA-VNS	mpce_iea36	270
IEA37	30,030	random	PSO-VNS	mpce_iea36p	30

References

- [1] A. Kusiak and Z. Song, “Design of wind farm layout for maximum wind energy capture,” *Renew. Energy*, vol. 35, no. 3, pp. 685–694, 2010, doi: 10.1016/j.renene.2009.08.019.
- [2] Y. Eroğlu and S. U. Seçkiner, “Design of wind farm layout using ant colony algorithm,” *Renew. Energy*, vol. 44, pp. 53–62, 2012, doi: 10.1016/j.renene.2011.12.013.
- [3] Y. Eroğlu and S. U. Seçkiner, “Wind farm layout optimization using particle filtering approach,” *Renew. Energy*, vol. 58, pp. 95–107, 2013, doi: 10.1016/j.renene.2013.02.019.
- [4] J. C. Bansal and P. Farswan, “Wind farm layout using biogeography based optimization,” *Renew. Energy*, vol. 107, pp. 386–402, 2017, doi: 10.1016/j.renene.2017.01.064.
- [5] S. Mirjalili *et al.*, “Salp swarm algorithm: A bio-inspired optimizer for engineering design problems,” *Adv. Eng. Softw.*, vol. 114, pp. 163–191, 2017, doi: 10.1016/j.advengsoft.2017.07.002.
- [6] P. Solanki and K. Deep, “Laplacian salp swarm algorithm for continuous optimization,” *Int. J. Syst. Assur. Eng. Manag.*, 2023, doi: 10.1007/s13198-023-01935-y.
- [7] R. Poli, “Mean and variance of the sampling distribution of particle swarm optimizers during stagnation,” *IEEE Trans. Evol. Comput.*, vol. 13, no. 4, pp. 712–721, Aug. 2009, doi: 10.1109/TEVC.2008.2011744.
- [8] D. Kraft, “A software package for sequential quadratic programming,” DFVLR, Inst. für Dynamik der Flugsysteme, Oberpfaffenhofen, Germany, Tech. Rep. DFVLR-FB 88-28, 1988.
- [9] DTU Wind Energy, “PyWake: Open-source static wake modeling framework,” Python package `py_wake`, ver. 2.6.20 (Horns Rev 1 site data in `py_wake/examples/data/hornsrev1.py`), MIT License, 2026. [Online]. Available: <https://gitlab.windenergy.dtu.dk/TOPFARM/PyWake>. Accessed: Sep. 25, 2026.
- [10] M. Bastankhah and F. Porté-Agel, “A new analytical model for wind-turbine wakes,” *Renew. Energy*, vol. 70, pp. 116–123, 2014, doi: 10.1016/j.renene.2014.01.029.
- [11] A. Benavoli, G. Corani, and F. Mangili, “Should we really use post-hoc tests based on mean-ranks?” *J. Mach. Learn. Res.*, vol. 17, no. 5, pp. 1–10, 2016.
- [12] N. F. Baker *et al.*, “Best practices for wake model and optimization algorithm selection in wind farm layout optimization,” in *Proc. AIAA SciTech 2019 Forum*, San Diego, CA, USA, 2019, Paper AIAA 2019-0540, doi: 10.2514/6.2019-0540.
- [13] N. F. Baker *et al.*, “IEA Wind Task 37 wind farm layout optimization case studies: Case files and submitted results.” [Online]. Available: <https://github.com/IEAWindTask37/iea37-wflo-casestudies>
- [14] C. W. Cleghorn and A. P. Engelbrecht, “Particle swarm stability: A theoretical extension using the non-stagnate distribution assumption,” *Swarm Intell.*, vol. 12, no. 1, pp. 1–22, 2018, doi: 10.1007/s11721-017-0141-x.
- [15] A. Benavoli, G. Corani, J. Demšar, and M. Zaffalon, “Time for a change: A tutorial for comparing multiple classifiers through Bayesian analysis,” *J. Mach. Learn. Res.*, vol. 18, no. 77, pp. 1–36, 2017.
- [16] D. Lakens, “Equivalence tests: A practical primer for t tests, correlations, and meta-analyses,” *Soc. Psychol. Personal. Sci.*, vol. 8, no. 4, pp. 355–362, 2017, doi: 10.1177/1948550617697177.